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DRESSER-RAND

USER MANUAL

SBM PETROBRAS
FPSO CIDADE DE PARATY
BRAZIL

MAIN-B COMPRESSOR

A view at the various installation is given below :

- General arrangement Tag No. : K-T7131/41 & K-T7132/42
  Volume 8, Project documentation, drawing HI.39490-AC-002.

Prior to proceeding further in the User Manual it is essential that personnel are familiar with the safety, warnings and procedures for this installation. Therefore proceed to read the safety, warnings and procedures before proceeding.

- Safety, warnings and precautions !
  Volume 1, Compressor Documentation, Part 2, chapter 105170B202

I have read and understood the Safety and Warnings Procedures and wish to proceed.

- **Start the user manual**

Contract/PO No.: 01.39490.0005, 6 & 7	Project Document No.: HI.39490-VDB-002	Area code: N/A	Date: APRIL 2011	Written by : SEGULA
Rev.0 : Volume 10, Technical Support Documentation Document 105170B198		SDL Code: N/A	DR Doc. No.: 105170-IOM	Approved by : JMD

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CHK: D. Elicsche, H. Bruin APP: L. Balkhoven on 05-Nov-2012
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TABLE OF CONTENT

COMPRESSOR DOCUMENTATION..... Volume 1

PART 1 - INTRODUCTION

PART 2 - MAIN DATA

PART 3 - TECHNICAL DESCRIPTION

PART 4 - INSTALLATION

PART 5 - OPERATING INSTRUCTIONS

PART 6 - MAINTENANCE

PART 7 - PARTS AND TOOLS

PART 8 - DOCUMENTS AND DRAWINGS

MANUFACTURER DOCUMENTATION. ..... Volumes 2 to 7

PROJECT DOCUMENTATION..... Volume 8 & 9

TECHNICAL SUPPORT DOCUMENTS..... Volume 10

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DRESSER-RAND

PART 1 - INTRODUCTION

1.1 DOCUMENTATION OVERVIEW 1-1

1.1.1 GENERAL INFORMATION 1-1

1.1.2 USER MANUAL STRUCTURE 1-1

1.1.3 THE USER MANUAL IN PAPER FORMAT 1-1

1.1.4 THE USER MANUAL IN ELECTRONIC FORMAT 1-2

1.1.4.1 DESCRIPTION 1-2

1.1.4.1.1 FORMAT 1-2

1.1.4.2 BROWSERS AND VIEWER 1-2

1.1.5 INFORMATION 1-3

1.1.5.1 DOCUMENTATION 1-3

1.1.5.2 COMPANY 1-3

1.2 CUSTOMER SUPPORT INFORMATION 2-1

1.2.1 GENERAL INFORMATION 2-1

1.2.2 CUSTOMER SERVICE ASSISTANCE 2-1

1.2.3 OVERHAUL AND REPAIR FACILITIES 2-1

1.2.4 CUSTOMER TRAINING 2-2

1.2.5 ORDERING SPARE PARTS 2-2

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1.1 DOCUMENTATION OVERVIEW

1.1.1 GENERAL INFORMATION

The instructions in this manual are directed toward personnel with a general training in operation and maintenance. The presentation of this manual and its information will not relieve the operator of the responsibility of exercising normal good judgement in the care and operation of the machine. The information provided will help to insure a safe, correct, and efficient operation and maintenance of the installed compressor train equipment.

The user manual written by Dresser-Rand Le Havre details the compressor operation and maintenance. Reference or links are made to manufacturers' information, as required, for the convenience of operation and maintenance personnel.

1.1.2 USER MANUAL STRUCTURE

The user manual overall structure is based on guidelines given in the Norsk Standard NS 5820 and recommendations from API 617 and 612 current editions. The structure of the user manual is divided into parts, according to the content of these standards. To make the documentation more user friendly, these parts are numbered 1 - 8. In order to control the documents within the Dresser-Rand document control system, each part is given a Dresser-Rand document number, visible at the document upper right corner.

Parts 1 - 7 include the user manual written by Dresser-Rand Le Havre. Each part is organized into logical chapters, following the product structure as illustrated in the documentation family tree. Each chapter is divided into sections to provide ease in locating information.

Part 8 is divided into 3 sections or files: Suppliers's Manufacturing Documentation, Project Drawing File and Technical Support Documents. The Technical Support File is defined by Dresser-Rand and includes documents considered to be of importance for future maintenance.

The illustration, Documentation Family Tree, provides a metaphorical overview of the user manual structure.

- 105170B198 User Manual - Revision Record Sheet, Chapter 8.3.
- 105170B199 "Family Tree" - User Manual Structure, Chapter 8.3.

1.1.3 THE USER MANUAL IN PAPER FORMAT

The user manual structure is specially designed for electronic format but is usable in a paper version. The paper version is based on print-outs from the documentation program and the illustrating program. The technical drawings are project drawing print-outs.

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1.1.4 THE USER MANUAL IN ELECTRONIC FORMAT

1.1.4.1 DESCRIPTION

The user manual in the electronic format is easy to use and enables the operator, when starting the user manual from the main index or “bookmarks”, to locate at a glance the particular topic from Part 1 to Part 8. Initiating one of the eight parts in the index will bring up the sub index which is the detailed list of the content showing the different chapters within each part. Selecting the part, chapter, or section will bring up the information related to the actual subject chosen.

Electronic links are established within Part 1 to Part 8 of the user manual. Links are made to various other sections of the manual, project documentation, technical drawings or illustrations, as well as to manufacturers’ documentation. A blue information field indicates where a link has been created. In addition to the blue information field, the text references the location (in the paper format) that the link can be found.

To activate a link, place the cursor over the blue information field. A “hand” will appear indicating that the field is electronically linked. Click with the mouse, and the information required will appear on the screen.

By using the “go back” button, the user can toggle between the blue information field and the link. Using the “next” or “previous” page button allows the user to continue paging up or down in the document. In addition to the bookmarks, these options provide the user flexibility in locating information within the manual.

1.1.4.1.1 FORMAT

Parts 1 to 8 are produced utilizing the Adobe Frame Maker 7.0 Documentation Program. The Frame Maker documents are then converted to:

- PDF .pdf

All Manufacturers’ Documentation and SMDL documents will be converted to:

- PDF .pdf

The User Manual will be delivered on CD ROM.

1.1.4.2 BROWSERS AND VIEWER

An Adobe Acrobat Reader, version 6, is provided to view the user manual.

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1.1.5 INFORMATION

1.1.5.1 DOCUMENTATION

For information, please contact:

Dresser-Rand S.A.
31, boulevard Winston Churchill
76080 Le Havre, cedex 7013
France

Telephone: (33)-(0)2-35-25-52-25
Telefax: (33)-(0)2-35-25-53-68

1.1.5.2 COMPANY

URL: http://www.dresser-rand.com

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1.2 CUSTOMER SUPPORT INFORMATION

1.2.1 GENERAL INFORMATION

The Dresser-Rand Service Center can provide aftermarket support functions. These functions include installation, preventive maintenance, factory or field overhaul, technical support and customer training.

Inspection and preventive maintenance of the driven compressor and accessories is strongly recommended at regular intervals. With preventive maintenance, as recommended and detailed within this user manual, the need for corrective maintenance will be reduced.

Note: If the recommended inspection and maintenance schedules are not adhered to any warranties, currently in force, may be invalidated.

1.2.2 CUSTOMER SERVICE ASSISTANCE

The Dresser-Rand Technical Support Department is part of a world wide corporation within Dresser-Rand to fulfil all aftermarket sales functions, with maximum focus on a customer's needs.

The Technical Support Department operates in close co-operation with the Dresser Rand Engineering Department in order to benefit from the experience gained when the equipment was designed.

The Dresser-Rand Service Center performs all types of turbine or compressor service including troubleshooting and solving problems. The service engineers are available on 24 hour basis, at standard service rates for customers all around the world. Long and short term service contracts are available through this department.

Request for personnel assistance should be directed to our nearest service station. Your nearest service station is:

Dresser-Rand Houston Service Center.
Field Project Manager. Tel:(713) 973-3407
1200 West Sam Houston Parkway Norh Fax:(713) 935-3486
HOUSTON, TX 77043
U.S.A.

1.2.3 OVERHAUL AND REPAIR FACILITIES

The Dresser-Rand Service Center is organized to provide all types of turbine repairs including factory overhaul of turbines and components, as well as overhaul of complete power systems.

Also compressor overhaul facilities are available within the Dresser Rand organization.

Request for overhaul or repair assistance should be directed to our nearest service station. Please see the previous section of this chapter for information related to your nearest service station.

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1.2.4 CUSTOMER TRAINING

The Dresser-Rand Service Center provides customer training of personnel on Dresser-Rand delivered equipment. A collaboration between Dresser-Rand and our subcontractors enables Dresser-Rand to arrange training courses on equipment within the Dresser-Rand scope of supply. Training of operating and maintenance personnel is strongly recommended to ensure proper handling of the equipment.

The Dresser-Rand Service Center is prepared to provide any form of training at the training center. This location is recommended for training of personnel since Dresser-Rand has the necessary equipment and knowledge at their factory. However, training courses for operation and maintenance personnel can also be held at site at customer‘s request.

Request for customer training should be directed to our nearest service station. Your nearest service station is:

Dresser-Rand Distribution Center.
Customer Training Manager Tel:(716) 375-3905
Paul Clark Drive Fax:(716) 375-3979
Olean, New York 14760
U.S.A.

1.2.5 ORDERING SPARE PARTS

In the User Manual, Part 7 compressor parts and tools, contains detailed information on how, and where, to order spare parts for the equipment within Dresser Rand’s scope of supply.

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DRESSER-RAND

PART 2 - MAIN DATA

2.1 INSTALLATION OVERVIEW 1-1
2.1.1 GENERAL INFORMATION 1-1
2.1.2 APPLICATION 1-1
2.1.3 DRESSER RAND RESPONSIBILITY 1-1
2.1.4 ARRANGEMENT DRAWINGS 1-1

2.2 SAFETY, WARNINGS AND PRECAUTIONS 2-1
2.2.1 GENERAL 2-1
2.2.2 WARNINGS AND PRECAUTIONS 2-2
2.2.3 CONFORMITY TO THE EUROPEAN REGULATIONS 2-3
2.2.3.1 GENERAL INFORMATION 2-3
2.2.3.2 REPLACEMENT PARTS 2-3
2.2.3.3 SAFETY PRECAUTIONS 2-3
2.2.4 IMPORTANT SAFETY WARNINGS 2-4
2.2.4.1 DEFINITION OF WARNING TERMS 2-4
2.2.4.2 SAFETY LABELS 2-5

2.3 EQUIPMENT MAIN DATA 3-1
2.3.1 GENERAL 3-1
2.3.2 D06R8B COMPRESSORS 3-1
2.3.3 DRIVER 3-2
2.3.4 GEARBOX 3-2
2.3.5 COUPLINGS 3-2
2.3.6 GAS SEAL PANEL 3-3
2.3.7 LUBE OIL SYSTEM 3-3
2.3.8 UCP 3-3

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2.1 INSTALLATION OVERVIEW

2.1.1 GENERAL INFORMATION

This chapter contains a brief overview of Dresser-Rand Le Havre scope of supply for this installation. Part 3 of the User Manual includes the technical description of the components in Dresser-Rand's scope of supply. Manufacturer's documentation is listed in Chapter 8.1. Project documents are listed in Chapter 8.2.

2.1.2 APPLICATION

Two (2) D06R8B compressor trains, driven by electric motor supplied by WEG purchased by SBM for PETROBRAS, FPSO.

2.1.3 DRESSER RAND RESPONSIBILITY

The following items are in Dresser-Rand Le Havre's scope of supply and are included within this User Manual :

1. Two (2) WEG electronic motors, model MGW900A,
   Tag No. KM-T7131/41.

2. Two (2) Dresser-Rand's compressors, model D06R8B, Tag No. K-T7131/32 & K-T7141/42, including auxiliaries.

3. Two (2) FLENDER gear, model TX71/6C, Tag No. KG-T7131/41.

4. Four (4) JOHN CRANE's couplings, model :

- Low speed Couplings Type HLFE-0320-0533-05080.
- High speed Couplings Type HLCE-0159-H533-06100.

2.1.4 ARRANGEMENT DRAWINGS

For a pictorial of the installation, refer to the following general arrangement drawings, Chapter 8.2. :

- HI.39490-AC-002 General Arrangement, D06R8B Compressor Train, K-T7131/41, Chapter 8.2.
- HI.39490-AD-006 General Arrangement, Low Speed Coupling, Motor/Gear, Chapter 8.2.
- HI.39490-AD-010 General Arrangement, High Speed Coupling, Gear/Compressor, Chapter 8.2.

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2.2 SAFETY, WARNINGS AND PRECAUTIONS

2.2.1 GENERAL

Dresser-Rand has a world-wide policy for the safe use of delivered equipment. Included of that policy is the labelling of all equipment and areas that could present a hazard. Hazard to equipment or personnel may occur when procedures relating to safety, operating and maintenance are not followed or protective equipments existence or use neglected. This labelling is considered of prime importance and therefore they should not be obscured, removed or defaced.

It is the responsibility of the management to ensure that all the required warning signs are prominently displayed and that all personnel with access to the installation can understand them. Supervision should ensure that these instructions are adhered to by the personnel. All personnel having control of, or access to, the equipment are to be familiar with the relevant documentation and competent in the use of equipment by having completed an authorized training procedure. This personnel should also be familiar with, and strictly adhered to all safety precautions.

The compressor and it's associated equipment are constructed, and installed, to provide maximum protection to personnel and the environment. Any hazards to the safety of personnel are normally the result of a disregard for normal communication and safety procedures, with the resultant activation of systems while maintenance procedures are being carried out.

An adequate supply of ear protectors, and any other desired protective clothing, must be available for all personnel in the turbine driven compressor vicinity, and their use strictly enforced.

First aid cabinets need to be stocked with extra items that relate to special hazards in that area. A supply of clean cold water will assist in the treatment of burns caused by accidental contact with hot components.

All safety precautions, detailed in this chapter of the user manual, and all other warnings and cautions, detailed in the documentation for the compressor set and associated equipment, are to be followed implicitly.

Where maintenance procedures are being carried out, ensure that all the recommended precautions are taken. Where component parts are from nominated suppliers, refer to any relevant information prior to commencing service operations.

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2.2.2 WARNINGS AND PRECAUTIONS

Only personnel of approved status should attempt to gain access to either the exposed or enclosed components of the installation.

Furthermore it is of vital importance that all personnel should be aware of the inherent dangers present in the installation, as mentioned briefly in the following.

Due to high-frequency sounds, generated by the compressor when operating, it should be mandatory for all personnel in the compressor proximity to wear ear protection. Continued exposure to high-frequency noise, without ear protection, may result in partial deafness and loss of balance.

The compressor when in stand-by mode should always be considered as an operational unit.

The compressor when in stand-by mode during maintenance should be considered as an operational unit, and should therefore be locked-out to prevent an automatic start in the event of a power failure.

Compressor outlet carry gases of very high temperatures, and exposed surfaces may have high temperatures. Care must be taken to prevent personnel from coming into contact with these surfaces. If a burn should be received, immediately apply clean cold water to the affected part for sufficient time to reduce the initial effects. It is vital that any physical injury, or burn resulting from contact with either high temperature surfaces or electricity is examined as soon as possible by a qualified medical attendant.

As a result of compressor test experience it is strongly recommended that the normally accepted operating and test standards, for the “closed loop” operation of the compressor, are modified in accordance with the following to greatly reduce the risk for explosions. Do not operate the compressor with gases containing a combustible amount of oxygen if the discharge temperature can exceed 121 ºC without the injection of inert gas into the intake and discharge end seal inner labyrinths. Any loop or bypass line must be designed and operated so that at least one of the combustion components (oxygen, fuel or heat) is eliminated. As heat will be generated by the compression of the gas, and fuel could be present in form of process gas or lubricating oil seepage from compressor bearings, then the oxygen should be eliminated. Use an inert gas, such as helium, nitrogen or carbon dioxide, during testing on closed loop circuit.

If a special inert gas injection system is used with the compressor, all alarm and shutdown devices must be periodically checked for proper operation. When the unit is operating on process gas, the bleed-down seal system and safety devices must be checked for proper operation.

Extreme caution should be exercised when inspecting or carrying out any operations within the control cabinets. All power to the area should be switched off and, if applicable, the circuit-breakers withdrawn and locked in the isolated position. If it is necessary to come in physical contact with any components of and electrical system that contains capacitors, for example the turbine ignition system, ensure first that the system is isolated and has had at least 10 minutes to de-energize. A voltage check on any exposed or enclosed electrical equipment should be performed as a precaution before any maintenance work is performed.

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The lubricating oil and fuels utilized in the installation are flammable substances. It is therefore of absolute importance to remove any leakage or spillage as soon as possible to avoid fire. Spillage of oil or other liquids onto walkways should be cleaned up immediately. Barriers and warning notices should be erected around the spillage area, to prevent injuries from slippery surfaces, until those surfaces have fully dried and no longer offer a safety hazard.

The lubricating oils, particularly those of synthetic origin, may be toxic in nature and could result in skin injury. Therefore protective gloves and clothing should be worn when dealing with leakage or spillage of lubricants and other fluids.

2.2.3 CONFORMITY TO THE EUROPEAN REGULATIONS

2.2.3.1 GENERAL INFORMATION

The Equipment supplied for this contact has been certified by Dresser-Rand in conformity with the European Community Rules 98/37/EEC and its attachments.

Consequently, any part or component replaced, repaired or modified without prior written approval by Dresser-Rand will definitely cancel this certification.

For equipment installed within the European Community, the E.C. Directive for Machinery and the E.C. label are legal requirements, which if not followed, may lead to an imposed shutdown.

2.2.3.2 REPLACEMENT PARTS

The use of repair parts other than those included within the DRESSER-RAND approved parts list may create hazardous conditions over which the company has no control. Such hazardous conditions can lead to accidents that may be life threatening, cause serious injury or result in damage to the equipment.

2.2.3.3 SAFETY PRECAUTIONS

As part of or implementation of European Norm 85/374 dated 25 July 1985, DRESSER-RAND is identifying on the equipment or material, the risks that have been incurred and other potential risks that may be incurred. The applicable product safety labels are placed in visible locations on the equipment to warn the operator and maintenance personnel of the potential risks.

DRESSER-RAND will provide keep free of charge to the user spare labels in the event that original labels become faded or unreadable.

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2.2.4 IMPORTANT SAFETY WARNINGS

Proper service and repair are important to the safe, reliable operation of the plant equipment. The recommended maintenance as described in Part 6, Maintenance, are effective methods for performing service operations. Some of the service and repair operations require the use of tools especially designed for the purpose. The special tools demanded should be used when and as recommended.

It is important to note that this user manual and other manufacturers information contain various Warnings, Cautions and Notices, which should be carefully read in order to avoid risks of injury to service personnel, or the possibility that improper service methods could be followed, which may damage the equipment or render it unsafe.

The manufacturer could not possible know, evaluate and advise the technician of all conceivable ways in which service might be done, or of the possible hazardous consequences of each way. Consequently, the manufacturer has not undertaken any such detailed broad evaluation, but relays on good technician skill and judgement.

Accordingly anyone who uses a service procedure or tool which is not recommended, must first satisfy themselves thoroughly that neither their or equipment safety will be jeopardized by the service methods selected.

Qualified personnel are persons who, on the basis of their training, experience and knowledge of the relevant standards, codes, accident prevention, regulations and operating conditions have been authorized, by the plant officer in charge, to perform the work required and who can firmly recognize and avoid potential work hazards. A sound knowledge of first aid and the local emergency equipment is also required.

2.2.4.1 DEFINITION OF WARNING TERMS

The terms “Danger”, “Warning”, “Caution” and “Note” indicate specific warnings or special information, which require particular attention:

Danger: Indicates possibilities for immediate hazards which WILL result in fatal or severe personnel injuries and substantial property damage, if the correct precautions are not taken.

Warning: Indicates possibilities for hazards or unsafe practices which COULD result in fatal or severe personnel injuries or substantial product or property damage, if the correct precautions are not taken.

Caution: Indicates possibilities for hazards or unsafe practices which COULD result in minor personnel injuries and/or property damage, if the correct precautions are not taken.

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Note: Draws attention to specific information of technical significance, which might not be obvious to specialist personnel, or points at certain procedures to follow.

All possible protection is afforded to personnel through the provision of guards and safety labels for both mechanical and electrical equipment. Also the provision of safety-interlock features to ensure that equipment cannot be operated to the detriment of personnel or equipment. All personnel in the environment of the installation must be informed about any potential hazards if safety procedures and labels are ignored.

2.2.4.2 SAFETY LABELS

- LHC-000-009 Safety Labels for Operators, Chapter 8.3.
- LHC-000-010 Safety Labels for Containers, Chapter 8.3.
- LHC-000-029 Safety Labels for Special Gas, Chapter 8.3.

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2.3 EQUIPMENT MAIN DATA

2.3.1 GENERAL

This section is giving an overview of the main components only. Referenced Project Data Sheets include product information related to the main components only.

Detailed information on maintenance and operation of the other equipment and instrumentation is provided in the Manufacturer's documentation, Chapter 8.1.

2.3.2 D06R8B COMPRESSORS

Type: Dresser-Rand Compressors Model D06R8B,
Tag Nos. K-T7131/32 & K-T7141/42
Serial No. 1000181442 & 1000181443.

Manufacturer: Dresser-Rand S.A., Winston Churchill Blvd., Le Havre, France.

- 523-264-344 Specification Plate Stamping - D06R8B Compressor
  Tag No K-T7141/42 - Serial No 1000181443, Chapter 8.3.
- 523-264-346 Specification Plate Stamping - D06R8B Compressor
  Tag No K-T7131/32 - Serial No 1000181442, Chapter 8.3.
- HI.39490-AK-002 Compressor Data Sheet, Chapter 8.2.
- HI.39490-AB-002 Compressor Performance Curves, Chapter 8.2.
- LHC-105170B-01 Hydraulic Fit Coupling Dimensions - D06R8B Compressors, Chapter 8.3.
- LHC-105170B-02 Rotors Dimensions - D06R8B Compressor, Chapter 8.3.
- LHC-105170B-03 Heads Connections - D06R8B Compressors, Chapter 8.3.
- LHC-105170B-04 Weights Records - D06R8B Compressors, Chapter 8.3.

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2.3.3 DRIVER

Type: WEG electric motor, Model MGW900L.
Tag No KM-T7111/21/31/41
Manufacturer's ref. No. 11449489

Manufacturer: WEG,
Av.. Frefeito Waldemar Grubba, 3000,
89256-900 - Jaragua do sul - SC - Brazil

- HI.39490-BL-001 Motor Data Sheet, Chapter 8.2.

2.3.4 GEARBOX

Type: FLENDER-GRAFFENSTADEN Gearbox,
Model TX71/6C.
Tag No KG-T7131/41

Manufacturer: FLENDER-GRAFFENSTADEN S.A.,
1, rue du Vieux Moulin
F-67400 ILLKIRCH-GRAFFENSTADEN

- HI.39490-AK-015 Gearbox Data Sheet, Chapter 8.2.

2.3.5 COUPLINGS

Type: John Crane High speed Couplings type HLCE and HLFE.

Manufacturer: John Crane France,
114, rue Jules Ferry
76250 Sotteville-les-Rouen, France

- HI.39490-AK-007 Low speed Coupling Data Sheet, Chapter 8.2.
- HI.39490-AK-011 High speed Coupling Data Sheet, Chapter 8.2.

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2.3.6 GAS SEAL PANEL

Manufacturer: Gesellschaft für OELTECHNIK mbH
Lessingstr. 32
68753 Waghäusel
Germany

2.3.7 LUBE OIL SYSTEM

Main Components:
Lube Oil Console: Tag No. A-T7131 & A-T7141
Lube Oil Reservoir: Tag No. A-T7131 & A-T7141
Lube Oil Pumps: Tag No. P-T7131A/B & P-T7141A/B
Lube Oil Cooler: Tag No. E-T7131 & E-T7141
Lube Oil Filters: Tag No. S-T7133A/B & S-T7143A/B

Manufacturer: Gesellschaft für OELTECHNIK mbH
Lessingstr. 32
68753 Waghäusel
Germany

2.3.8 UCP

Manufacturer: DRESSER-RAND Control Systems
1200 West Sam Houston Parkway North,
Houston, USA - (713) 467-2221

OFFICIAL COPY 00P39490100490.00003--FB003 Reviewed with comments(C)--05-Nov-2012 08:31 AM

DRESSER-RAND

PART 3 - TECHNICAL DESCRIPTION

3.1 COMPRESSOR 1-1
3.1.1 GENERAL INFORMATION 1-1
3.1.2 SERVICE 1-1
3.1.3 COMPRESSOR DESCRIPTION 1-1
3.1.3.1 MODEL EXPLANATION 1-1
3.1.3.2 PRINCIPLE OF OPERATION 1-2
3.1.3.3 ROTOR THRUST 1-2
3.1.4 COMPRESSOR CASE VARIATIONS 1-2
3.1.4.1 COMPOUND COMPRESSOR 1-2
3.1.4.2 SIDESTREAM INLET 1-3
3.1.4.3 INTERSTAGE OUTLET 1-3
3.1.4.4 BRIDGEOVERS 1-3
3.1.5 GENERAL CONSTRUCTION 1-3
3.1.5.1 COMPRESSOR ASSEMBLY 1-3
3.1.5.2 COMPRESSOR CASE 1-4
3.1.5.3 BUNDLE ASSEMBLY 1-4
3.1.5.4 ROTOR ASSEMBLY 1-4
3.1.5.5 LABYRINTH SEALS 1-4
3.1.5.6 HEADS 1-5
3.1.5.7 INNER SEALS 1-5
3.1.5.8 THRUST BEARING 1-5
3.1.5.9 JOURNAL BEARINGS 1-6
3.1.6 OTHER FEATURES 1-6
3.1.6.1 AXIAL MOVEMENT DETECTORS 1-6
3.1.6.2 VIBRATION DETECTORS 1-6

3.2 DRIVER 2-1
3.2.1 GENERAL INFORMATION 2-1
3.2.2 DESCRIPTION 2-1
3.2.3 SPECIFICATIONS 2-1

OFFICIAL COPY 00P39490100490.FB003--FB003 Reviewed with comments (C)--05-Nov-2012 08:15 AM

3.3 GEARBOX 3-1
3.3.1 GENERAL INFORMATION 3-1
3.3.2 DESCRIPTION 3-1
3.3.3 SPECIFICATIONS 3-1

3.4 COUPLINGS 4-1
3.4.1 GENERAL INFORMATION 4-1
3.4.2 DRIVER / GEARBOX COUPLING 4-1
3.4.3 GEARBOX / D06R8B COMPRESSOR COUPLING 4-1
3.4.4 SPECIFICATIONS 4-2
3.4.5 COUPLINGS ALIGNMENT 4-2

3.5 GAS SEAL CARTRIDGE 5-1
3.5.1 GENERAL INFORMATION 5-1
3.5.2 DESCRIPTION 5-1
3.5.2.1 MODEL EXPLANATION 5-1
3.5.2.2 PRINCIPLE OF OPERATION 5-1
3.5.2.3 GAS SEAL ARRANGEMENTS 5-2
3.5.2.4 DRY GAS SEAL CHARACTERISTICS 5-3
3.5.2.4.1 GROOVE PATTERN 5-3
3.5.2.4.2 ANTI HANG-UP “L” SLEEVE 5-3
3.5.2.4.3 SHROUDED SEAT DESIGN 5-3
3.5.3 OTHER FEATURES 5-4

3.6 GAS SEAL PANEL 6-1
3.6.1 GENERAL INFORMATION 6-1
3.6.2 DESCRIPTION 6-1

3.7 LUBE OIL SYSTEM 7-1
3.7.1 GENERAL INFORMATION 7-1
3.7.2 DESCRIPTION 7-1
3.7.2.1 LUBE OIL CONSOLE 7-1
3.7.2.1.1 LUBE OIL RESERVOIR 7-2
3.7.2.1.2 LUBE OIL PUMPS 7-2

OFFICIAL COPY 00P39490100490.FB003--FB003 Reviewed with comments(C)--05-Nov-2012 08:31 AM

3.7.2.1.3 LUBE OIL COOLERS 7-3
3.7.2.1.4 LUBE OIL FILTERS 7-3
3.7.2.2 LUBE OIL RUNDOWN TANK 7-4

3.8 UNIT CONTROL PANEL 8-1
3.8.1 GENERAL INFORMATION 8-1

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3.1 COMPRESSOR

3.1.1 GENERAL INFORMATION

Only subjects related to the technical description of the Dresser-Rand D06R8B compressor, are provided within this chapter. Additional information relating to compressor maintenance is described in this User Manual, Part 6 maintenance, Chapter 6.1 Compressor.

3.1.2 SERVICE

The compressor is for Main B Compressor service.

- HI.39490-AK-002 Compressor Data Sheet, Chapter 8.2.
- HI.39490-AB-002 Compressor Performance Curves, Chapter 8.2.

3.1.3 COMPRESSOR DESCRIPTION

3.1.3.1 MODEL EXPLANATION

D06R8B COMPRESSOR	MODEL DESIGNATION
D	“DATUM” Centrifugal Compressor (Product line)
06	Frame Size (Based on case inside diameter)
R	Radial (Vertical) Split Casing
8	Actual number of impellers
B	“Back-To-Back” Configuration*

* There are two types of rotor configurations available for each model and selected for the specific operating conditions. A unit with a “Straight Thru” configuration (“thru-rotor”) is one which has all the impellers facing the same direction and is provided with a balance drum (piston) at the discharge end. The balance drum is used to reduce the rotor thrust to a nominal value that can be carried on the thrust bearing shoes. A unit with “Back-To-Back” rotor is one which has the second section end impellers facing the opposite direction of the first section end which eliminates the need for a conventional balance drum.

Additionally the compressor may be a double flow unit type with two process inlet connections and a single discharge. The configuration is then denoted by adding the letters “DF” following the casing designations as for example, D29RDF6.

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3.1.3.2 PRINCIPLE OF OPERATION

The rotor impellers, when rotating at operating speed, impart energy into the process gas which is flowing through them. The process gas enters the intake (suction) volute of the compressor and flows through the inlet wall and first stage inlet guide which directs the gas at a suitable angle into the "eye" of the first stage impeller. The gas is then discharged from the periphery of the impeller at a high velocity by centrifugal force into the passage formed by the inlet wall and first stage diaphragm. The gas then flows around the return bend, through a vaned, annular passage in the diaphragm and into the second stage impeller. The annular passage through the diaphragm is diverging (increasing in area as the gas flows toward the impeller eye) thereby converting part of the velocity energy of the gas into pressure energy. After subsequent stages of compression, the gas flows from the compressor discharge nozzle to the customer's process.

When a "Back-To-Back" rotor is provided, the impellers are designated as first section and second section. The process gas is compressed by the first section impellers, is discharged from the first section to 2nd section intake, or to customer process for cooling or process requirements then returned to the compressor 2nd section intake. The gas is then further compressed by the second section impellers from where it flows into the discharge volute and exists to the customer's process.

3.1.3.3 ROTOR THRUST

Because of a difference in area between the side walls of each impeller, the axial thrust of the rotor assembly is toward the inlet "eye" of the impellers. In order to limit this thrust to a reasonable but still positive value for long thrust bearing life, a balance drum (piston) is secured to the shaft at the discharge end following the last stage impeller. A mating labyrinth seal is bolted to the discharge volute. The impeller side of the balance drum is subject to compressor discharge pressure (approximate). The area at the opposite side of the drum is externally piped to a lower pressure area by the balance lines. The difference in pressure on each side of the balance drum produces thrust which opposes the natural thrust of the rotor toward intake. Thrust is thereby reduced to a nominal value that can be absorbed by the active thrust bearing shoes.

When a compressor is provided with a rotor having "Back-to-Back" mounted impellers, the balance drum is eliminated. The second section end impellers, being mounted in the opposite direction from the first section, produces thrust which opposes the thrust generated by the first section impellers. The impeller configuration nearly cancels overall rotor thrust with the residual thrust easily absorbed by the active thrust bearing shoes.

3.1.4 COMPRESSOR CASE VARIATIONS

3.1.4.1 COMPOUND COMPRESSOR

"Compound Compressor" is essentially two compressor sections in one case, with two sets of impellers mounted on a common shaft. Gas enters the compressor intake, is compressed, and discharges from the first section to process. Usually, the same gas (at different pressure and temperature due to process) re-enters the second section intake for further compression.

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3.1.4.2 SIDESTREAM INLET

"Sidestream Inlet" is a second intake connection at an interstage location in the compressor. In this application, gas at higher pressure than compressor suction is piped from process and passed into the sidestream inlet to mix with the gas already passing through the compressor, for further compression.

3.1.4.3 INTERSTAGE OUTLET

"Interstage Outlet" is an insterstage tap or discharge connection from which a percentage of the total gas flow, at some intermediate pressure, is removed for process or other purpose.

3.1.4.4 BRIDGEOVERS

"Bridgeovers" are provided to reduce the number of compression stages in a compressor for several reasons :

- To provide a given pressure rise across the compressor to suit available horsepower or process requirements.
- To provide a casing which, with minimum changeouts of internal components (including rotor), can be converted to a higher pressure ratio.
- To provide a means of bringing (with a sidestream "in" connection) a cooled (similar or compatible) gas stream into the process under compression and hold the overall temperature rise across compressor to a nominal value while increasing capacity.
- To provide a means of bypassing (with an interstage outlet connection) a percentage of the total gas flow process requirements.
- To provide an internal area in compound compressors for "out" and "in" connections, whereby process gas is cooled and returned to compressor, or "in" gas may be mixed with a similar or compatible in greater volume for increased overall capacity.

3.1.5 GENERAL CONSTRUCTION

3.1.5.1 COMPRESSOR ASSEMBLY

The "Compressor Assembly" consists of a case, an internal bundle assembly, a rotor, bearings and seals. The unit is supported on both sides of the case at approximately the rotor centerline and shims are used under the supports to obtain correct alignment.

- 1000183022 Unit Assembly Drawing - Compressor S/N 1000181442, Chapter 8.3.
- 1000183412 Unit Assembly Drawing - Compressor S/N 1000181443, Chapter 8.3.

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3.1.5.2 COMPRESSOR CASE

The “Compressor Case” is a one-piece cast or forged steel barrel which contains the bundle and rotor assembly.

- 1000183026 Case Assembly Drawing - Compressor S/N 1000181442, Chapter 8.3.
- 1000183415 Case Assembly Drawing - Compressor S/N 1000181443, Chapter 8.3.

3.1.5.3 BUNDLE ASSEMBLY

The “Bundle Assembly” is built up around, and with, the rotor assembly and consists of the rotor, heads, bearings, inner seals, and internal assembly including return bends, diaphragms, inlet wall, inlet guide, etc...All components are in halves, split at the horizontal centerline. Capscrews at the splitline secure mating halves of the bundle. The complete bundle assembly (including rotor) is removed and installed through the service end of the compressor case and retained in the case by the service end head.

- 1000183025 Bundle Assembly Drawing - Compressor S/N 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing - Compressor S/N 1000181443, Chapter 8.3.

3.1.5.4 ROTOR ASSEMBLY

The “Rotor Assembly” consists of a shaft on which are assembled the impellers, impeller spacers, balance piston (when used), impeller shroud ring and split rings, thrust bearing thrust disc and coupling hub (two hubs if a through drive compressor). The balance piston (when used), impellers and spacers are “moderate” shrink fit on the shaft. They are located axially and retained by split rings fitted in annular grooves at each end of the shaft. The split rings are secured by shrink fit shroud rings. The coupling hub and thrust disc are “heavy” shrink fit on the shaft and are hydraulically expanded at assembly and removal. The rotor can be driven from either end through a “flexible” type coupling. The coupling hub is secured to the tapered end of the shaft with a locknut.

- 1000183031 Rotor Assembly Drawing - Serial No. 10-01-050, Chapter 8.3.
- 1000183417 Rotor Assembly Drawing - Serial No. 10-01-051, Chapter 8.3.

3.1.5.5 LABYRINTH SEALS

“Labyrinth Seals” are installed in the inlet guide vanes and in the interstage diaphragms. The labyrinth seals are in halves, secured by splitline holding screws, and are replaceable. The clearance between the seals and impeller covers and impeller spacers is small to limit gas leakage from discharge to intake pressure areas of the impellers.

- 1000183030 Internal Assembly Drawing - Compressor S/N 1000181442, Chapter 8.3.
- 1000192540 Internal Assembly Drawing - Compressor S/N 1000181443, Chapter 8.3.

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3.1.5.6 HEADS

"Heads" are secured to the compressor case with shear rings and retaining rings to form a pressure-tight vessel . The heads contain the inner seals and support the thrust bearing housing and journal bearing housings bolted onto the outer face.

- 1000183025 Bundle Assembly Drawing - Compressor S/N 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing - Compressor S/N 1000181443, Chapter 8.3.

3.1.5.7 INNER SEALS

"Inner Seal Assemblies" are installed in the heads cavity, adjacent to first stage inlet guide vane, and the balance piston (when used). The seals are designed to positively prevent gas leakage into the bearing chambers. Refer to User Manual, Part 5 operating instructions, Chapter 5.1 operating procedures, for description of inner seals function used in subject compressor.

- 1000183034 Gas Seal Assembly - 1ST Section, Drawing, Chapter 8.3
- 1000183035 Gas Seal Assembly - 2ND Section, Drawing, Chapter 8.3.

3.1.5.8 THRUST BEARING

"Thrust Bearing" is JC Sartorius self equalizing type. Rotor thrust is normally toward the intake end of compressor. However, under certain operating conditions, it is possible to have a reversal of thrust which necessitates a double direction thrust bearing. Lubricating oil from a force-feed system at approximately 20 psig (1.4 kg/cm^2, 138 kPA) is supplied to the bearings. The active shoes pivot against the leveling plates and base ring to form an oil film "wedge" to protect the thinly babbitted shoe faces from excessive wear. The thrust bearing are horizontally split for easy maintenance and can be removed /installed with thrust disc in place.

- 1000183032 Thrust & Journal Bearing Assembly - 1ST Section End, Drawing, Chapter 8.3.
- 1000183046 JC SARTORIUS Thrust Bearing Drawing, D06R8B Compressor, Chapter 8.3.
- LHA-105170B-89 JC SARTORIUS Thrust Bearing Instructions Manual, Chapter 8.1.

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3.1.5.9 JOURNAL BEARINGS

"Journal Bearings" are DRESSER-RAND self-aligning tilting pad type comprising a bearing cage and five babbitt faced pads or shoes which tilt to form an oil wedge in a manner similar to that of the thrust bearing shoes. The journal bearing housings are in halves and are bolted to the heads. Lubricating oil is supplied to these bearings at approximately 20 psig (1.4 kg/cm^2, 138 kPA).

- 1000183032 Thrust & Journal Bearing Assembly - 1ST Section End, Drawing, Chapter 8.3.
- 1000183045 DRESSER-RAND Journal Bearing Drawing - 1ST Section End, Drawing, Chapter 8.3.
- 1000183033 Journal Bearing Assembly - 2ND Section End, Drawing, Chapter 8.3.
- 1000183051 DRESSER-RAND Journal Bearing Drawing - 2ND Section End, Drawing, Chapter 8.3.
- LHA-105170B-88 DRESSER-RAND Journal Bearings Instructions Manual, Chapter 8.1.

3.1.6 OTHER FEATURES

3.1.6.1 AXIAL MOVEMENT DETECTORS

Electronic proximity sensing probes are used to detect and indicate rotor shaft axial movement. Correctly calibrated, the sensor and indicating components will "read-out" axial movement of approximately 0.001 inch (0.025 mm). The indicating member may be connected to either alarm or shutdown (or both) to warn of possible thrust bearing failure. The alarm is to be set at 0.005 inches (0.127 mm) rotor movement from its "thrusted" position toward the sensor. Shutdown is to set at 0.010 inches (0.254 mm) movement.

3.1.6.2 VIBRATION DETECTORS

Electronic proximity sensing probes are used to detect and indicate rotor shaft radial vibration. Setpoint actuation for shutdown is at a value which will provide ample protection against possible damage. A momentary time delay is usually incorporated in such controls to prevent inadvertent shutdown during start-up when accelerating to rated speed and passing through the critical speed ranges of the equipment .

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3.2 DRIVER

3.2.1 GENERAL INFORMATION

Only subjects related to the technical description of the WEG motor, are provided within this chapter. Additional information relating to the motor maintenance is described in this User Manual, Part 6 maintenance, Chapter 6.2 Driver. Refer also to the WEG documentation for informations pertaining to installation procedures and maintenance, Part 8, Chapter 8.1.

- LHA-105170B-01 WEG Instruction Manual, Motor MGW900A, Chapter 8.1.

3.2.2 DESCRIPTION

The WEG motor MGW900A, KM-T7111/21/31/41, is installed on a common baseplate with the gearbox and the compressor.

- HI.39490-AC-002 Compressor Train General Arrangement, Chapter 8.2.

The motor drives the low speed input gearbox shaft, through a disc coupling, at the design mechanical power and speed in order to match the compressor requirements. The WEG motor operates at a constant speed motor.

The electrical motor is equipped with air cooling system for the internal winding.

- HI.39490-AD-017 WEG Motor Outline Drawing, Chapter 8.2.

The motor bearings are lubricated through a common lube oil system supplied by Dresser-Rand.

3.2.3 SPECIFICATIONS

- HI.39490-BL-001 WEG Electrical Motor Data Sheets, Chapter 8.2.

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3.3 GEARBOX

3.3.1 GENERAL INFORMATION

Only subjects related to the technical description of the FLENDER gearbox, model TX71/6C, are provided within this chapter. Additional information relating to the gearbox maintenance is described in this User Manual, Part 6 maintenance, Chapter 6.3, gearbox. Refer also to the FLENDER documentation for informations pertaining to installation procedures and maintenance, Part 8, Chapter 8.1.

- LHA-105170B-56 FLENDER Operating & Maintenance Instructions, Chapter 8.1.

3.3.2 DESCRIPTION

The FLENDER gearbox model TX71/6C is installed on common baseplate, between driver motor and the compressor to increase the output motor speed in order to match the required compressor speed.

- HI.39490-AK-002 Compressor Data Sheet, Chapter 8.2.

The gearbox is built with parallel shafts, and is using a double helical design for high speed pinion and low speed gear.

- LHA-105170B-56 FLENDER Operating & Maintenance Instructions, Chapter 8.1.

The gearbox is driven by the motor through a low speed disc coupling, and is driving the compressor through a high speed disc coupling. Refer to Part 3 technical description, Chapter 3.4 couplings, for details.

The gearbox is lubricated through a common lube oil system supplied by Dresser-Rand.

3.3.3 SPECIFICATIONS

- HI.39490-AK-015 Gearbox Data Sheet, Chapter 8.2.

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3.4 COUPLINGS

3.4.1 GENERAL INFORMATION

Only subjects related to the technical description of the J.CRANE couplings are provided within this chapter. Additional information relating to the couplings maintenance is described in this User Manual, Part 6 maintenance, Chapter 6.4 couplings.

3.4.2 DRIVER / GEARBOX COUPLING

The coupling transmits the torque from the Driver to the Gearbox.

The ambient environmental conditions will affect the thermal growth of the bearing supports. The coupling maintains train alignment within limited angular and offset misalignment tolerances between the Driver and the Gearbox. However, a correct alignment is essential for reliable service and equipment life. These must be within the specified close tolerance during the installation (cold) and then re-confirmed when the train is still hot (no more than 10 minutes after a shutdown).

The low speed coupling, between Driver and Gearbox is John Crane coupling, model HLFE-0320-0533-05080. For detailed information on coupling weights and coupling transmission torque, refer to John Crane instructions, User Manual, Part 8 documents and drawings, Chapter 8.1 manufacturer documentation.

- LHA-105170B-02 J.CRANE Couplings Fitting and Maintenance Instructions, Between Driver & Gearbox, Chapter 8.1.

- HI.39490-AD-006 J.CRANE Low Speed Coupling Drawing (Driver/Gearbox), Chapter 8.2.

This low performance type coupling is a precision built coupling comprising two flexible diaphragm assemblies connected by a torque tube called a “coupling spacer”. The couplings hubs are hydraulically fitted on the compressors shafts ends. The low speed coupling has a total enclosed type coupling guard.

- LHA-105170B-91 Coupling Guard Drawing & Parts List, Between Driver & Gearbox, Chapter 8.3.

3.4.3 GEARBOX / D06R8B COMPRESSOR COUPLING

The coupling transmits the torque from the Gearbox / Driver to the D06R8B compressor.

The ambient environmental conditions will affect the thermal growth of the bearing supports. The coupling maintains train alignment within limited angular and offset misalignment tolerances between the Gearbox and the compressor. However, a correct alignment is essential for reliable service and equipment life. These must be within the specified close tolerance during the installation (cold) and then re-confirmed when the train is still hot (no more than 10 minutes after a shutdown).

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The high speed coupling, between Gearbox and D06R8B compressor is John Crane coupling, model HLCE-0159-H533-06100. For detailed information on coupling weights and coupling transmission torque, refer to John Crane instructions, User Manual, Part 8 documents and drawings, Chapter 8.1 manufacturer documentation.

- LHA-105170B-03 J.CRANE Couplings Fitting and Maintenance Instructions, Between Gearbox & Compressor, Chapter 8.1.

- HI.39490-AD-010 J.CRANE High Speed Coupling Drawing (Gearbox/Compressor), Chapter 8.2.

This high performance type coupling is a precision built coupling comprising two flexible diaphragm assemblies connected by a torque tube called a “coupling spacer”. The couplings hubs are hydraulically fitted on the compressors shafts ends. The high speed coupling has a total enclosed type coupling guard.

- LHA-105170B-92 Coupling Guard Drawing & Parts List, Between Gearbox & Compressor,Chapter 8.3.

3.4.4 SPECIFICATIONS

- HI.39490-AK-007 Low Speed Coupling Data Sheet (Driver & Gearbox), Chapter 8.2.

- HI.39490-AK-011 High Speed Coupling Data Sheet (Gearbox & D06R8B Compressor), Chapter 8.2.

3.4.5 COUPLINGS ALIGNMENT

A train alignment chart drawing is provided defining the offset alignment requirement between the driver, gearbox, and the compressor. In addition to the alignment drawing, installation requirements are included in couplings manufacturers documentation, Chapter 8.1.

- ALN-1000194939 Compressor Train Alignment Chart, Chapter 8.3

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3.5 GAS SEAL CARTRIDGE

3.5.1 GENERAL INFORMATION

Only subjects related to the technical description of the Dresser-Rand gas seal, model L200W 160.2 is provided within this chapter. Additional information relating to the gas seal cartridge maintenance is described in this User Manual, Part 6 maintenance, Chapter 6.5 gas seal cartridge. Refer also to the compressor maintenance for information pertaining to installation procedures and maintenance, Part 6, Chapter 6.5.

- 1000184143 Gas Seal General Arrangement, 1ST Section End, Chapter 8.3.
- 1000184144 Gas Seal General Arrangement, 2ND Section End, Chapter 8.3.

3.5.2 DESCRIPTION

3.5.2.1 MODEL EXPLANATION

GAS SEAL MODEL	DESIGNATION
L200	Dresser-Rand Product Line Designation
W	“Tandem with Intermediate Labyrinth” Arrangement.
160.2	Seal Code

3.5.2.2 PRINCIPLE OF OPERATION

The Dresser-Rand gas seal utilizes the principle of non-contacting mechanical seals to replace "oil film seal" technology.

There are three main components of the dry gas seal.

A **rotating ring,** which is attached to the shaft. In rotation, special grooves on the face of the ring, capture the gas. It is typically comprised of tungsten or silicon carbide.

A **stationary ring,** or static ring, is attached to the compressor casing. It is typically made from a carbon material, and has a highly polished seal face.

A **pusher sleeve,** also called an “L” sleeve, is spring loaded.

- LHC-000-069 Components of a Gas Seal, Chapter 8.3.

As the compressor shaft and the attached seal ring rotate, gas is drawn inward to the root of the groove. The shape of each groove is such that the gas is forced into the closed end, or sealing dam, where a pressure rise occurs.

- LHC-000-070 Description of Gas and Groove Operation, Chapter 8.3.

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The pressure build-up between the rings forces them apart by a few microns. Some process gas leakage does occur; it is predictable and a small quantity. This low level leakage may be sent to flare, vented to a safe area or reused. The quantity of leaking gas varies according to pressure, speed and temperature.

In the dynamic condition, there is a fine balance between the closing and the opening forces. The closing forces result from the spring load plus the pressure applied on the outboard face of the stationary ring. The opening forces result from the pressure gradient (due to the effect of the grooves) existing between the stationary and the rotating rings.

There must be a sufficient high film stiffness to prevent the faces from coming into contact during rotation. High stiffness of the gas film also enables the seal to withstand transient axial displacement and compressor surge.

3.5.2.3 GAS SEAL ARRANGEMENTS

Although the principle of operation remains the same for all dry gas seals, various configurations of seal arrangements exist. Thus the particular gas seal arrangement for any contract is determined by the application requirements.

- LHC-000-072 Gas Seal Arrangements, Chapter 8.3.

A **tandem seal** has two pairs of rotating/stationary rings in series, which face in the same direction. In this configuration, the inboard side seal operates against the full pressure to be sealed, leaving the outboard side seal as a back-up. It is the most common arrangement for Dresser-Rand gas seals.

A **tandem with intermediate labyrinth** is similar to a tandem seal except that the inboard and outboard seals operate with two different gases. In this case, a labyrinth is installed between the inboard and outboard seals.

A **single seal** has only one pair of rotating/stationary rings. This type of seal is used where the leakage of process gas is not critical (i.e. failure of the seal would not release dangerous gases).

A **double seal** has two pairs of rotating/stationary rings, positioned back-to-back. The rings operate in the opposite directions. This type of seal is used for very low pressure applications, and/or where the process gas is too toxic, corrosive or volatile and cannot be used to pressurize the gas seal. For this application the sealing gas must be compatible with the process gas, as the sealing gas flows through both the inboard and outboard seals.

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3.5.2.4 DRY GAS SEAL CHARACTERISTICS

The Dresser-Rand gas seal design incorporates several features that are unique to this product.

3.5.2.4.1 GROOVE PATTERN

In order to minimize the leakage rate of a gas seal, the gap between the two faces of the gas seal must be minimized. However, the smaller the gap, the risk for accidental contact between the two faces of the seals, is increased. The Dresser-Rand groove pattern (patented) is designed to generate an optimized pressure distribution and high gas film stiffness.

- LHC-000-073 Pressure Profile in the Groove Area, Chapter 8.3.

3.5.2.4.2 ANTI HANG-UP "L" SLEEVE

The risk of "hang-up" or increased leakage rates that prevent pressurization of the seal is the result of a combination of several factors or events, that occur simultaneously. A single factor will not cause "hang-up syndrome" to occur; typically it is identified in situations where there is low or lack of gas pressure to the gas seals, and the O-rings in the gas seal "stick" or extrude.

The following example is used to explain how "hang-up" may occur:

> In the event of a very low sealing pressure, (at the start of the compressor pressurization process), only the gas seal spring force acts to close the seal gap. If simultaneously, the shaft of the compressor has a small axial displacement and if the friction between the O-ring and the seal housing, or deformation of the O-ring occurs, which prevent the springs to close the seal, then a large gap (50 -100 times the normal gap) may appear between the two faces of the seal. The leakage through this interface then may be so high that pressurization of the the compressor may be impossible.

- LHC-000-075 O-Ring Location, Chapter 8.3.

This is known as the "hang-up". The only remedy is to disassemble the compressor end, remove the seal cartridge, fix the gas seal and reinstall it.

Dresser-Rand developed and incorporated in the D-R Gas Seal, an "L" sleeve design which reduces the risk of "hang-up' significantly.

3.5.2.4.3 SHROUDED SEAT DESIGN

Typically, the rotating seats of gas seals are made of tungsten carbide or silicon carbide. In the unlikely event of a severe rub (implying a thermal shock and/or "heat crazing") or in case of a scratched face breaking at high speed, the brittle material of the seat may "burst" into fragments. These rotating fragments could damage the seal cavity and shaft.

- LHC-000-076 Shrouded Seat Design, Chapter 8.3.

The standard Dresser-Rand gas seal design is with shrouded seats. The shrouds are designed to retain the fragments of a "burst" seat, thus reducing the potential for damage to the seal and/or the compressor.

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3.5.3 OTHER FEATURES

The Dresser-Rand gas seal cartridge is used with a barrier seal (supplied by Flowserve).

- 1000183034 Gas Seal Assembly - 1st Section, Drawing, Chapter 8.3.
- 1000183035 Gas Seal Assembly - 2nd Section, Drawing, Chapter 8.3.

In the compressor, the barrier seal is located between the bearing and the gas seal cartridge.

- 1000183025 Bundle Assembly Drawing, Compressor S/N 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, Compressor S/N 1000181443, Chapter 8.3.

The barrier seal is installed to prevent any bearing oil vapors from passing along the shaft into the gas seal.

The barrier seal is delivered as complete assembly, “ready-to-install”. For barrier seal description, please refer to Flowserve documentation, User Manual, Part 8 documents and drawings, Chapter 8.1 manufacturer’s documentation.

- LHA-105170B-04 Flowserve Barrier Seal Instructions, Chapter 8.1

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3.6 GAS SEAL PANEL

3.6.1 GENERAL INFORMATION

Only subjects related to the technical description of the gas seal panel, is provided within this chapter. Additional information relating to the instrumentation is described in this User Manual, Part 8 documents and drawings, Chapter 8.1 supplier's manufacturer documents.

3.6.2 DESCRIPTION

A gas seal panel is supplied attached onto the edge of compressor skid, and is equipped with instrumentation to control seal gas injection .

- HI.39490-AC-002 Compressor Train General Arrangement , Chapter 8.2.
- HI.39490-AD-002 Gas Seal Panel General Arrangement, Chapter 8.2.

These instruments can send alarm and shutdown signals to the unit control panel for the protection of Dresser-Rand gas seal cartridges fitted in the compressor. Refer to Part 5, chapter 5.1, Operating Procedures, for operating details of instrumentation .

For complete information on the various instruments of gas seal panel, refer to Part 8 documents and drawings, Chapter 8.1 supplier's manufacturer documents.

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3.7 LUBE OIL SYSTEM

3.7.1 GENERAL INFORMATION

Only subjects related to the technical description of the OELTECHNIK lube oil system, including main equipments, lube oil console and rundown tank are provided within this chapter. Additional information relating to lube oil system maintenance is described in this User Manual, Part 6 maintenance, Chapter 6.6 lube oil system. Refer also to OELTECHNIK documentation for informations pertaining to installation procedures and maintenance, Part 8, Chapter 8.1.

- LHA-00-3677 Oeltechnik Lube Oil Console Instruction Manual, Chapter 8.1.
- LHA-105170B-41 Oeltechnik Lube Oil Console Instruction Manual, Chapter 8.1.

3.7.2 DESCRIPTION

3.7.2.1 LUBE OIL CONSOLE

The lube oil console is a part of the complete lubrication system that includes the oil distribution and return, to the oil reservoir.

The OELTECHNIK lube oil console is a stand alone unit, built on a baseplate.

- H.39490-AC-017 Lube Oil Console General Arrangement, Chapter 8.2.

Main components of OELTECHNIK lube oil console are:

1. The lube oil reservoir.
2. The lube oil pumps.
3. The lube oil coolers.
4. The lube oil filters.

- HI.39490-AF-006 Lube Oil Console P&ID, Chapter 8.2.
- HI.39490-AS-007 Lube Oil Console List Of Materials, Chapter 8.2.
- HI.39490-AF-007 Lube Oil System P&ID, Chapter 8.2.
- HI.39490-AS-010 Lube Oil System List Of Materials, Chapter 8.2.

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3.7.2.1.1 LUBE OIL RESERVOIR

The lube oil reservoir A-T7131 & A-T7141 is designed to provide sufficient capacity of oil at a minimum operating level. The minimum retention time is based on a steady state flow, during normal operation. The reservoir is to be filled with oil at the correct specification.

- LHC-000-037 Lubricating Oil Recommendations - Iso Grade 32, Chapter 8.3.

To maintain an acceptable oil viscosity level during start-up, the oil reservoir is equipped with electric heaters, Tag No. H-T7131 & H-T7141, supplied by Elmess.

- LHA-105170B-20 Elmess Heater Specification & Drawing, Chapter 8.1.
- LHA-105170B-21 Elmess Heater Instructions, Chapter 8.1.

Instrumentation, to monitor the entire system, is installed onto a gauge board fixed on the skid edge.

For complete information on the various components of lube oil reservoir, refer to Part 8 documents and drawings, Chapter 8.1 supplier's manufacturer documents.

3.7.2.1.2 LUBE OIL PUMPS

The lube oil pumps Tag No. P-T7131A/B & P-T7141A/B are Leistritz screw pumps, type IFOKTO-G.

- LHA-105170B-35 Leistritz Oil Pump Drawing, Chapter 8.1.
- LHA-105170B-36 Leistritz Oil Pump Drawing, Chapter 8.1.
- LHA-105170B-37 Leistritz Oil Pump Drawing, Chapter 8.1.
- LHA-105170B-38 Leistritz Oil Pump Drawing, Chapter 8.1.
- LHA-105170B-39 Leistritz Oil Pump Instructions, Chapter 8.1.

The oil pumps are driven by ABB electrical motors, Tag No. PM-7131A/B & PM-7141A/B.

- LHA-105170B-07 ABB Motors Drawing, Chapter 8.1.
- LHA-105170B-08 ABB Motors Instruction Manual, Chapter 8.1.

The main oil pump circulates oil to bearings of the rotating units. The main oil pump discharges the lube oil to the oil coolers for temperature regulation, then into the lube oil filter for cleanliness. This operation is programmed in the control logic.

The stand-by oil pump is designed to supply oil when the main oil pump discharge pressure is too low. The stand-by oil pump operation is programmed in the control logic as follows:

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When the unit is running, if the lube oil pressure (upstream the oil coolers) decreases down, a low pressure signal is sent, then the stand-by oil pump is automatically started.

Automatic stop and pump changeover sequence is not foreseen. When the stand-by oil pump is running, it is the operator's decision to stop the main oil pump, assuming the root cause of auto-start of stand-by pump is identified, and the pumps placed again in auto-start condition, once the problem is solved.

3.7.2.1.3 LUBE OIL COOLERS

The lube oil coolers Tag No. E-T7133 & E-T7143 are Thermowave lube oil coolers, type plate.

The lubricating oil is fed to the oil coolers, to remove heat, preventing unacceptable high system temperatures. A temperature control valve is installed in conjunction with the oil coolers to maintain the correct reservoir temperature.

For complete information on the lube oil coolers, refer to Part 8 documents and drawings, Chapter 8.1 supplier's manufacturer documents.

- LHA-105170B-45 Thermowave Oil Coolers Instruction Manual, Chapter 8.1.

3.7.2.1.4 LUBE OIL FILTERS

The lube oil filters Tag No S-T7133A/B & S-T7143A/B are Indufil duplex oil filters, type IDGH 3-0700.

The lubricating oil is fed to the oil filters to insure oil quality requirement and oil cleanliness .

For complete information on the components of lube oil filters, refer to Part 8 documents and drawings, Chapter 8.1 supplier's manufacturer documents.

- LHA-105170B-33 Indufil Duplex Oil Filters Drawing & Parts Lists, Chapter 8.1.
- LHA-105170B-34 Indufil Duplex Oil Filters Instructions, Chapter 8.1.

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3.7.2.2 LUBE OIL RUNDOWN TANK

The lube oil rundown tank, Tag No. T-T7131 & T-T7141, supplied by Oeltechnik is designed to supply compressor with sufficient quantity of lubricating oil during shutdown, in case of lube oil pumps failure. The tank must be filled prior to start the unit, and is equipped with level instruments.

This horizontal rundown tank is installed on site, at sufficient height above the compressor centerline to ensure lubrication by gravity. Integrity checking must be performed during maintenance periods. Refer to operating routine instructions in Part 5, Operation, chapter 5.1

- HI.39490-AC-015 Rundown Tank General Arrangement, Chapter 8.2.
- HI.39490-AF-006 Lube Oil Console P&ID, Chapter 8.2.
- HI.39490-AS-007 Lube Oil Console List Of Materials, Chapter 8.2.

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3.8 UNIT CONTROL PANEL

3.8.1 GENERAL INFORMATION

Only subjects related to the technical description of the Dresser-Rand Houston, unit contol panel, are provided within this chapter.

Refer to the Dresser-Rand Houston, separate CD for informations pertaining to installation procedures and maintenance.

- LHA-105170B-05 Dresser-Rand Houston, Unit Control Panel User Manual, Chapter 8.1.

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DRESSER-RAND

PART 4 - INSTALLATION

4.1 INSTALLATION 1-1

4.1.1 GENERAL INFORMATION 1-1

4.1.1.1 DELIVERY OF EQUIPMENT TO CUSTOMER 1-1

4.1.1.2 UNCRATING OF EQUIPMENT 1-1

4.1.1.3 STORAGE OF EQUIPMENT 1-1

4.1.1.4 SAFETY PRECAUTIONS 1-2

4.1.2 COUPLING ALIGNMENT 1-2

4.1.2.1 GENERAL INFORMATION 1-2

4.1.3 EXTERNAL PIPING AND INSTRUMENTATION 1-3

4.1.4 CLEANING AND FLUSHING PIPING 1-5

4.1.4.1 GENERAL INFORMATION 1-5

4.1.4.2 GENERAL PIPE CLEANING 1-5

4.1.4.2.1 SAFETY PRECAUTIONS 1-5

4.1.4.2.2 CLEANING PROCEDURE 1-5

4.1.4.3 OIL SYSTEM FLUSHING PROCEDURE 1-8

4.1.4.4 INITIAL CHECKS FOR START-UP 1-11

4.1.4.5 GENERAL INSTALLATION PRACTICES 1-13

4.1.4.5.1 CORROSION PROTECTION 1-13

4.1.4.5.2 PROTECTION DURING NORMAL OPERATION 1-13

4.1.4.5.3 OUTDOOR INSTALLATION 1-13

4.1.4.5.4 STORAGE BEFORE ERECTION OF UNIT, IF LESS THAN 3 MONTHS ... 1-14

4.1.4.5.5 STORAGE AFTER THE ERECTION OF UNIT 1-14

4.1.4.5.6 ACCESSORY EQUIPMENT 1-16

4.1.4.5.7 CARE AND PROTECTION OF BUNDLE AND (OR) ROTORS 1-16

4.1.4.5.8 STORAGE OF SPARE BUNDLE AND (OR) ROTORS 1-17

4.1.4.5.9 UNIT RECOMMISSIONING AND PRIOR TO OPERATION CHECKS 1-17

4.1.4.6 DRESSER-RAND STANDARD SPECIFICATIONS 1-18

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4.1 INSTALLATION

4.1.1 GENERAL INFORMATION

The following installation instruction are provided as a guide to good practice and do not necessarily include every item of consideration for any given train of equipment; however, if these recommendations and those of the main drive equipment manufacturer are observed, many of the problems associated with installation will be eliminated.

4.1.1.1 DELIVERY OF EQUIPMENT TO CUSTOMER

Dresser-Rand centrifugal compressors are assembled and tested at the factory then are disassembled, thoroughly cleaned and the internals prepared for short-term storage.

All external piping connection points for inner seals and journal bearings which are not piped at the factory are suitable plugged or provided with blanking plates. Journal and thrust bearings are fully installed. All rotor labyrinth seals (inner end and end cover) are fully installed. EXCEPT FOR THE INNER GAS SEAL CARTRIDGES, WHICH ARE SHIPPED SEPARATELY. The coupling hubs on both the driving and driven equipment could be not fitted on the shafts, and would have to be installed on site.

4.1.1.2 UNCRATING OF EQUIPMENT

Before uncrating any equipment, it is strongly recommended that adequate storage facilities are provided to avoid mechanical damage, atmospheric corrosion or loss of the various components.

A complete shipping list is provided with every crate; carry out a careful inventory and notify Dresser-Rand of any shortages or missing items. If our Service Representative is available, it is recommended that he be present during all uncrating to facilitate identification of equipment and determine any shipment shortages.

4.1.1.3 STORAGE OF EQUIPMENT

If the equipment is to be stored for some time before installation, it is strongly recommended that a clean, dry location be provided. All surfaces subject to corrosion such as bearing and seal journals, gear teeth, etc... should be inspected at frequent intervals especially if the atmosphere of the storage area is subject to contamination by industrial smoke, temperature and humidity cannot be controlled.

If the equipment is to be installed but not operated for some time, the oil system should be flushed at frequent intervals. The oil heaters have to be connected to power supply and the oil pumps should be turned on once a week and oil circulated through the systems and back to the reservoir for 15 minutes.

Caution: If the compressor is furnished with gas seals and the seals are installed, then the separation gas injection to the outer barrier seals or labyrinths must be turned on and correct pressure established before turning on the oil pump.

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4.1.1.4 SAFETY PRECAUTIONS

There are number of basic safety precautions that should be followed by service personnel and operators. The following list of safety precautions has been compiled to help service personnel and equipment operators use the equipment in a safe and efficient manner at all times and to assure their safety. Remember that failure to allow just one safety precaution can cause an accident to man and/or machine. Observe all safety precautions called out.

1. Protect personnel from injury (burns, etc...) by wearing protective clothing, using gloves and providing adequate eye and ear protection.

2. Provide and maintain proper fire fighting equipment at all times and execute all fire precautions.

3. Cover open lines and access ports while servicing unit to prevent entrance of foreign objects.

4. Do not make any type of weld repairs without supervision by trained personnel and appropriate equipment. Do not attempt welding repairs during operation of unit or use unit for grounding of welding equipment.

5. Do not reuse o-rings and gaskets. Discard and replace o-rings and gaskets with exact equivalent, same size and material. Cover threads with paper or other suitable material, and apply a petroleum jelly (Vaseline, etc...) to o-rings and relative surfaces prior to reassembly of parts.

6. Use only lubricants and oil recommended avoiding contamination and unit failure and/or excessive wear.

7. Do not mark on unit with anything other than a grease pencil to avoid carbonizing of stainless steel parts at high temperatures from graphite.

8. A Dresser-Rand Field Service representative should be present whenever unit is disassembled for inspection and repair and for reassembly.

9. Be sure all motive power is turned off and locked out and compressor cases are completely de-pressurized before starting any disassembly work on bearings, seals and/or internals.

4.1.2 COUPLING ALIGNMENT

4.1.2.1 GENERAL INFORMATION

Correct alignment is mandatory for the successful operation of high speed equipment. A "flexible" coupling is no excuse for misalignment; its purpose is to eliminate transmission of end thrust and to compensate for slight changes in alignment that occur due to equipment temperature of operation (thermal growth or contraction) and piping stresses affecting the machine pedestals.

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Correct coupling alignment may be determined either by using a dial indicator on the face and outside diameter, O.D. of the coupling hubs, or by a reverse reading method using the indicator only on O.D. of hubs.

A "laser" optical alignment system would be preferably applied, using the devices on both sides of compressor casing.

- ALN-1000194939 Compressor Train Alignment Chart, Chapter 8.3
- LHC-000-473 Maximum Allowable Misalignment Chart, Chapter 8.3.

4.1.3 EXTERNAL PIPING AND INSTRUMENTATION

Install all piping, electrical conduit, instruments and controllers per installation drawings provided by Dresser-Rand and others.

1. Observe the following:

   a. All carbon steel and stainless steel lube oil and gas seal piping supplied with the unit is pickled and the exposed flanges sealed with blanking plates or tape before transportation to the job site. It is strongly recommended that customer supplied piping be pickled before installation; this will reduce flushing time and ensure a clean system which is vital for protection of bearings, seals and pumps.

   b. Do not install oil piping with loops (bends) such, that dirt or sediment can settle in them. If loops are unavoidable, install drain plugs in the bottom of each loop for removal of sediment during the flushing operation.

Note: Do not under any circumstances, install vent lines with loops which could cause oil pockets to form

   c. The same reasoning applies to instrument air, seal oil, seal air, etc... piping.

2. The piping must be installed and supported to avoid strains on the casing. Sources of piping strains are:

   a. Dead weight of piping itself.

   b. Improperly installed and/or supports, which causes strains on the casing. Misalignment, which is a frequent cause of machine vibration, can very often be traced directly to piping strains.

   c. Expansion or contraction of the piping as it undergoes temperature change.

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d. Pressure within the piping.

3. Adequate piping supports are necessary to prevent excessive dead loads on the flanges of rotating machinery. They should normally be of the flexible type and located as close as possible to the flange connections on the machinery.

4. The flange loading caused by pressure and temperature in the piping can be reduced satisfactorily by the use of a balanced expansion joint in an elbow as close to the flange as possible. Elbows should be long-radius type.

5. For best performance, gas piping should be arranged for streamline flow over the entire area of compressor inlet.

6. It is frequently necessary to increase pipe diameter just beyond compressor discharge flange. Where such conversion is necessary, transition should be gradual.

7. Summarizing, a satisfactory piping arrangement can normally be obtained by giving proper attention to:

   a. Provide adequate support for all parts of piping system.

   b. Allowing for expansion in a manner that will avoid piping strains on compressor.

   c. Installing a sufficient number of anchors in piping system so that direction and magnitude of expansion are controlled.

   d. Designing inlet and discharge piping so as to provide substantially laminar flow with uniform velocity over entire area of piping.

   e. All process pipelines to and from the unit should be inspected and cleaned before starting the first time. This is very important as any foreign material left in the inlet line may cause serious damage to the rotor. Also, provisions should be made to prevent foreign matter of any kind from entering the machines.

8. After installation of piping, observe the following:

   If compressor inner or end cover rotor seals use carbon rings, the rings (or entire seal assembly) are shipped separately. Disassemble unit and install seals/carbon rings; refer to Maintenance Section in this manual for procedure.

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4.1.4 CLEANING AND FLUSHING PIPING

4.1.4.1 GENERAL INFORMATION

In some cases, if so designated by the other, Dresser-Rand will provide some or all interconnecting oil piping between the compressor and oil systems. Any interconnecting oil piping for completing the oil circuit, whether provided by Dresser-Rand or by the customer, shall be cleaned and pickled prior to placing into flushing service.

Clean all bearing shells, seals and check entire bearing assembly to be sure all is clean. Clean the oil reservoirs thoroughly, removing all deposits that may have accumulated on the bottom. After the system has been completely cleaned and installed, the next step is to flush it. It is recommended that the same type of oil be used for flushing as will be used for lubrication. This will eliminate the possibility of contamination if any flushing oil remains in the system.

4.1.4.2 GENERAL PIPE CLEANING

These procedures are applicable to batch, submerged tank, carbon and stainless steel pipes.

4.1.4.2.1 SAFETY PRECAUTIONS

The alkaline and acid cleaners are injurious to the eyes and skin upon contact. the following precautions should be followed:

1. If the chemicals accidentally come in contact with the eyes or skin, flush immediately with plenty of clean cold water for at least 15 minutes and obtain medical treatment.

2. Rubber aprons, rubber gloves, boots and protective face shield are to be worn by workmen handling acids and other corrosive chemicals.

3. Cleaning operations are to be performed in an adequately ventilated area.

4. Adequate drainage shall be provided for any solutions which may escape.

5. The spent acid solution must be neutralized before being disposed of disposal must be in accordance with applicable environmental regulations.

Caution: the following procedure is not to be used on non-ferrous metals.

4.1.4.2.2 CLEANING PROCEDURE

The following procedure should be used to remove paint, oil, grease, rust mill scale and other foreign matter from carbon steel and stainless steel pipes.

Note: For stainless steel pipes, Steps 1. thru 6. of this procedure should be used.

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1. Remove all non-ferrous and painted stainless steel valve identification tags.

2. Loosen welding debris from inside and outside of piping by brushing, chipping or grinding.

3. Remove loose debris from inside of pipes by vacuum or compressed air.

4. Refer to table below for proper chemicals and inhibitors to be used.

Recommended Cleaning Chemicals And Corrosion Preventives

AGENT	REPRESENTATIVE SUPPLIES
Alkaline Cleaner For Ferrous Metals	• Houton Auto Clean 3501. • Calgon P-2000, Heavy Duty, Alkaline Hot Immersion Tank Cleaner. • Turco Type C, Alkaline, Hot Tank Cleaner. • Magnus 61-DRX, High Strength Alkaline Detergent. • Oakite, Energy Saving, Heavy Duty Alkaline Detergent. • Ashland Globrite 34 Alkaline Descaler.
Descaling Acid	• Houton Auto Clean 3501. • Turco 4368, Inhibited Descaling Acid. • Magnus D-Scale-R 100 Acid Rust and Scale Remover. • Oakite 32, Inhibited Acid Descaler. • Ashland Globrite 570 Pickling Compound.
Corrosion Preventive	• Houton Cerfa Clean 5380. • Ashland Tectyl 894 Water Displacing Corrosion Preventive. • Oakite, Water Displacing Special Protective Oil. • Magnus Magnafilm 31 Water Displacing Rust Preventive.

Warning: Mix alkaline solution in accordance with manufacturer's instructions. See table. Specifies recommended alkalines.

5. Remove paint, oil, grease, rust, scale as follows:

   a. Fill the immersion tank with alkaline solution.

   b. Submerge pipe completely in tank.

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c. Agitate solution or pipe to eliminate air pockets from pipe.

d. Pipe should remain submerged in tank until clean.

6. Rinse pipe in clean water to remove loose debris and all traces of alkaline as follows:

a. Fill rinse tank with clean water at ambient temperature. Make certain rinse tank has an overflow.

b. Immerse pipe completely in tank and agitate for approximately one minute.

c. Lift pipe from tank, and let water drain from pipe.

d. Visually inspect pipe for paint, oil, grease and rust.

e. Pipe showing any signs of contamination should be recycled through the cleaning process, Steps a. through d.

7. Remove mill scale from pipe using acid solution as follows:

Warning: Mix acid solution in accordance with manufacturer's instruction. See table. Specifies recommended acids. Pipe not previously cleaned in alkaline, and rinsed in clear water should not be immersed in acid solution.

a. Refer to table for recommended acid. Make certain acid solution is at ambient temperature.

b. Immerse pipe completely in water and agitate for approximately one minute.

c. Inspect inside and outside of pipe for foreign matter. If pipe is not clean repeat cleaning procedure.

d. Suspend pipe over rinse tank and thoroughly clean using high pressure water or steam with an oil soluble rust inhibitor.

8. Remove loose debris, and all traces of residual acid solution with water soluble rust inhibitors as follows:

a. Make certain inhibitor in tank is at ambient temperature and tank overflow is operational.

b. Immerse pipe completely in tank and agitate for approximately one minute.

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c. Inspect inside and outside of pipe for foreign matter. If pipe is not clean repeat cleaning procedure.

d. Suspend pipe over rinse tank and thoroughly clean using high pressure water or steam with an oil soluble rust inhibitor.

Note: If high pressure water or steam does not have a rust inhibitor additive, immediately coat the inside and outside of the pipe with corrosion preventive to prevent flush rusting. Refer to table, for recommended corrosion preventives.

9. Clean stainless steel pipes, using Steps 1. thru 6. Do not use Steps 7. and 8. for cleaning stainless steel.

10. Protect inside of pipes from dirt, grit, moisture and foreign matters, as follows:

    a. Seal pipe openings with a vapour barrier. The vapour barrier is to consist of polyethylene film wrapping, tied securely with wire or commercially available caps.

Caution: Waxed fabric is not to be used since the wax coating can be scraped off the fabric and enter the interior of piping. Wax can cause plugging of small clearances in seal and bearing assemblies.

    b. To insure against subsequent entrance of dirt and foreign matter, the vapour barriers are not to be removed until just prior to final assembly of piping.

    c. Parts are not to be stored outdoors after cleaning.

4.1.4.3 OIL SYSTEM FLUSHING PROCEDURE

The bearings of compressor is lubricated by the lube oil console. Refer to the lube oil system and lube oil console P&I.D's in the project documents file.

Note: For flushing operation, as the flushing oil should not be passing through the compressor bearings, a temporary piping will be installed, preferably a flexible pipe, to bypass the bearings and avoid any dirt coming into the bearings.

1. When flushing the lube oil systems, observe the following:

    a. The oil piping has been pickled at the factory before assembly so that the main and auxiliary oil pumps can be used for flushing.

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b. Inspect the oil reservoir for cleanliness prior to filling. Inspect the oil cooler tube bundles and interior of shells; clean if necessary.

c. The flushing oil is not to be passed through the compressor bearings. The connections of bearing housings will need to be blinded. Use “on site” flushing connections to bypass the bearings.

d. Since the piping has been pickled, the same type of oil that will be normally used in the system may also be used for flushing. Heat the oil to 140°F (60°C) before flushing, using the reservoir heaters.

Check the oil pump data plate for the maximum permissible operating temperature.

Caution: It is not good practice to pass steam through lube oil coolers to heat oil, especially if they are of the "two pass" type, since uneven heating of the tubes may cause them to pull excessively on the tube sheets; the use of steam coils or other heating device in lube oil reservoir is recommended.

Caution: Prior to start the flushing, insure flushing screens are installed at inlet AND outlet connecting point of each equipment, that are connected together for the flushing. (compressor bearings excepted). Refer also to manufacturers instruction to check if the driver and gearbox can be connected or not for flushing. If so, flushing screens must be fitted as well. As an exemple, inlet and outlet connections of lube oil console, same connections for the separate fin fan lube oil cooler (if this type is supplied) . This is mandatory after piping works, to avoid pollution of a clean equipment, this is also to check where the dirts come from. By following this procedure, flushing time can be optimised as much as possible.

e. Before starting the oil pump, be sure the flushing screens are installed at each customer connection point. The screens are 100 mesh stainless steel and are supplied by Dresser-Rand to protect the equipment against damage by dirt, welding beads, etc., that may be in field installed interconnecting piping. Once the flushing is completed, the screens must be removed from the piping before start-up of the compressor or other equipment serviced by the system. When the screens are removed, inspect the upstream side of the piping for any accumulation of dirt or other foreign material.

Caution: Before starting either oil pump, check the alignment between the pump and its driver. The pump sets are final aligned at the factory but normal transportation may have caused them to become misaligned. Refer to the pump, driver and coupling manufacturer's instructions in the auxiliary equipment documentation for the maximum allowable misalignment limits.

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Note: Incorrect bolt down on separately mounted auxiliary systems baseplates can cause excessive strain on the systems piping and components resulting in pump mislagnment. It is preferred that the baseplates be "grouted in" before tightening down hold-down bolts but is not mandatory. As a minimum requirement, however, stainless steel shims must be installed, as required, at the hold-down bolt locations to prevent deformation or twisting of the base plate when the hold-down bolts are tightened. See recommendations for separately mounted oil consoles.

f. One method of determining system cleanliness is to install either a 50 to 60 mesh screen or a 1/16 inch clean white felt pad, suitably supported on both sides by a fine mesh perforated steel plate (or similar), between any convenient flanges of "oil in" piping to bearings; then continue flushing. By frequently inspecting the screen or felt pads, the cleanliness of the system may be readily ascertained.

g. Install a suitable sized orifice in the oil pump discharge line with the oil filter characteristics being the sizing factor.

h. Set the filter transfer valve to mid-position, allowing flow through both filters.

i. Check that the differential pressure gauge valves to the gauges are fully open.

j. Flush the system, frequently checking to insure that the differential pressure across the filters does not exceed 20 psig (1.4 kg/cm^2). While flushing, occasionally tap the piping with a soft face hammer.

k. Flush the system until clean (minimum of 4 hours). The oil temperature should be allowed to increase to 160 to 180°F (71 to 82°C). Check the pump data plate for the maximum allowable operating temperature. Check the flushing screens, Step f., at four to six hour intervals to determine the system cleanliness.

l. After flushing, drain and clean entirely the reservoir then refill with clean oil above the operating level .

m. Remove the oil filter cartridges; inspect and clean the filter case. Thoroughly drain the flushing oil from all the system components.

n. Remove temporary by-pass piping for flushing . Install operating lube oil piping. Check for oil leaks at first start of lube oil console.

Warning: The drive shaft end of the compressor rotor could be coated with a rust preventive when shipped. Rust preventive is to be removed with a solvent such as naphtha, varsol or equivalent.

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2. Instrument air piping

   a. Disconnect all instruments, controllers, etc..., which are supplied with air and cover exposed pipe openings in the instruments with tape to prevent ingress of dirt or moisture.

   b. Bend disconnected piping away from instruments and adjacent relays, etc...

   c. The air compressor should be supplied with a suction filter.

   d. If a system air drier is used, it should be checked for correct operation per manufacturer's instructions.

   e. Open all drain valves in system.

   f. Back off all pressure regulating valves to provide maximum volume of air flow through the piping.

   g. Start the air compressor (or open main header block valve), then close drain valves. Blow air through piping for one hour.

   h. Open the drain valves; if any appreciable amount of water is present, recheck the air drier (if used) for correct operation.

   i. Remove any filter elements used and inspect for cleanliness. Clean elements as necessary.

   j. Repeat blow-through for one hour.

   k. Reconnect all instruments and set pressure regulators to recommended values. Check fittings with soap and water solution for leaks.

3. Set all the lube and seal system pressure and temperature switches, pressure regulators, controllers, transmitters, etc..., to the recommended values.

4. Inner Labyrinth Buffer Gas And Outer Labyrinth Separation Gas Piping

   This piping can be cleaned in the same manner as the instrument air piping outlined in Step 3. above.

4.1.4.4 INITIAL CHECKS FOR START-UP

1. Set all controllers, transmitters, valves, pressure switches, etc..., to recommended values.

2. Check all auxiliary and emergency pumps for correct actuation.

3. All alarm and shutdown devices should be checked as accurately as possible.

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Caution: Never start any equipment with shutdown devices inoperative.

4. Check compressor sealing system for correct operation.

Caution: Never start any equipment with sealing devices inoperative.

5. It is recommended that a cone style in-line strainer be installed in the compressor suction piping during initial start-up and the first year of operation. The strainer should be located as close as possible to the compressor intake flange but in no case closer than two pipe diameters minimum. The strainer location should permit adequate access for inspection and maintenance including removal of the strainer for cleaning or replacement. It is also recommended that a differential pressure indicator be installed to give an indication of pressure loss across the strainer.

Note: The strainer must be designed and constructed such, that there will be no possibility of it being drawn into the compressor. Strainers are usually made of perforated steel plate, welded to form a cone and installed with the cone tip pointing upstream (i.e. away from compressor intake).

6. Observe drive equipment manufacturer's instruction.

7. Bar over compressor rotor to determine if there are any tight labyrinths or impeller/diaphragm rubs. If the rotor does not turn freely, do not start unit until cause is determined and rectified.

8. Be sure that any main suction and discharge valves used, are operative.

9. It is normally recommended that the main drive equipment be started and checked out, uncoupled from the compressor. The compressor may then be started. If there is no undue vibration, compressor surge, or high bearing temperature, the unit should be brought up to design operating temperature and speed. It is recommended that the compressor is operated at design intake and discharge temperatures (if possible) for at least eight hours and a coupling alignment "hot" check carried out.

10. It all alignment and operating conditions are satisfactory, the equipment may be put into regular operation.

11. Observe Operating Routine outlined in Part 5, Operating Instructions.

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4.1.4.5 GENERAL INSTALLATION PRACTICES

4.1.4.5.1 CORROSION PROTECTION

After the cleaning procedure has been accomplished the metal has a bright, clean finish and is prone to rust very quickly. The following procedure should be carried out immediately:

1. Protect internal surface of pipes from corrosion.
2. Coat all internal surfaces of pipes with a corrosion preventive.
3. Circulate a fresh supply of corrosion preventive for at least 5 minutes.

4.1.4.5.2 PROTECTION DURING NORMAL OPERATION

The exposed parts of exposed, critical or unpainted areas should be covered with a suitable rust preventive compound.

All electrical wiring and connections should be made suitable for local weather conditions. Check electrical equipment instructions.

Where low temperatures are encountered, all water and supply oil lines should be adequately heat traced to prevent freezing or overcooling. Care must be taken to prevent localized overheating of the oil, as this may be injurious to the oil and oil cooler. Before starting unit when low temperatures are encountered, it may be necessary to heat the lube oil to approximately 50°F, and then circulate through the system until the whole system, including bearings, is warmed. Depending upon the temperature encountered, stirring or circulation alone may be sufficient.

Warning: If motor-driven oil pump is started with the oil temperature below 50°f, the pump motor may be overloaded.

4.1.4.5.3 OUTDOOR INSTALLATION

Outdoor installations present additional problems during erection and operation since the unit is exposed to the elements, which impose variable conditions on the equipment. The following covers some of the more important problems.

1. Keep rain, dust, dirt or any other foreign material from entering the casing; lube oil reservoir, all pipe connections or associated auxiliary equipment. Keep all exposed connections either taped over or covered with tarpaulins.
2. Do not align or assemble major parts that are exposed to the hot rays of the sun. This work should be done when this effect is at a minimum.
3. Do not attempt to handle any of the equipment by crane in a high wind to avoid possible damage through uncontrolled movements.

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4.1.4.5.4 STORAGE BEFORE ERECTION OF UNIT, IF LESS THAN 3 MONTHS

It is recommended that the assistance of the Dresser-Rand Serviceman be requested in preparing the unit for extended storage prior to erection.

If the unit and its components are to be stored before being installed, place them in a clean and dry area sheltered from the elements and free from temperature extremes. Crates are to be positioned as level as possible to avoid distortion or warping.

Storage should consist of an inspection of the crates and packages at least once a month during this period of dormancy for the effectiveness and durability of the protective coatings on all parts subject to corrosion. If the slushing is to be renewed or the quality changed, care should be taken not to use a coating having harmful acid contact that will attack the materials. The rotor should be turned over on its support to inspect the shaft journals, which are in contact with the supports.

Care should be exercised at these periods of inspection, while handling and unpacking to be sure no damage occurs to exposed parts or attachments. If lifting is required in order to make the inspection, be sure it is done in accordance with the sling markings put on the crates at the factory to avoid possible damage to parts.

Louvers in the crates are to be kept free and unobstructed to allow a natural circulation of air over the crated part, thus eliminating possible moisture traps.

The piping, as prepared before shipping, has been cleaned and pickled by the factory. Immediately after the pickling operation, the piping is given an internal dip of corrosion preventive and the ends are sealed off to keep dust, dirt or any other foreign material from entering the pipe. If this storage duration is in excess of two to three months, the corrosion preventive must be renewed. If the corrosion preventive is not renewed, it is strongly recommended that piping be repickled before installing on the erected unit.

All accessories to the unit are to be prepared for and inspected during storage in line with the particular manufacturer's instructions.

4.1.4.5.5 STORAGE AFTER THE ERECTION OF UNIT

When a unit is taken out of operation, prompt action is to be taken to protect it from harmful deterioration due to corrosion. The extent of preparation deemed necessary is based on atmospheric conditions and the length of time the unit is to be out of service.

Main consideration is to be given to the protection of internal parts directly affecting the operation and useful life of the unit, such as rotors, diaphragms, seals, bearings, etc...

Note: Site conditions, discrimination and judgment should be exercised as to the amount of the amount of preparation required. The following instructions cover procedures for the most severe conditions.

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Blank off steam lines to assure protection against steam leakage into unit. Also, remove all drain and sewer connections to units so that no air or vapors can gain entrance to the casings. When there is a likelihood of freezing, drain oil coolers to avoid damage. This would also apply to gauge lines filled with condensate and the accessories of a condenser.

Partial dismantling of the compressor will be necessary and separate storage of the detachments is recommended. Reliable brands of corrosion preventives should be used.

Remove rotor from casing and air dry internal parts.

Warning: Removal of rotor requires heavy duty equipment and skilled manpower. A Dresser-Rand superintendent of erection should be consulted.

It is essential that the internal components of the casing be cleaned with a spray of safety solvent (such as Tower Chemical Company N° 399 or equal) and blown dry with air hose to take a spray solution of rust preventive, meeting the standard MIL-C 16163B, Grade 3, or better. An alternate preventive measure is to protect the internals with VPI-250 if all openings in the casing are completely sealed off, thus preventing a recirculation of air.

It is advisable, at this point, to inspect for worn parts so that replacements can be ready when it is time to reassemble.

Drain and clean oil reservoir and refill with clean oil. Keep the oil system clean and free of water. It is suggested that a quantity of oil be drained from the bottom of the reservoir after standing awhile, during which time sediment and water will have had a chance to settle to the bottom. It is recommended, for an exceptionally prolonged shutdown period, that the oil reservoir and all lubricating lines be drained and slushing compound be poured into the oil reservoir to circulate through lines, thus protecting the entire lubrication system. The turbine oil pump driver casing should be opened, dried out, cleaned, slushed and sealed for storage. The main oil pump should be protected from moisture while idle.

Preparation of the rotor removed from the casing for extended storage should be as follows: Completely clean rotor with safety solvent (such as Tower Chemical Company N° 399), dry and slush with rust preventive of MIL-C-16173C, Grade 2, or equal.

Journals are to be given extra attention in that they are to be cleaned off with lintless rags, soaked in 90% methanol solution.

Warning: Do not touch journal with hands after cleansing with methanol.

After methanol has dried, liberally slosh Grade 2 rust preventive to journals and thrust collar (if integral part of shaft). The rotors are to be stored on the original shipping cradles with their journals separated from direct contact with the wood by lead sheets. This arrangement of rotor storage must allow the rotors to be rotated 135° monthly. The rotor is to have a suitable cover to protect against dust and dirt between these monthly rotor positionings.

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Bearing shells, thrust bearings and removable thrust collars are to be removed from bearing bodies, cleaned and dried. Coat machined surfaces with rust preventive and wrap in plastic for storage in a protective place separate from unit.

The casing should be reassembled to prevent dust and dirt, etc..., from accumulating in the slush internals of the unit.

4.1.4.5.6 ACCESSORY EQUIPMENT

Any accessory equipment not already mentioned should be prepared for storage in line with foregoing recommendations for D-R equipment. Whenever available, consult the manufacturer's instructions for complete details.

4.1.4.5.7 CARE AND PROTECTION OF BUNDLE AND (OR) ROTORS

The following recommendations apply to the care and protection, after shipment, for all D-R Company manufactured bundle / rotors, which are shipped separately from the main equipment:

Caution: When rotors are separately shipped in crates, it is a standard practice to mount these rotors in a horizontal position, these crates should be stored horizontally.

1. Care and protection of Rotors :

   a. Rotate rotors 135° once every month, if the rotors are mounted in a horizontal position, to prevent formation of a permanent sag in the rotors.

   b. Rotors may be stored in a vertical position. Vertical storage eliminates the necessity of periodic rotation to the rotor, as required with a horizontal mounted rotor. If the customer would intend to store rotors in vertical position, special design of rotor container is required. It is recommended that Dresser-Rand Engineering Department either design the container or review and approve the customer's design..

2. Protection of bundles with rotors : It is now a standard practice to separately ship spare bundle with rotor fitted in, in pressurized containers, in horizontal position. These containers could be stored either in horizontal or vertical position, as the rotor in position in the bundle, is locked for vertical storage. Container maintenance instructions will be supplied by Dresser-Rand.

- LHC-000-292 Rotor container, N_2 Storage, Instruction Manual, Chapter 8.3.

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4.1.4.5.8 STORAGE OF SPARE BUNDLE AND (OR) ROTORS

Rotors in crate : When rotors are to be stored in crate, following procedure could be used.

The rotor shall be thoroughly and completely cleaned with a solvent cleaner to be free of dirt, rust, grease and oil, drained and dried.

The rotor shall then be sprayed with a Grade 2 corrosion preventive (soft film) so that all impellers, blades, disc and stub shafts shall be coated thoroughly.

The bearing journals shall be cleaned to remove fingerprints with a fingerprint remover (such as Mark Remover, Plunkett Chemical Co. or a rag soaked in 90% methanol). The journals shall be allowed to dry and should not be touched with hands after cleaning. The journals shall be coated with a Grade 2 (soft film) corrosion preventive. The journals shall then be wrapped and taped with:

1. A layer of heavy duty oil manila paper.

2. A layer of heavy duty Grade C barrier wrap (such as Marvelpak # 12) applied cloth side out.

3. A layer of VPI (# 60 Crepe). This is a three layer wrapping, which shall protect the journals from the atmosphere.

Pressurized containers : The recommended storage procedure for spare rotor or spare bundles with rotors fitted in, is using a pressurized container. Vertical storage of rotors (secured in position) in nitrogen pressurized container eliminates the risk of corrosion and necessity of periodic rotation of the rotor, providing that Dresser-Rand maintenance procedure is applied.

- LHC-000-292 Rotor Containers (N_2 Storage), Instructions Manual, Chapter 8.3.

4.1.4.5.9 UNIT RECOMMISSIONING AND PRIOR TO OPERATION CHECKS

1. It is recommended that a Dresser-Rand Field Service Representative be present to supervise recommissioning and inspection of all parts.

2. All parts, which are coated with a corrosion preventive, should be thoroughly cleaned and inspected prior to operation. The interior surfaces of casings should be thoroughly cleaned to be completely free of dust or grit.

3. Any extended storage over six months shall require the casing to be disassembled and machine internals inspected prior to operation.

4. All headers and connecting blind flanges must be removed prior to operation. To insure this, it is usually desirable to have the man installing the blinds sign the red tag and allow only that man to remove the flanges, keeping an accurate count.

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5. All VPI bags, desiccant bags, pipe plugs, packing materials, preservative materials, etc..., must be removed prior to operation.

6. Machine alignment must be rechecked prior to any operation. All vendor(s) installation manuals should be consulted at this time to verify proper installation and setup.

7. Any parts with evidence of damage from extended storage should have extent of damage carefully documented and photographed and D-R Turbo Engineering should be consulted before using them.

4.1.4.6 DRESSER-RAND STANDARD SPECIFICATIONS

- 003-067-001R39 Installation & Commissioning Requirements for Dresser-Rand Turbomachinery, Chapter 8.3.
- 003-074-001R6 Protection of non Running Compressors and Systems, Chapter 8.3.
- 003-171-001R22 Preparation for Shipment and Short term Storage, Chapter 8.3.
- 003-021-001R13 Export Packaging Specification, Chapter 8.3.

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PART 5 - OPERATING INSTRUCTIONS

5.1 OPERATING PROCEDURES 1-1
5.1.1 GENERAL INFORMATION 1-1
5.1.2 COMPRESSOR INNER GAS SEAL FUNCTION 1-1
5.1.3 PRE-START INTRODUCTION 1-3
5.1.4 PRE-START INSPECTION 1-3
5.1.4.1 COMPRESSOR GENERAL INSPECTION 1-3
5.1.4.2 LUBE OIL SYSTEM PRE-START INSPECTION 1-4
5.1.4.3 GAS SEAL SYSTEM PRE-START INSPECTION 1-5
5.1.5 COMPRESSOR START-UP SYSTEM 1-5
5.1.6 COMPRESSOR SHUT-DOWN SYSTEM 1-7
5.1.7 OPERATING ROUTINE INSPECTION 1-8
5.1.7.1 DAILY 1-8
5.1.7.2 WEEKLY 1-9
5.1.7.3 QUARTERLY 1-9
5.1.7.4 PRESERVATION DURING UNIT SHUTDOWN PERIOD 1-10
5.1.7.4.1 LUBE OIL RUNDOWN TANK INTEGRITY CHECK 1-10

5.2 OPERATING PRECAUTIONS 2-1
5.2.1 SURGE IN A CENTRIFUGAL COMPRESSOR 2-1
5.2.1.1 EXPLANATION OF SURGE 2-1
5.2.1.2 METHOD OF PREVENTING SURGE 2-3
5.2.2 CONTROLS AND SAFETY DEVICES 2-3
5.2.3 SYNTHETIC LUBRICANTS 2-3
5.2.4 CASING SUCTION DRAIN PLUGS AND VALVES 2-4
5.2.5 CRITICAL OPERATING SPEEDS 2-4
5.2.6 CORROSION (WET CO2) 2-4
5.2.7 OPERATION IN A CLOSED LOOP 2-4

OFFICIAL COPY 00P39490100490.FB003 Reviewed with comments (C) 05-Nov-2012 08:15 AM

5.1 OPERATING PROCEDURES

5.1.1 GENERAL INFORMATION

Prior to compressor package operation or maintenance always familiarize with the Installation and Operating Manual, safety warnings and precautions. Ensure that personnal protective clothing and safety equipment is available and worn by personnel as demanded by the safety standards. Pay attention to warnings listed in each manufacturer's documentation.

5.1.2 COMPRESSOR INNER GAS SEAL FUNCTION

- SD105170B01-D6R8B Contract Seal Diagram - D06R8B Compressors, Chapter 8.3.
- HI.39490-AF-008 Gas Seal P&ID, Chapter 8.2.

The D06R8B compressor is equipped with DRESSER-RAND tandem gas seals cartridges. The seals together with the sealing system minimize process gas loss via the primary and secondary seal gas vents and/or contamination of the tandem gas seals with oil vapours from the bearing chambers.

Note: All TAG numbers are for train 1 - A-T7130 -; for train 2 -A-T7140 -; increase sequence number by 1000. eaxmple T71-PIT-3767 for train 1, T71-PIT-4767 for train 2.

Seal area "N_1" (2nd Section End inner labyrinth seal) is connected back to Intake 1st Section End (in order to reduce the axial load applied to the thrust bearing).

Seal area "D_2" at 2nd Section End is connected to a pressure tap in order to supply information about the balance connection pressure.

Gas supply ports "L_1", 1st Section & 2nd Section Ends are connected to common sealing gas piping.

In normal operating conditions, sealing gas injection is taken from compressor discharge (1st Section End), and is filtered through S-T7132A and S-T7132B (train 1), S-T7142A and S-T7142B (train 2).

This flow can be automatically adjusted by valve FCV3769 to obtain 227 Nm^3/H (each side). Low flow alarm signals are sent by FIT3770 & FIT3771.

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Gas leakages from primary seals, ports "U_2", are connected to LP Flare system. They are vented individually through flow orifices, FO3776 & FO3777.

If a seal failure occurs in primary seal, the seal leakage increases as well as the pressure upstream to FO3776 & FO3777. With leakage increasing, the protection of gas seal is activated as follows:

1. When the Delta P increase up a high differential pressure alarm signal is activated by PDIT3776 & PDIT3777.

2. If the pressure across FO3776 & FO3777 increases up, a high/high pressure signal is initiated and an emergency unit trip is activated by PIT3778 and/ or PIT3779 with a automatic sequence of unit depressurization.

3. If leakage pressure still increases up to 5 Bar.g, the rupture discs PSE3774 and PSE3775 are bursting in order to unload the secondary seal in case of primary seal total failure.

Secondary Gas supply ports "W_1", 1st & 2nd Section ends are not connected. (Provision only).

Secondary seals vent, ports "V_2" are connected to atmosphere vent.

Separation Seal, ports "F_3" at the outboard ends of the seals are connected to common Air system. Dry and clean Nitrogen is injected in these ports to avoid oil migration from bearing chambers to seal cartridges. Injection is controlled by PCV3766 to maintain a pressure of 7 Bar.g at each side. This line is equipped with pressure elements PDIT3765. Pressure transmitters send an alarm signal if separation nitrogen differential pressure reaches to 0.5 Bar.d.

Air injection line is also equiped with a pressure transmitter PIT3767 connected to the U.C.P. Lube oil pumps have permissive start when air pressure is above 0.25 Bar.g. If pressure falls down below, a signal to cut out lube oil pump is immediately sent and the starting sequence is aborted.

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5.1.3 PRE-START INTRODUCTION

Before starting the compressor train for the first time, it is necessary to clean and flush the lube oil piping prepared and installed on site. Refer to User Manual, Part 4, Chapter 4.1 Installation.

It is necessary as well to check the position of the hand valves, and to adjust the control valves, pressure and temperature transmitters, and all instruments which normally survey the operation of the train.

- HI.39490-AF-008 Gas Seal P&ID, Chapter 8.2.
- HI.39490-AF-007 Lube Oil System P&ID, Chapter 8.2.
- HI.39490-AF-006 Lube Oil Console P&ID, Chapter 8.2.
- HI.39490-AF-005 Temperature and Vibration P&ID, Chapter 8.2.

It is important that the operator reads this chapter thoroughly and observes all the cautionary notes given. Also observe, User Manual, Part 5, Chapter 5.2 Operating Precautions, before attempting to start the unit. Refer to User Manual, Part 8, Chapter 8.1 Supplier's Manufacturer Documents. Refer also to procedures to be observed before starting the lube oil console.

All pressure and temperature gauges should be checked for correct calibration against gauges of known accuracy.

Always verify that check valves are installed according to their function. Check valves installed in the wrong direction of flow may easily cause damage to pumps, gauges, etc..., when the system is pressurized.

Observe that all block and by-pass valves (hand operated) are opened or closed according to their function in the system. It is often necessary to open a normally closed valve or vice-versa during initial setting-up a system, and sometimes during start-up to prevent over-pressurizing oil filter elements, etc... The operator must be certain of the purpose of all hand valves before starting the lube oil pumps, etc...

5.1.4 PRE-START INSPECTION

5.1.4.1 COMPRESSOR GENERAL INSPECTION

1. Open compressor casing drains and remove any liquid accumulation. Be sure drain valves are closed before start-up.
2. Observe all instructions provided and manufacturers instructions for the prestart operation of auxiliary equipment .
3. Insure availability of instrument air to control valves, nitrogen (or gas) at separation barrier seal , sealing gas injection at gas seal cartridge ports
4. Observe availability of electrical power to motors .

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5. Observe availability of electrical power to instrumentation requiring electrical connection, such as transmitters, vibration signals etc....

6. Prepare a fault finding checking for all alarm and shutdown detection signals.

5.1.4.2 LUBE OIL SYSTEM PRE-START INSPECTION

1. Check cleanliness of the reservoir and fill the reservoir with oil of required oil grade at the maximum level.

- LHC-000-037 Lubricating Oil Recommandations, ISO grade 32, Chapter 8.3.

2. Connect electrical power to lube oil reservoir heaters and start lube oil temperature heating regulation. Lube oil recommendations are as follows:

   - Minimum oil temperature for oil circulation through the bearings: 70 °F.

   - Minimum oil temperature to start the compressors: 70 °F.

3. Prior to start the lube oil console, turn on separation air supply at the barrier seal to obtain lube oil pump permissive to start.

- HI.39490-AF-008 Gas Seal P&ID, Chapter 8.2.

4. When minimum oil temperature is obtained, select one filter and cooler. Establish lube oil flow to compressor lube oil system and fill the overhead rundown tank with lube oil . Proceed as follows :

   a. Open the rundown tank quick filling valve, and start one lube oil pump. Run the lube oil console and fill the rundown tank until the operating level is reached (level alarm signal is off) .

   b. Close the quick filling valve .

- HI.39490-AF-007 Lube Oil System P&ID, Chapter 8.2.

- HI.39490-AF-006 Lube Oil Console P&ID, Chapter 8.2.

5. Keep the lube oil console running for few times while observing correct operation of temperature regulation system, and the pump running without abnormal vibrations. Check availability of lube oil pumps change over (manual or automatic sequence, etc.).

6. Insure there is a steady flow through flow glasses in the lube oil drain lines. Check all connecting lines and fittings for leaks.

7. Observe the lube oil pressure indicators, check actuation of lube oil pressure transmitters, and pressure regulating valve.

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5.1.4.3 GAS SEAL SYSTEM PRE-START INSPECTION

- HI.39490-AF-008 Gas Seal P&ID, Chapter 8.2.

The gas seal supply line drains must be inspected for fluid accumulation, prior to pressurization (prior to start-up) of the compressor and after depressurization (after shutdown) of the compressor. Gas seal supply line low point drains must not be opened during pressurized periods . See cautionary explanation note in operating routine.

1. Insure availability of air injection at gas seal connections for barrier seal. Insure availability of air power at control valves air connections.

2. Open drain valves at low point of sealing gas filters to remove liquid accumulation.

3. Open the drain valves at the low point of sealing gas supply piping. Drain instrument lines and block manifold to remove liquid accumulation.

4. Open drain valves at low point of primary and secondary seal vent lines to remove liquid accumulation.

5. Check actuation of all flow transmitters, pressure and temperature transmitters and ΔP transmitters.

5.1.5 COMPRESSOR START-UP SYSTEM

Prior to spin the compressor shaft, to avoid oil migration to seal cartridges, and gas leakages to bearing chambers, it is mandatory to respect the logic starting sequence summarized as follows :

1. Start air injection through separation barrier seal, first.

2. Place the lube oil system in operation.

3. Start the gas sealing injection system.

4. Once above are in operation, process gas can be admitted to the compressor.

IMPORTANT NOTE :

If the process piping is to be purged before start-up and the total purge gas flow directed through the compressor inlet piping, with seal cartridges and bearings installed, then the barrier seal injection, and lube oil system must be in operation. This is required to prevent damage to the bearings and seal cartridges, since the purge gas flow could cause rotation of the compressor rotor.

For information, compressor rotors are mainly equipped with gas seal cartridges which CANNOT run in reverse rotation (excepted for “Flowserve“ bidirectional gas seal cartridges).

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The following documents referred to below, are a guide to help the operator in starting and subsequent operation of the compressor package:

- HI.39490-AF-008 Gas Seal P&ID, Chapter 8.2.
- HI.39490-AF-007 Lube Oil System P&ID, Chapter 8.2.
- HI.39490-AF-006 Lube Oil Console P&ID, Chapter 8.2.
- HI.39490-AF-005 Temperature and Vibration P&ID, Chapter 8.2.
    - The compressor casing drain valves must be opened to remove any condensate, then should be fully closed before pressurizing compressor casing.
    - Observe the separation air injection is in operation and set at correct pressure.
    - Observe the lube oil system is in operation and is supplying oil to the compressor at correct pressure and temperature. Lube oil reservoir and overhead rundown tank must have been filled with oil at correct level.
    - Operation of the compressor is controlled by anti-surge, insure the anti surge system is in operating conditions and anti-surge valves in position for start-up.
    - HI.39490-AJ-002 Anti-Surge Valve Data Sheet, Chapter 8.2 .
    - Insure that all alarm and shut down signals are set and ready for operation. Refer to above “ Inner gas seal function “ paragraph for gas seal protection instruments. See “temperature & vibration “diagram and B.O.M for vibration protection. Refer to lube oil system and console P.I.D ‘s for lube oil protection instruments.

Compressor must not be operated in a condition of surge. Violent surging is easily detected by an audible "thumping" from the compressor, vibration and large fluctuations in discharge pressure and flow. Partial surging is not always audible, but may be detected by unsteady discharge pressure/flow condition. Refer to chapter 5.2 for detailed explanation.

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5.1.6 COMPRESSOR SHUT-DOWN SYSTEM

1. When practicable, reduce the load for the main driver cooling out as required in driver‘s documentation. If the speed is reduced, avoid bringing the compressor into critical speed range and into surge.

2. Shutdown the driver as advised by manufacturer’s manual. Keep the lube oil, instrument air, barrier seal air and sealing injection in operation until after equipment coastdown.

3. With the process pressurized, avoid bringing the compressor into surge. The operation of compressor is controlled by anti-surge valves. Refer to following documents :

- HI.39490-AJ-002 Anti-Surge Valve Data Sheet, Chapter 8.2 .

4. Close the compressor suction and discharge valves if so directed by the plant operator's manual. If left open and the process is still pressurized, the sealing system and the lube oil system must be left in operation.

5. The compressor is subject to settling out pressure:

   a. Keep the sealing system in operation as well as separation air injection, and lube oil system.

   b. Do not allow the pressure to exceed the value listed.

- 523-264-344 Specification Plate Stamping - D06R8B Compressor
  Tag No K-T7141/42 - Serial No 1000181443, Chapter 8.3.

- 523-264-346 Specification Plate Stamping - D06R8B Compressor
  Tag No K-T7131/32 - Serial No 1000181442, Chapter 8.3.

   c. Maintain an "in-operation" watch of the gas seal system, as well as separation air injection through carbon rings between seals and bearings.

6. The air separation injection can only be shutdown after complete shutdown of lube oil and gas pressure sealing systems.

- HI.39490-AF-008 Gas Seal P&ID, Chapter 8.2.

- HI.39490-AF-007 Lube Oil System P&ID, Chapter 8.2.

- HI.39490-AF-006 Lube Oil Console P&ID, Chapter 8.2.

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Caution: During a shut down, if the gas sealing system and the lube oil system are still in operation, the separation air injection at barrier seals must be left in operation as well. If necessary, a back up air source must be supplied by the plant user.

If an automatic shut-down is caused by a gas seal failure, a unit depressurization is required.

5.1.7 OPERATING ROUTINE INSPECTION

A regular inspection routine during operation, depends on a great extent on the type of gas being compressed and whether or not the unit is running continuously. The rate of fouling, erosion or corrosion of the compressor internals cannot be reliably estimated. Inspection to determine rate of fouling must be considered, especially in case of progressive drop in compressor performance.

The following proposal may assist in developing an operating inspection routine suitable for the particular application of the equipment.

5.1.7.1 DAILY

Observe and log ALL pressures and temperatures that are not automatically recorded by instruments. The following lists some of the items which should be logged; however each installation will differ, and from a knowledge of the installation, the operator should determine the vital items for the safe operation of the equipment.

1. Compressor process gas intake and discharge temperatures and pressures.
2. Sealing gas differential pressure at the intake gas filters.
3. Lubricating oil temperature and pressure before and after passing through coolers and filters.
4. Lubricating oil pressure at the compressor bearings.
5. Compressor bearing pad detectors temperatures .
6. Lubricating oil temperatures at the compressor drain lines.
7. Sealing gas differential pressure at the compressor seals. Primary sealing gas leakage on the F.I.T at the primary seal drain lines.
8. Separation nitrogen air injection pressure at separation barrier seal.

The operator is advised to list all normal operating pressures, temperatures, pressure drops, etc., specified in this manual and other manufacturer's manuals for associated equipment. In addition, the operator should list the maximum allowable deviation from normal conditions.

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Note: A continual log or record of these values, whether by visual or automatic means, is not of any use unless they are checked and corrective measures taken where necessary. The life of any piece of machinery operating under normal circumstances, is determined by the careful and thoughtful actions of the operators.

Gas seal supply line low point drains must not be opened during pressurized period.

Caution : **Opening of the primary gas seal supply line drains during pressurized operation could result in the backflow of unfiltered process gas accross the inboard seal. Such backflow can result in contamination of the primary gas seal faces**.

In addition to the major equipment, the lubricating oil pumps, coolers and filters must be checked for correct performance per manufacturer's recommendations.

If instrument air lines are provided with low point traps for condensate draining, open the drain valves and close after draining.

Gas seal rupture disk system are to be visually checked for integrity, at any time that the primary vent monitor indicates no or very low flow, as compared to values recorded during commissioning, or operation when rupture disk integrity confirmed.

5.1.7.2 WEEKLY

Duplex type filters are used in the system. Check the pressure drop (which is an indication of fouling) and switch over to the filter not in use. Do not transfer the units unless the pressure drop is excessive. Before transferring to the stand-by unit, be sure that it is full of oil and that all the air has been removed. Duplex units with a single transfer valve can be changed over without any need for shutting down the train of equipment. Be sure to have unit which was in use, cleaned and readiness for future service.

Check the actuation of the auxiliary oil pump if the operating routine permits.

5.1.7.3 QUARTERLY

Carry out a test of all the safety shut-down devices if the plant operation permits.

Calibrate any gauges whose accuracy is doubtful.

Take an oil sample, from lube oil tank for analysis to insure that its properties are within the limits of the specified oil grade recommandations. If the results are out of the limits, lube oil must be renewed or added until original properties are met.

Inspect the gas seal system instrumentation. Visually check the rupture disks integrity, if the rupture disks are not equipped with an electrical alert sensor, for shutdown purposes.

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5.1.7.4 PRESERVATION DURING UNIT SHUTDOWN PERIOD

Have the necessary equipment on hand to carry out an immediate coupling “hot”alignment check when compressor package is shut-down. Record the readings taken and adjust the alignment shimming if necessary during shutdown period.

Dismantle the compressor and check the condition of bearings, seals, impellers, diaphragms, etc... Check all running clearances. Clean all internals and renew all O-rings, gaskets, etc..., disturbed during disassembly.

Reassemble the compressor, per maintenance instructions, adjust the alignment if necessary.

Inspect the couplings for wear, fretting and replace them if necessary.

Tighten all foundation bolts, flange bolts, instrument air line connections, etc...

Inspect all the auxiliaries, pump couplings, control valves, etc...

Open and inspect the oil coolers. Clean if necessary.

Open ans inspect the oil filters. Clean if necessary, and install new cartridges.

Drain the lube oil reservoir and clean out if possible. Refill the reservoir with new oil. In any case, the lube oil tank must be refilled with new oil, if analysis cannot be performed, or if lube oil properties cannot be obtained as required.

- LHC-000-037 Lubricating Oil recommendations, ISO Grade 32, Chapter 8.3

All system rupture disks are to be visually checked for integrity, especially if they are not equipped with electrical sensor.

Carry out the driver manufacturer's (if used) preservation instructions.

If unit is shut-down due to an automatic safety tripping device, be sure to determine true cause and correct it before re-starting.

If the shutdown is due to excessive gas seal leakage, all system rupture disks are to be visually checked for integrity. If broken, rupture disk must be replaced, to avoid non or mal-function of gas seal alarm and shutdown instrumentation. If the rupture disks are equipped with a burst alert sensor, this device, connected to the control system, will lock the compressor operation in stop position until the bursting disk is renewed.

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5.1.7.4.1 LUBE OIL RUNDOWN TANK INTEGRITY CHECK

Following procedure is applying when a lube oil rundown tank is used in the lube oil system. To verify correct operation of rundown tank, following procedure is recommended to check integrity of the rundown tank check valve:

It is strongly recommended to apply this procedure at each time, (or once a year) the unit is shutdown for overhaul inspection . Hot oil temperature would rather be used to meet operating conditions.

1. First of all, the unit need to be in stop position. The lube oil rundown tank is to be empty.

2. Then open the quick filling valve, and start one lube oil pump.

3. Run lube oil pump until lube oil rundown tank is at operating level (level alarm signal is off).

4. Close the quick filling valve, and stop the lube oil pump.

5. Monitor the time needed to empty the rundown tank.

Repeat this procedure 1 to 2 times, to insure reliability of the system.

If this time is more longer than the expected rundown time, the check valve is to be considered stuck. Then this item will need to be replaced.

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5.2 OPERATING PRECAUTIONS

The following operating precautions must be understood and observed before starting and during the operation of the centrifugal compressor. These precautions are presented in this chapter for emphasis and a readily available tabulation of important procedures or warnings to be observed.

5.2.1 SURGE IN A CENTRIFUGAL COMPRESSOR

In a Centrifugal Compressor there is, for every speed, a certain capacity below which the operation can become unstable. This breakdown point is called the surge point (or minimum stable flow point). The unstable, pulsating operation after passing this point is called surge, though the unstable operation may actually be caused by various forms of rotation stall (i.e., impeller rotating stall, diffuser rotating stall, etc...). The following is an explanation of surge, its effects and ways of preventing surge.

5.2.1.1 EXPLANATION OF SURGE

- LHC-000-006 FIG. A - Typical Impeller Head Diagram, Centrifugal Compressor Surge Illustrations, Chapter 8.3.

Fig. A, illustrates a typical centrifugal compressor impeller with backward curved blades. These are used for most industrial and process applications. A vector diagram has been constructed at the impeller exit. The head developed in a stage of a centrifugal compressor is a function of the impeller peripheral speed (U2) the tangential component of the absolute velocity leaving the impeller (V2U) and the hydraulic efficiency at a given rotative speed. As the capacity of the stage is reduced from the design value at constant speed, the value of the relative velocity leaving the impeller is reduced very nearly in proportion to the reduction in stage capacity. The angle of this relative velocity remains very nearly constant, except for the effect of "slip" which results from a circulation effect at the impeller tip. It is therefore apparent that, as the capacity is reduced at constant speed, the value of the absolute velocity (V2) leaving the impeller increases, and at a much greater rate than the hydraulic efficiency is reduced. The result is an increase in head developed by the stage as the capacity is progressively reduced below the design value, and a typical centrifugal compressor head flow characteristic results.

As this capacity is reduced, the velocities in the impeller flow channel also decrease and become more non-uniform in the blade to blade and hub to shroud directions. Eventually, the flow will separate from the passage surfaces and some localized recirculation occurs. The more the flow is reduced, the larger this circulation effect becomes.

At some point, the separation or recirculation zones become so large that the impeller can no longer overcome the downstream pressure level (i.e., in the diffuser, volute, process piping, pressure vessels, etc.). At this point, the flow reverses direction, (i.e., goes backward through the impeller) and the unit surges. The actual percentage of design flow is dependent on the level of speed, Mach number and molecular weight.

- LHC-000-006 FIG. B - Impeller Flow Path, Centrifugal Compressor Surge Illustrations, Chapter 8.3.

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- LHC-000-006 FIG. C - Typical Performance Curves, Centrifugal Compressor Surge Illustrations, Chapter 8.3.

The capacity continues to reduce until the downstream reservoir pressure is low enough that the impeller can once again overcome it. This is indicated by points (3) in Fig. C. When the downstream pressure has dropped to a level corresponding to point (3), the impeller can again begin to deliver, and it immediately handles a capacity corresponding to point (4).

If the condition which has caused surge has not been eliminated, the compressor capacity will again be progressively reduced along the constant speed curve through point (1) to point (2), and the surge cycle will be repeated.

The frequency of the surge is obviously a function of the time required for the compressor capacity to change from point (2) thru point (3) and point (4), and finally back to point (2).

If the downstream receiver capacity is large, the frequency of the surge will be low and the violence of it audible. If the downstream receiver capacity is small, the frequency will be high and the violence considerably reduced and perhaps just barely audible. This also depends on the pressure involved. The higher the pressure, the greater the energy.

A check valve may be installed in the compressor discharge pipe, as close to the compressor flange as possible to reduce the magnitude of surge if it occurs.

Although a centrifugal compressor stage, operating at constant speed, will surge at some percentage of its design capacity,(i.e.60%), the stability of multi-stage units is appreciably lower. This follows from the basic characteristic of each compressor stage, since at capacities less than the design, a greater volume reduction is obtained. Thus, if each stage of a multi-stage compressor surges at 60% of design capacity the overall unit may surge at 70% of design capacity of the inlet of the unit. This flow corresponds to 60% of the design capacity of the last impeller because of the increased volume reduction encountered in the earlier stages.

A compressor will inevitably surge when the capacity through it is reduced too much for a given speed, or if a greater pressure differential is imposed across it than it can develop at a given speed.

Violent or prolonged surging may cause the thrust bearing to fail. Surging for extended period of time will also cause the compressor to overheat, and may cause damage to the impellers

If the molecular weight of the gas being handled increases above the molecular weight for which the compressor was designed, then the stability of the compressor is decreased for a given speed. Stability here is defined as the difference of the design capacity at the surge point at design speed divided by the design capacity and is given as a percent of the design capacity.

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5.2.1.2 METHOD OF PREVENTING SURGE

Several steps may be taken to bring a compressor out of surge:

1. Increase the flow from the process to the compressor suction

2. Increase the flow through the compressor by-passing some of the gas being pumped from the compressor discharge through a cooler, back to the suction of the unit.

3. Decrease the pressure ratio across the compressor by decreasing the discharge pressure, or increasing the suction pressure, or both, if the flow is to remain constant.

What may be done to bring a compressor out of surge is dependent upon what can be done with the process involved. if the speed remains constant, then the flow in the compressor must be increased, or the pressure ration across the compressor reduced, to eliminate surge.

If the process requires a higher pressure ratio than the compressor is able to develop at a given speed, an increase in pressure ratio may be obtained by increasing the speed of the compressor. This speed increase will be limited by the maximum allowable operating speed of the compressor and driver.

If the process requires a lower capacity than the compressor can develop at a given speed, the speed may be reduced if the pressure ratio can be reduced. When the speed is lowered, the first critical speed of the compressor and driver may be the limiting factor.

Surge is a phenomenon that, when better understood, will permit the centrifugal compressor to be properly installed and operated.

5.2.2 CONTROLS AND SAFETY DEVICES

All controllers and control valves must be accurately adjusted. All hand valves must be in their normally open, "NO", or normally closed, "NC", position unless specifically noted in the User Manual, Part 5, Chapter 5.1 operating procedures. It is advisable to tag each valve either "NO" or "NC" according to its function.

All safety switches must be accurately checked and set for consistent reliable actuation. When the lube oil system is pressurized, the oil cooler, filters, gauges, etc..., must be vented to remove all air from the systems. Refer to the recommendations in the User Manual, Part 5, Chapter 5.1 operating procedures.

5.2.3 SYNTHETIC LUBRICANTS

If a synthetic lubricant (fire resistant fluid) is to be used instead of a normal turbine type oil for lubrication of bearings, gears, etc..., the material of all gaskets and o-rings in contact with the lubricant must be compatible with the lubricant. Normal gasket materials will usually swell and deteriorate when immersed in synthetic lubricants; also, normal paints will peel from interior walls of reservoirs and bearing chambers.

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5.2.4 CASING SUCTION DRAIN PLUGS AND VALVES

The compressor is provided with valved drain connections to remove any liquid accumulation from the case. Both drains should be opened (one at a time) before start-up or pressurization to remove this accumulation. Once a compressor is on-stream, both drain valves should be closed before operation.

5.2.5 CRITICAL OPERATING SPEEDS

If compressor is driven by a variable speed driver, avoid operating at or near a critical whirling speed which would cause the rotor to bow with resultant wiping of interstage labyrinth seals and vibration. Refer to compressor and driver specification nameplates for critical speeds and driver overspeed trip setting.

- 523-264-344 Specification Plate Stamping - Serial No. 1000181443, Chapter 8.3.
- 523-264-346 Specification Plate Stamping - Serial No. 1000181442, Chapter 8.3.

5.2.6 CORROSION (WET CO_2)

If a compressor is to be used for compression of gas containing CO_2 which is saturated with H_2O at inlet conditions, the unit must be purged as soon as possible after shutdown because condensed H_2O and CO_2 will form H_2CO_3 which is very corrosive and will attack iron, steel and copper. Any inert gas, air or natural gas may be used for purging; probably the same type of gas that is used for purging other vessels in the process.

5.2.7 OPERATION IN A CLOSED LOOP

As a result of explosions which have occurred while operating centrifugal compressors in a closed loop on air, Dresser-Rand is including this announcement for your guidance while operating centrifugal compressors.

The explosions occurred while the units were on test, the tests being conducted in accordance with standard and accepted procedures successfully employed throughout the industry for many years.

To prevent explosions of this type, any loop or bypass line which returns compressor discharge back to intake must be designed and operated such that at least one of the three elements which form the combustion triangle (oxygen-fuel-heat) will be eliminated. Air is commonly used for carrying out hot checks of coupling alignment, etc...,and contains oxygen.

Oil flows through the bearing chambers and is often used in seals adjacent to the gas stream. Under normal conditions, the oil will not enter the compressor gas stream; however, if due to mal-operation, improper sealing or parts failure, it does, then oil under the right conditions will provide fuel (second element of combustion triangle). Heat (the third element) can be form heat of compression.

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Do not operate a compressor in a closed loop on air or any other gas containing a combustible amount of oxygen. If such loops are unavoidable during prestart checks, use an inert gas such as helium, nitrogen or carbon dioxide depending on the required molecular weight.

If a special inert gas injection system is used with the compressor, all differential alarms and shutdown devices must be periodically checked for proper operation. When the unit is operating on process gas, the bleed-down seal system and safety devices must be checked for proper operation. See User Manual, Part 5, Chapter 5.1 operating procedures, describing seal system for detailed instructions.

Warning: Do not operate the compressor with gases containing a combustible amount of oxygen if discharge temperature can exceed 250°F (121°C) without an inert gas injection into intake and discharge end inner seal labyrinth.

The foregoing has been provided for the information of the customer. It may not be feasible for the customer to maintain the prescribed conditions under all circumstances. Further, explosions remain possible even when operating within these limitations, although the probability is greatly reduced. All operation of the compressor with gases containing oxygen regardless of conditions involves some risk of fire or explosion and this risk is the customer's. Dresser Rand disclaims all liability for operation of the compressor with gases containing oxygen. The instructions above are based upon the best engineering opinion available, but Dresser Rand disclaims all responsibility for the accuracy and effectiveness of such information and operating limitations.

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PART 6 - MAINTENANCE

6.1 D06R8B COMPRESSOR 1-1

6.1.1 GENERAL INFORMATION 1-1

6.1.2 GENERAL RECOMMENDATIONS 1-1

6.1.2.1 DISASSEMBLY GENERAL RECOMMENDATIONS 1-2

6.1.2.2 REASSEMBLY GENERAL RECOMMENDATIONS 1-3

6.1.3 HYDRAULIC FIT TYPE COUPLING 1-4

6.1.3.1 COUPLING HUB DISASSEMBLY 1-4

6.1.3.2 COUPLING HUB REASSEMBLY 1-6

6.1.4 BUNDLE ASSEMBLY REMOVAL & INSTALLATION 1-8

6.1.4.1 PULLER BLOCKS, RETAINING & SHEAR RINGS DISASSEMBLY 1-8

6.1.4.2 BUNDLE REMOVAL FROM CASE 1-9

6.1.4.3 BUNDLE INSTALLATION IN CASE 1-12

6.1.4.4 RETAINING & SHEAR RINGS REASSEMBLY 1-14

6.1.5 BUNDLE ASSEMBLY - UPPER HALF REMOVAL 1-15

6.1.6 THRUST & JOURNAL BEARINGS 1-17

6.1.6.1 THRUST BEARING AXIAL CLEARANCE CHECK 1-17

6.1.6.2 THRUST BEARING DISASSEMBLY 1-18

6.1.6.3 HYDRAULIC FIT THRUST DISC REMOVAL 1-19

6.1.6.4 JOURNAL BEARINGS DISASSEMBLY 1-22

6.1.6.4.1 1ST SECTION END 1-22

6.1.6.4.2 2ND SECTION END 1-23

6.1.6.5 BEARING INSPECTION 1-24

6.1.6.6 CHECKING JOURNAL BEARING CLEARANCES 1-24

6.1.6.7 JOURNAL BEARINGS REASSEMBLY 1-25

6.1.6.7.1 1ST SECTION 1-25

6.1.6.7.2 2ND SECTION 1-26

6.1.6.8 HYDRAULIC FIT THRUST DISC REASSEMBLY 1-27

6.1.6.9 THRUST BEARING REASSEMBLY 1-29

6.1.7 INNER GAS SEALS 1-31

6.1.7.1 BARRIER SEAL DISASSEMBLY 1-31

6.1.7.2 GAS SEAL CARTRIDGE DISASSEMBLY 1-33

OFFICIAL COPY 00P39490100490.FB003--FB003 Reviewed with comments(C)--05-Nov-2012 08:31 AM

6.1.7.3 INSPECTION 1-34
6.1.7.4 GAS SEAL CARTRIDGE REASSEMBLY 1-35
6.1.8 BUNDLE DISASSEMBLY AND REASSEMBLY 1-36
6.1.8.1 BUNDLE DISASSEMBLY 1-36
6.1.8.1.1 ROTOR REMOVAL: 1-37
6.1.8.1.2 INTERNAL PARTS DISASSEMBLY: 1-38
6.1.8.2 BUNDLE REASSEMBLY INSTRUCTIONS 1-39
6.1.8.2.1 BUNDLE PARTS REASSEMBLY 1-40
6.1.8.2.2 INTERNALS REASSEMBLY 1-41
6.1.8.2.3 ROTOR INSTALLATION 1-42
6.1.8.2.4 BUNDLE COMPLETION 1-43
6.1.8.3 BUNDLE ASSEMBLY - UPPER HALF INSTALLATION 1-44
6.1.9 INTERNALS OR ROTORS REPLACEMENT- AXIAL SHIMMING 1-45
6.1.10 RADIAL & AXIAL PROXIMITY PROBES 1-47
6.1.10.1 DISASSEMBLY 1-47
6.1.10.2 REASSEMBLY 1-47
6.1.10.3 PROXIMITY PROBE SETUP AND CALIBRATION PROCEDURE 1-48
6.1.11 EMBEDDED TEMPERATURE DETECTORS 1-50
6.1.11.1 DISASSEMBLY 1-51
6.1.11.2 REASSEMBLY 1-51
6.1.11.3 TEMPERATURE DETECTOR REPLACEMENT 1-51

6.2 DRIVER 2-1
6.2.1 GENERAL INFORMATION 2-1
6.2.2 MOTOR MAINTENANCE 2-1

6.3 GEARBOX 3-1
6.3.1 GENERAL INFORMATION 3-1
6.3.2 GEARBOX MAINTENANCE 3-1

6.4 COUPLINGS 4-1
6.4.1 GENERAL INFORMATION 4-1
6.4.2 COUPLING GUARDS MAINTENANCE 4-1
6.4.2.1 DRIVER / GEARBOX COUPLING GUARD 4-1

OFFICIAL COPY 00P39490100490.FB003--FB003 Reviewed with comments(C)--05-Nov-2012 08:31 AM

6.4.2.1.1 DISASSEMBLY 4-1
6.4.2.1.2 REASSEMBLY 4-2
6.4.2.2 GEARBOX / D06R8B COMPRESSOR COUPLING GUARD 4-2
6.4.2.2.1 DISASSEMBLY 4-2
6.4.2.2.2 REASSEMBLY 4-3
6.4.3 COUPLINGS MAINTENANCE 4-3
6.4.3.1 DRIVER / GEARBOX COUPLING 4-3
6.4.3.2 GEARBOX / D06R8B COMPRESSOR COUPLING 4-4

6.5 GAS SEAL CARTRIDGE 5-1
6.5.1 GENERAL INFORMATION 5-1
6.5.2 GAS SEAL CARTRIDGE MAINTENANCE 5-1
6.5.2.1 INTRODUCTION 5-1
6.5.2.2 GAS SEAL CARTRIDGE INSTALLATION 5-2
6.5.2.3 BARRIER SEALS MAINTENANCE 5-4
6.5.2.4 GAS SEAL CARTRIDGE REMOVAL 5-4
6.5.2.5 COMMINSSIONING PROCEDURE 5-5
6.5.2.5.1 STATIC TEST 5-5
6.5.2.5.2 DYNAMIC TEST 5-5
6.5.2.6 STORAGE 5-5

6.6 LUBE OIL SYSTEM 6-1
6.6.1 GENERAL INFORMATION 6-1
6.6.2 LUBE OIL SYSTEM PREVENTIVE INSPECTION 6-1
6.6.2.1 COMPRESSOR LUBE OIL SYSTEM PREVENTIVE MAINTENANCE 6-1
6.6.2.2 LUBE OIL CONSOLE PREVENTIVE MAINTENANCE 6-1
6.6.3 PREPARATION OF MAINTENANCE ACTIVITIES 6-2
6.6.3.1 COMPRESSOR LUBE OIL SYSTEM MAINTENANCE 6-2
6.6.3.2 LUBE OIL CONSOLE MAINTENANCE 6-2

OFFICIAL COPY 00P39490100490.FB003 -- FB003 Reviewed but not commented (C) -- 05-Nov-2012 08:31 AM

6.1 D06R8B COMPRESSOR

This chapter is a composite maintenance for the DRESSER-RAND D06R8B "Datum" vertically split centrifugal compressors.

6.1.1 GENERAL INFORMATION

Prior to performing any maintenance always familiarize with the User Manual, Part 2 Main Data, Chapter 2.2 Safety Warnings and Precautions. Also pay attention to warnings listed under each manufacturer's documentation.

Good preventive maintenance practices are essential to secure a long operating life for each compressor. The information given within this chapter is considered as guidelines only. Each installation requires different inspection intervals depending on the operating and environmental conditions present at site. DRESSER-RAND Client Support Centre has long experience in maintenance programs design. A close co-operation between the Customer and DRESSER-RAND Customer Service Center when designing a preventive maintenance program is recommended. Please refer to User Manual, Part 1 Introduction, Chapter 1.2 Customer Support Information.

The information in this chapter is maintenance information related to the DRESSER-RAND built systems. For detailed information related to a specific component of the compression train please refer to this User Manual, Part 8 Documents and Drawings, Chapter 8.1, Suppliers Manufacturers Documents.

Project drawings from project drawing list are located in this User Manual, Part 8 Documents and Drawings, Chapter 8.2 Project Documents.

Technical drawings required outside the project drawing list are located in this User Manual, Part 8 Documents and Drawings, Chapter 8.3 Technical Support Documents.

6.1.2 GENERAL RECOMMENDATIONS

It is recommended that a Dresser-Rand Service Representative be present when major disassembly is undertaken.

Warning: Before starting any maintenance work, be sure that the equipment has been locked out and tagged out using the equipment operator's procedures. Be sure that all live motive power sources or stored energy such as electric, pressurized gas, steam, fuel, pressurized liquid, spring tension or others have been de-energized. Also be sure the equipment is isolated from its process or other associated processes which may be operation while maintenance is being performed.

Warning: After de-pressurization, before starting any maintenance work, compressor, vessels, piping, and all other parts of equipment which may contain hydrocarbons, shall need to be properly vented in order to insure that hydrocarbon gas are effectively removed before opening the equipment.

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Warning: Always wear proper safety equipment such as hard hats, safety glasses and safety shoes as a minimum when working on DRESSER-RAND equipment. Additional safety equipment may be required depending upon the working environment and local regulations to protect personnel. Always use special power and hand tools for their intended purposes. Remember...Safety first!

Warning: Always use safe rigging practice when lifting of the components is associated with the maintenance of DRESSER-RAND Turbo Equipment.

6.1.2.1 DISASSEMBLY GENERAL RECOMMENDATIONS

Warning: Before pulling the bundle assembly off the casing, both parts must be stabilized at the same temperature. It is not allowed to remove the bundle assembly when the casing temperature is lower than the bundle temperature. Same requirement must be observed for installation of the bundle.

1. Care should be exercised in keeping dirt or any foreign material out of working parts of the compressor.

2. All piping, tubing, and compressor openings should be covered during maintenance periods. Tools and components should not be rested on nozzles or above compressor where they may drop into any openings.

3. Extreme care must be used in handling bearing and seal parts, as well as the rotor assembly. Internal parts having shaft labyrinths must be handled and stored with care.

4. SPECIAL RECOMMENDATION : During maintenance periods, the rotor and the thrust and journal bearing assembly must be protected against corrosion (particularly for the protection of probe track areas). Then, it is recommended to apply a generous coat of Valvolyne Tectyl 506 for more protection. Furthermore, the advantage is that after start-up Valvolyne Tectyl 506 dissolves in the hot lube oil.

5. Studs and bolts have sharp threads and should be worked around with care.

6. Before starting any maintenance work, ensure that all electric power if off and fuel, steam valves, etc.., closed on the driving equipment.

7. Inspection and replacement of the journal and thrust bearings and inner seal may be conducted with the bundle assembly installed <u>in</u> the case or with the bundle assembly <u>removed</u> from the case.

8. The gaskets and O-ring seals are of importance for the best performance of the compressor. They must be inspected closely and new ones installed if there is the slightest question about their condition.

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9. All matchmarks on mating pieces, dowels and fitted keys must be returned to their original locations when the compressor is reassembled to insure correct assembly. All diaphragms are numbered in mating halves and numbered to the case. Use care to insure the diaphragms are not mixed during reassembly.

10. All threads are right unless otherwise noted in text.

11. Whenever the compressor is dismantled and replacement parts installed, check all clearances and record for future reference. Refer to design clearances specifications, in bundle drawing, throughout the compressor.

12. Refer to User Manual, Part 7, Chapter 7.1, for detailed cross-sections of the compressor before starting any dismantling of unit.

6.1.2.2 REASSEMBLY GENERAL RECOMMENDATIONS

1. The need for cleaning the internal components or casing will be determined by inspection. Normally, washing all parts with a good grade of solvent and blowing out with clean, dry compressed air should be adequate.

Note: Do not wash parts protected per special instruction or recommendation. Do not wash O-rings in a solvent. This will cause them to swell and deteriorate.

2. Inspect all the O-rings and gaskets; they must be in first class condition. If there is any doubt it is strongly recommended that they be renewed.

Note: When AFLAS O-rings and glass filled Teflon back-up rings are used (heads and internal bundle assembly), the back-up rings require special handling to prevent over-stretching. Place the back-up ring on the head or bundle so that the contoured face will be toward the O-ring. Starting at top center and working both ways in small increments, carefully work the back-up ring over the head or bundle. Again working the full diameter of the back-up ring, move it axially in small increment until it is seated in the groove.

3. Inspect all labyrinths, machined surfaces, etc..., for nicks or burrs before and during reassembly.

4. All pipes, tubing and volute openings should be covered during any dismantling or opening of the case. If this has not been adhered to a careful cleaning of all such pipes and openings must be done.

5. Good attention must be given to all matchmarking of keys, diaphragms, inlet guides, coupling bolts, etc..., to ensure correct reassembly.

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6. Measure and record all possible running clearances . Determine that they are within recommended limits given in the bundle assembly drawing . Record for future reference.

- 1000183025 Bundle Assembly Drawing, (Sheet 2), s/n 1000181442, Chapter 8.3.

- 1000183414 Bundle Assembly Drawing, (Sheet 2), s/n 1000181443, Chapter 8.3.

7. Apply Loctite 515 to seal horizontal splitline surface of bearing housings.

6.1.3 HYDRAULIC FIT TYPE COUPLING

Refer to the manufacturer's data for the model, size and shaft taper dimension of the coupling.

- LHA-105170B-03 J.CRANE Couplings Fitting and Maintenance Instructions, Between Gearbox & Compressor, Chapter 8.1.

- HI.39490-AD-010 J.CRANE Coupling Drawing, Between Gearbox and Compressor, Chapter 8.2.

Remove the coupling spacer per recommendations shown in manufacturer’s instructions.

6.1.3.1 COUPLING HUB DISASSEMBLY

1. The coupling spacer has been removed per above instructions. Observe whether the spacer bolts and nuts are numbered to a given location. Store the nuts on the bolts from which they were removed. Unless specifically instructed by the manufacturer’s instructions, do not remove a disc pack from the hub.

2. Protect the coupling spacer and other removed components from corrosion or other damage and store safely.

3. Loosen the setscrew which secure the coupling locknut. Remove the locknut with a face spanner (not supplied). The coupling hub can now be removed.

4. Wipe the coupling hub face and shaft end and clean carefully to remove all oil. With a micrometer depth gauge, measure and record the hub overhang on the shaft to four decimal points. The hub must be returned to this dimension (+0.010/-0.000 in; +0.254/-0.000 mm), when reinstalled.

5. The coupling hub is a heavy interference fit on the shaft taper and must be hydraulically expanded for installation and removal. In order to remove the coupling hub proceed as follows:

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Note: Use extreme care during the following steps to keep dirt, filings, etc..., from the hydraulic oil and all components used for disassembly. Once the coupling hub has been removed, wrap the shaft taper in heavy well greased paper. Apply the same protection procedure to the hub and store safely.

- LHC-000-001 Hydraulic Fit Type Coupling Tools Illustration, Chapter 8.3.

6. Refer to Fig. A - Coupling Hub Tools. Obtain the coupling hub removal components from the Special Tools. Clean the tools carefully and install them as illustrated.

   a. Be certain that the back-up rings 5 and 8 are placed on the low pressure side of the O-rings 4 and 7, as shown.

   b. Be certain that the coupling nut 3 is seated against the shaft shoulder.

   c. The holder 13 is Teflon lined. It is provided to handle the larger sized components to keep them square with the shaft during assembly.

7. Refer to Fig. B, and procure the expander pump 14 and all the tubing/fittings illustrated from the Special Tools provided. Blow out all the tubing, fittings, etc..., with solvent/compressed air, then blow dry. Clean and install the connector 5 to the shaft center.

Warning: These tool assemblies incorporate high energy apparatus. Extreme care must be used in assembly, maintenance and operation of the tools. Never substitute other parts for those provided. Coupling nut and pusher ring are safely code color "yellow".

8. Install the tubing between the pusher ring 6, Fig. A, and pump 4, Fig. B. Refer to pump supplier manual for instruction, and fill the pumps with hydraulic oil and purge all the air from the lines.

- LHA-00-3463 RIVERHAWK Hydraulic Hub Installation-Removal, Chapter 8.1

9. Pressurize the pusher ring with pump 4 so that the ring is just touching the coupling hub. Pressurize the hub bore with expander pump 14 to 10,000 psig (703 kg/cm², 68.9 MPa) initially then in 1,000 psig (70 kg/cm², 6.9 MPa) increments until the coupling hub is loose on the shaft-end taper. It may be necessary to maintain the bore pressure for fifteen to twenty minutes to allow the coupling hub expansion. Do not exceed 35,000 psig (2 460 kg/cm², 241.3 MPa) maximum bore pressure.

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Note: Record the hydraulic bore pressure at which the coupling hub becomes "loose" on the shaft-end taper. For subsequent removal or assembly, the bore can be initially pressurized to this value to reduce maintenance time.

10. Release the bore pressure. Disconnect the tubing from the shaft and pusher ring. Return the pumps to tools storage. Remove the nut and pusher ring from the shaft and return to tools storage.

11. Pull the coupling hub from the shaft. Wrap in heavy paper to protect the hub bore and store safely.

12. Reinstall the coupling locknut. Wrap the shaft taper with heavy paper or sheet gasketing, and tape or wire in place.

6.1.3.2 COUPLING HUB REASSEMBLY

- LHC-000-001 Hydraulic Fit Type Coupling Tools Illustration, Chapter 8.3.

1. All the coupling components are on hand and are clean and serviceable. Procure the pump assembly illustrated on Fig. B, and the assembler/remover tools shown on Fig. A. Clean these components carefully.

2. Remove the protective cover from the shaft taper and the hub locknut. Clean the taper and drillings carefully.

3. Inspect the bore of the coupling hub and the shaft taper for high spots, burrs or signs of fretting from the previous operation. If evident, carefully remove with a stone or fine emery paper. Do not use a file.

4. Check the contact of the coupling hub on the shaft taper with Prussian Blue or an equivalent; at least 80% contact should be obtained.

5. Assemble the coupling hub on the shaft without O-rings 9, 11 and back-up rings 10, 12 (Fig. A). The hub must be metal-to-metal and "square" with the shaft.

6. Very carefully measure the distance from the face of the coupling hub to the end of the shaft. Record the dimension found. This is the coupling hub shaft overhang with the hub metal-to-metal on the taper without the O-rings, mentioned Step 5.

7. Pull the hub from the shaft and then reassemble, with O-rings 9 & 11 and back-up rings 10 & 12 installed as shown on Fig. A.

8. Refer to the User Manual, Part 7 Part & Tools, Chapter 7.2 Special Tools and assemble the hydraulic pump components, Fig. B, and coupling hub pusher components, Fig. A, as illustrated. Check that the coupling nut is seated firmly against the end of the shaft. Vent all air from the tubing.

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9. Use the pusher pump and move the coupling hub to the position found in Step 6. The O-rings will extrude if the hub bore is pressurized before the coupling is pushed metal-to-metal on the shaft taper.

10. Install a dial indicator to observe the coupling hub axial travel. Set the indicator dial to zero. Refer to “Hydraulic Fit Coupling Dimensions” for the axial travel required.

- LHC-105170B-01 Hydraulic Fit Type Coupling Dimensions, Chapter 8.3.

11. Set the pusher relief valve (integral part of pump) at 8 000 psig (562.4 kg/cm², 55.12 MPa).

12. Increase the pusher pump pressure up between 500 and 1 000 psig (35 to 70 kg/cm², 3.4 to 6.9 MPa).

13. Pressurize the coupling hub bore to 10,000 psig (703 kg/cm², 68.9 MPa), or to the pressure recorded in Step 9. of the disassembly procedure, while maintaining the pusher pump pressure.

Caution: Do not allow the coupling hub to move off the shaft taper.

14. Gradually increase the bore pressure and pusher pressure to move the coupling hub to its assembled position. The maximum allowable bore pressure is 35,000 psig (2461 kg/cm², 241.3 MPa, 2413 Bar).

15. Record the axial travel observed on the indicator.

16. Release the expander pump bore pressure gradually (at 5 000 psig, 351 kg/cm², 34.5 MPa, intervals) while maintain the coupling hub position with the pusher pump pressure. Wait a minimum of 10 minutes between intervals, while releasing the bore pressure. When the bore pressure reads zero, maintain the pusher pressure for an additional 5 minutes then release the pusher pressure. Final coupling hub position should be within the limits shown on the “Hydraulic Fit Coupling Dimensions”.

- LHC-105170B-01 Hydraulic Fit Type Coupling Dimensions, Chapter 8.3.

17. Remove the pumps and pusher components and return to the tool store.

18. Re-measure the coupling hub to shaft-end overhang; which should agree with the dimension recorded during disassembly procedure. As a second check, the dimension obtained in Step 6, minus the dimension measured in Step 18, should equal the axial travel, dimension "C", listed in the “Hydraulic Fit Coupling Dimensions”.

19. Install coupling locknut with a face spanner and lock with provided setscrew.

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20. Once the coupling hub is reassembled, install the coupling guard and coupling spacer. Refer to Part 6 Maintenance, Chapter 6.4, Couplings, and refer to the coupling manufacturer's instructions.

- LHA-105170B-03 J.CRANE Couplings Fitting and Maintenance Instructions, Between Gearbox & Compressor, Chapter 8.1.
- HI.39490-AD-010 J.CRANE Coupling Drawing, Between Gearbox and Compressor, Chapter 8.2.

6.1.4 BUNDLE ASSEMBLY REMOVAL & INSTALLATION

The entire bundle will be removed and installed, as a “cartridge” including internals, rotor and heads, with seals and bearings assemblies.

Disconnect all piping interfaces from 1st and 2nd Section head. Disconnect interface piping from bearing vents and drains as well as temperature and vibration wiring. Cover standing ends of pipe to prevent ingress of dirt, etc...Coupling, coupling guard and adapter have been removed.

- 1000183026 Case Assembly, s/n 100181442, Drawing, Chapter 8.3.
- PAGE 1000183026 Case Assembly, s/n 100181442, Parts List, Chapter 8.3.
- 1000183415 Case Assembly, s/n 100181443, Drawing, Chapter 8.3.
- PAGE 1000183415 Case Assembly, s/n 100181443, Parts List, Chapter 8.3.

6.1.4.1 PULLER BLOCKS, RETAINING & SHEAR RINGS DISASSEMBLY

The service end head is retained against internal pressure by means of segmented retaining and shear rings seated in an annulus turned in the casing bore. The following sequence must be followed closely, since some of the segmented components could be loose to drop onto the exposed bearing housing.

1. Place a protective cover over the exposed head connections.

   - LHC-000-007 Puller blocks, retaining and shear rings illustrations, Chapter 8.3.

2. Use the pusher nut and force the bundle "in" away from the shear ring, then remove the puller blocks, studs and nuts.

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3. Remove the capscrews through upper retainer ring segment 'A'. Use a jackscrew in hole found in segment radial center. Screw in to force retainer segment out of annulus. It may be necessary to back off on jackscrew and drive retainer segment into annulus to free up. Apply penetrating oil if found to be rusty. Use a hardwood block for driving segment back into annulus. Remove segment when loose. During the removal of this retainer segment, observe tightness of top shear ring segment 'A' which will be completely uncovered by retainer segment removal. If shear ring top segment appears to be loose, hold in installed position with a pry bar. DO NOT ALLOW TO FALL OUT OF ANNULUS.

   Start to remove the shear ring segment ‘A’. If necessary, use a jackscrew to free the segment up. When free, pull the segment straight back and out of the casing. Handle very carefully as the shear ring segment is heavy.

4. Remove remaining retaining ring to shear ring capscrews. Work remaining retainer segments ‘B, C and D’ out of annulus as described in step. 3.

5. Remove right and left hand shear ring segments ‘B and D’ as described in step. 4.

6. Remove bottom shear ring segment ‘C’. This segment is provided with a dowel at bottom center engaging a slot in both the bundle and casing. Handle carefully.

7. The bundle may now be pulled from the case.

6.1.4.2 BUNDLE REMOVAL FROM CASE

With the bundle retaining and shear ring segments removed, the bundle assembly is accessible. Inspect the case bore and bundle assembly for matchmarks which ensure correct bundle to case alignment when the bundle is reinstalled in the case. Provide a clearly visible matchmark if none can be found. The bundle assembly is removed / installed with the bearings and inner seals in place. With all piping disconnected, cover standing end of piping to prevent ingress of dirt, etc...

Caution: The vent and drain piping must be removed from the bearing housings before the bundle is removed.

Note : There is no rollers supplied for this size of compressor. Refer to special tools list, for specific tools supplied.

1. With a depth micrometer, measure from the face of the casing to the service end head in at least four places and record. Bundle assembly must be returned to this position (+0.0030 to 0.0050 in./0.076 to 0.127) to permit fitting of the shear and retaining ring segments in the case annulus.

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2. Clean the exposed case bore carefully and apply a coat of "Lubri - Silk" synthetic grease to reduce friction while pulling the bundle assembly. Using the first section service end puller blocks and studs, evenly pull the bundle assembly out of the case bore as far as possible breaking the bundle to case fit. Remove puller studs and blocks from case.

- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.

Note: The Second Section pusher/puller blocks (refer to tools list section) can be reinstalled and aid in pushing the module out using the inboard pusher nuts. Be sure to remove these items once module is "loose" in case bore.

Note: *For the "monohead" casing, there is no puller blocks at discharge end (second section) , but a special pusher/ puller plate is supplied to be attached at the second section end of the bundle (after removing the bearing baffle) to aid in pushing the module out, using the pusher nuts. Refer to tools list section.*

- PAGE 1000176623 Bundle Pusher Assembly, Tool List, Chapter 8.3.
- 685-325-203 Bundle Cradle Assembly, Drawing , Chapter 8.3.
- PAGE 685-325-203 Bundle Cradle Assembly, Tool List, Chapter 8.3.
- HI.39490-AC-002 Compressor Train General Arrangement,Chapter 8.2.

3. Obtain the bundle cradle support (supplied by D-R). Refer to general arrangement drawing. Install the cradle support onto the end of the skid, and on additional supporting foot (supplied by D-R), as shown on general arrangement drawing, and secure with provided bolts and washers.

4. Obtain the bundle cradle, and set the cradle down on the support, adjacent to the service end of the compressor case. Block up under the cradle as required. Position blocking under the jackscrews so that the cradle rail can be aligned to the case bore and cradle leveled. Blocking must be rigid and capable of supporting the cradle and module assembly. Check approximate weights. Install capscrews and washers which attach the cradle to the compressor case and just snug up.

- LHC-105170B-04 Weights Records - D06R8B Compressors, Chapter 8.3.

5. Adjust bundle cradle jackscrews, so that the cradle rail is in same plane as the case bore then drive up capscrews which attach the cradle to the compressor case.

- 685-325-203 Bundle Cradle Assembly, Drawing , Chapter 8.3.

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6. Apply a liberal coat of Lubri-Silk Grease on the rails of bundle cradle.

7. Install a safety hoist ring at the bottom of the head. Attach a come-a-long to the safety hoist ring and cradle stop pin.

8. Using the come-a-long, slowly pull the bundle assembly out of the compressor case. While pulling-out the bundle, observe there is no friction between the bundle and the casing. If necessary, adjust the position of the cradle. Continue to pull the bundle out of the case until the second section of the rotor shaft is well clear of the compressor case.

9. *When high pressure monohead casing is supplied, the first section head diameter could be much bigger than the return bend, then additional tools are required to guide the bundle for removal and installation. Adapter tool is supplied to maintain the bundle axis horizontally, and other tool blocks to support the return bend during bundle extraction. Bundle cradle will be installed, first, without the radial sliding block pads assembly.*

- PAGE 1000176623 Bundle Pusher Assembly, Tool List, Chapter 8.3.
- PAGE 139-670-007 Bundle Pusher Block, Tool List, Chapter 8.3.
- PAGE 1000192219 Extension Case, Tool List, Chapter 8.3.
- PAGE 1000192221 Bundle Extension Assembly, Tool List, Chapter 8.3.

10. *In such cases, remove the special pusher / puller plate, and install the special bundle tool at the end of bundle second section (discharge) end, with the long cylindrical bar resting on the bottom of casing bore, coated with grease, to guide the bundle for extraction.*

11. *Pull the bundle out until the first section head is well clear of the compressor case. Stop pulling and install the sliding blocks pads assembly each side of the cradle just after the adapter plate bolted on the casing face. Observe carefully the notes and dimensions shown on the cradle drawing, sheet 2, to insure correct position.*

- 685-325-203 Bundle Cradle Assembly, Drawing , Chapter 8.3.

12. When the bundle is far enough on the cradle, bundle lifter can be used for final bundle extraction. Remove the “come-a-long” from the cradle.

13. Rig chain falls to hoist and bundle lifter. Move and bolt bundle lifter to the top of bundle assembly. Carefully remove slack from hoist gear and adjust chain falls to keep all movement of the bundle assembly level.

- PAGE 1000190567 Bundle Lifter, Tool List, Chapter 8.3.

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14. Slowly lift the bundle assembly, observing the bundle is hanging level. If necessary, set bundle down and adjust the chain falls. Raise the bundle off the cradle and move it axially until the non-service end of the rotor shaft is well clear of the compressor case. Set the bundle down on solid wooden blocks.

15. Now, the bundle is ready to start the disassembly procedure. The bundle can be disassembled in maintenance area or on the bundle cradle, if so, the bundle must be chocked to prevent any radial rolling.

NOTE : If the bundle is to be moved, it can only be lifted with the contractual bundle lifter.

6.1.4.3 BUNDLE INSTALLATION IN CASE

1. The bundle assembly has been completely reassembled, carefully cleaned and new O-rings and back-up rings installed on the heads and return bend O.D (outer diameter).

2. The bundle is resting on the bundle stands, unless it has been assembled directly on the bundle cradle .

3. The bundle cradle is still resting on the cradle support, in position against the casing, aligned to the casing bore.

4. Carefully clean the case bore and inspect for nicks and burrs especially around the case annulus for the shear ring which could tear or abrade the bundle O-rings. Carefully note that the process gas openings in the bundle are correctly oriented to the casing inlet and discharge nozzle openings. Use care not to drop cleaning rags, tools, etc… into the case nozzle openings.

5. Apply a coat of Lubri-Silk Grease to the case bore and bundle assembly (straddling bottom center) and to cradle rail.

6. If the bundle comes from the maintenance area, it must be transported only with the contractual bundle lifter. If the bundle is built on the cradle, remove the anti-rolling devices placed under the bundle.

- PAGE 1000190567 Bundle Lifter, Tool List, Chapter 8.3.

- 685-325-203 Bundle Cradle Assembly, Drawing , Chapter 8.3.

7. Install the stop pin, at the end of the cradle, allowing enough space to install a hydraulic jack between the pin and the service end head. Provide some wooden piece to avoid any damage to the bundle and stop pin.

8. Using the hydraulic jack, slowly push the bundle into the case bore. Check alignment frequently between module and case bore to keep it centered.

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9. As the bundle enters the compressor case and hydraulic jack is extended its full length, remove the jack and move the stop pin inward on the cradle. Repeat this steps, until the bundle assembly is within the casing as far as possible.

10. Install puller blocks, studs and nuts at the First Section (service end) and Second Section (drive end) of the compressor case. Using the pusher nuts at the service end and the puller nuts at the drive end, fully seat the bundle assembly in the case bore.

11. With a depth micrometer, measure from the face of the case to the service end head in at least four different places. Compare the record with those found at the beginning of "Cartridge Removal From Case" paragraph.

- 1000183026 Case Assembly, s/n 100181442, Drawing, Chapter 8.3.
- PAGE 1000183026 Case Assembly, s/n 100181442, Parts List, Chapter 8.3.
- 1000183415 Case Assembly, s/n 100181443, Drawing, Chapter 8.3.
- PAGE 1000183415 Case Assembly, s/n 100181443, Parts List, Chapter 8.3.

12. The First Section shear and retaining ring segments may now be installed.

13. *When high pressure monohead casing is supplied, first section head diameter could be much bigger than the return bend, then additional tools are required to guide the bundle for installation. Adapter tool is supplied to maintain the bundle axis horizontally, and other tool blocks to support the return bend during insertion of the bundle. Bundle cradle is installed, and equipped with the radial sliding block pads assembly.*

- 685-325-203 Bundle Cradle Assembly, Drawing , Chapter 8.3.

14. *Install the special bundle tool at the end of bundle second section end, with the long cylindrical bar resting on the bottom of casing bore, coated with grease, to guide the bundle for insertion.*

- PAGE 1000176623 Bundle Pusher Assembly, Tool List, Chapter 8.3.
- PAGE 139-670-007 Bundle Pusher Block, Tool List, Chapter 8.3.
- PAGE 1000192219 Extension Case, Tool List, Chapter 8.3.
- PAGE 1000192221 Bundle Extension Assembly, Tool List, Chapter 8.3.

15. *Push the bundle in until the first section head is close to the compressor case. Stop pushing and remove the sliding blocks pads assembly each side of the cradle to allow the head entering the case.*

- 685-325-203 Bundle Cradle Assembly, Drawing , Chapter 8.3.

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16. *Continue to push the bundle in the case until the second section of the rotor shaft is well clear of the compressor case.*

17. *The special puller plate can be attached at the second section end of the bundle (after removing the bearing baffle) to aid in pulling the module in the casing, using the puller nuts. Refer to tools list section.*

- PAGE 1000176623 Bundle Pusher Assembly, Tool List, Chapter 8.3.

6.1.4.4 RETAINING & SHEAR RINGS REASSEMBLY

1. Install the puller blocks, studs and nuts if not already done previously.

- LHC-000-007 Puller Blocks, Retaining & Shear Rings Illustrations, Chapter 8.3.

2. Obtain the segmented shear and retaining rings. Clean carefully and remove all nicks and burrs. Remove all jackscrews used for disassembly.

3. Install shear ring segment 'C' at the bottom of casing groove.

4. Adjust axial position of lifting support 2 on the stud 1, depending on dimension between head and casing. Align bottom center dowel to metal contact.

5. Obtain a retaining ring segment ‘C’ and try fit at bottom center. If difficulty is experienced, note the puller nuts on the studs are loose then drive up the pusher nuts to seat the head in the case bore. Then, try the retaining ring segment fit, which should be now, satisfactory.

6. Install the remaining shear ring segments 'A, B and D' and then the retaining ring segments. Install the capscrews which attach the retaining ring segments to the shear ring and tighten evenly.

7. Use the pusher nuts to bottom the head in the case bore then check the clearance between the outboard end of the retaining ring and case annulus; should be within 0.003 to 0.005 in (0,08 to 0.13 mm). If a replacement retaining ring is to be installed, it must be ground at assembly to provide this clearance. Finish grind must ensure the end faces are parallel within 0.001 in (0.0025 mm). F.I.R. This axial clearance check should be done at each overhaul. It provides a safety double check on the bundle overall length. Installation of a spare bundle assembly may require sizing of retaining ring .

8. Loosen the inboard pusher nuts, and evenly tighten the outboard puller nuts to pull the retaining and shear ring/head up square against the case annulus.

- LHC-000-007 Puller Blocks, Retaining & Shear Rings Illustrations, Chapter 8.3.

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9. If compressor location is not under a shelter, it is strongly recommended that a high quality preservative compound be applied to shear rings. It is further recommended that jackscrews be coated with water resistant grease and reinstalled in all lower half segments to prevent entry of rain water, etc...

10. Carefully clean all exposed shafting. Do not remove protection on shaft taper.

Caution: All work done and equipment used while installing the bundle is taking place around shafting. While covered and protected to some degree, the shaft taper is also in a concentrated work area. Use forethought and carefully step-by-step planning while installing the bundle and related retaining rings. It may require many hours of painstaking works to make journal bearing operable.

11. Piping disturbed for bundle removal can now be assembled. Take care to clean all removed piping sections carefully, and use sound gaskets.

6.1.5 BUNDLE ASSEMBLY - UPPER HALF REMOVAL

The bundle assembly has been pulled out of the compressor case and is resting on bundle cradle.

- 1000183030 Internal Assembly, Drawing, s/n 1000181442, Chapter 8.3.
- PAGE 1000183030 Internal Assembly, Part List, s/n 1000181442, Chapter 8.3.
- 1000192540 Internal Assembly, Drawing, s/n 1000181443, Chapter 8.3.
- PAGE 1000192540 Internal Assembly, Part List, s/n 1000181443, Chapter 8.3.
- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.
- PAGE 1000183025 Bundle Assembly, Parts List, s/n 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.
- PAGE 1000183414 Bundle Assembly, Parts List, s/n 1000181443, Chapter 8.3.

Caution: The upper half of the bundle and the bottom half of the bundle are NOT identical.

Note: Before removing the bearings, measure the overlaps at the splitline for each impeller / diffuser stage. Proceed as follows :

1. Remove O-rings and back-up rings from bundle assembly outside diameter.

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2. Before removing the bundle clamps, it is necessary to tighten the whole bundle to facilitate the clamp removal. Proceed as follows: Refer to special tools and install the 2 tightening tools on the end faces of the bundle head and 2nd section cartridge, then tighten the tools on both sides.

- PAGE 121-593-202 Bundle Clamp, Tools List, Chapter 8.3.

3. At the first section of the bundle, remove the capscrews and the upper clamp segments only, attaching the return bend to the head.

4. Install safety hoisting rings onto the internal assembly. Lifting holes are provided at the top of 1st and 2nd section return bends. Attach chain falls to overhead hoist and internal assembly hoisting rings.

5. Remove the splitline taper pins and capscrews attaching the upper half assembly to the lower half.

6. Take up evenly on the chain falls and taking care to keep all movement truly vertical, hoist the upper half return bend assembly off the lower half.

7. Continue hoisting the upper half return bend assembly until well clear of the rotor impellers, observing the splitline is still horizontal, to avoid damaging the rotor. Division wall labyrinth carrier should be resting on the rotor.

8. Set the upper half return bend assembly down in the storage area on solid level planks. Turn upper half return bend assembly over so that the splitline is up and parallel to the floor line. Chock to prevent rolling. *Install the bundle upper half on supplied top bundle support tool with the splitline facing up.*

- PAGE 123-985-201 Craddle Bench Top Bundle Support, Tool list, Chapter 8.3

9. At the splitline of the bundle, measure the overlap on each impeller at each side of the bundle splitline. Record the results as a reference for reassembly.

- LHC-000-008 Impeller/Diffuser Overlap Illustration, Chapter 8.3.
- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.

10. Measure and record all labyrinth clearances.

- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.

11. Thrust & Journal Bearings can be removed.

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6.1.6 THRUST & JOURNAL BEARINGS

The thrust and journal bearings are considered as an integral part of the compressor internals module and are to be removed/installed once the module is removed from the compressor case and transferred to the special bundle assembly/disassembly stands.

However, if a bearings and inner seals inspection and/or replacement is required, then these assemblies can be removed/installed without having to remove the internals module. The following procedure is for thrust and journal bearings disassembly/reassembly and is applicable whether or not the module is removed from the case.

The complete bundle has been removed from compressor casing and is resting on bundle stands.

Note: Insure that impeller overlaps have been recorded prior to start thrust bearing removal.

6.1.6.1 THRUST BEARING AXIAL CLEARANCE CHECK

1. Coupling guard, coupling spacer and coupling hub have been removed.

2. Observe that the capscrews attaching the top halves of the thrust bearing housing and the journal bearing housings to the bottom halves, and capscrews attaching thrust bearing housing to the compressor head are tight.

- 1000183032 Thrust & Journal Bearing Assembly, 1ST Section - Drawing, Chapter 8.3.

- PAGE 1000183032 Thrust & Journal Bearing Assembly, 1ST Section, Parts List, Chapter 8.3.

3. The thrust bearing clearance is the actual distance that the rotor will travel from one extreme to the other in an axial direction.

4. To check the amount of thrust clearance, attach a dial indicator to the coupling guard adapter at the second section end of the compressor. With the spindle of the indicator contacting the end of the rotor shaft, apply sufficient force to the opposite end of the shaft to move it one direction its total unit of travel. Set indicator to zero reading and thrust the shaft in the opposite direction. The amount read on the dial indicator is the total clearance. Record the clearance. Remove dial indicator.

Note: The total design clearance is given by the bundle assembly drawing. If clearance is excessive, the thrust bearing components must be inspected to determine the “worn” pieces and new parts installed.

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- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.

6.1.6.2 THRUST BEARING DISASSEMBLY

- 1000183032 Thrust & Journal Bearing Assembly, 1ST Section - Drawing, Chapter 8.3.
- PAGE 1000183032 Thrust & Journal Bearing Assembly, 1ST Section, Parts List, Chapter 8.3.

1. Disconnect piping, conduit and wiring for radial and axial probes in thrust and journal bearings. Cover ends of piping to prevent ingress of dirt, etc...

2. Remove the end cover and remove axial and radial proximity probes from the bearing housings. Radial probes can be threaded directly out of the journal bearing housing. Axial probes can be removed directly out of the thrust bearing housing once the end cover has been removed.

Caution: Vent adapter, and drain adapters must have been removed previously to allow removal of the bundle.

3. Remove capscrews attaching thrust housing upper half to the bearing housing.

4. Remove the housing splitline capscrews. Install safety hoist ring at top of the thrust housing and rig to hoisting equipment. Carefully lift the upper half housing straight up until clear of the thrust bearings and then move to storage area. Tapped holes are provided in the upper housing for jackscrews to break the splitline sealant if necessary.

Caution: Be sure the temperature detector lead wires are disconnected from the junction box and packing glands are “loose”. Feed the wire through the lower half bearing housing while removing the thrust bearing housing upper half.

5. Push the rotor to the inboard side. Carefully rotate JC Sartorius outboard base ring assembly, until splitline is aligned with horizontal split of the housing.

- 1000183046 JC Sartorius Thrust bearing assembly drawing, Chapter 8.3.
- LHA-105170B-89 JC Sartorius Thrust Bearing Instructions Manual, Chapter 8.1.

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6. Carefully lift and remove top half of Kingsbury outboard base ring assembly. Roll bottom half base ring to top position; use care not to crimp or damage temperature detector wiring. Remove bottom half outboard base ring from bearing housing.

7. Push the rotor to the outboard side. Repeat the same procedure to remove Kingsbury inboard base ring halves from thrust bearing housing. Use care not to damage the babbitt faced thrust shoes during removal.

8. The lower half thrust bearing housing can now be removed. Using eyebolts in the tapped holes in splitline surface, slowly lower the housing until clear of the thrust disc, then remove to storage. Bolt the housing halves together while in storage to protect the splitline surfaces.

9. Coat thrust bearing components with oil to protect against corrosion and store safely.

6.1.6.3 HYDRAULIC FIT THRUST DISC REMOVAL

1. Refer to rotor drawing. Loosen setscrews which secure the thrust disc retainer nut. Remove the retainer nut .

- 1000183031 Rotor Assembly, Serial No.: 10-01-050 - Drawing, Chapter 8.3.
- PAGE 1000183031 Rotor Assembly, Serial No.: 10-01-050 - Parts List, Chapter 8.3.
- 1000183417 Rotor Assembly, Serial No.: 10-01-051 - Drawing, Chapter 8.3.
- PAGE 1000183417 Rotor Assembly, Serial No.: 10-01-051 - Parts List, Chapter 8.3.

Caution: The retainer nut and the thrust disc bushing are RIGHT hand threads.

2. The split ring 23 halves will be uncovered with nut removal. Remove the split ring.

3. Measure the dimension “B” from the outboard end of the bushing to the face of the thrust disc with a depth gauge. Record the dimension. The thrust disc must be returned to this position when reassembled and using the same rotor and internals.

- LHC-000-124 Hydraulic Fit Thrust Disc Axial Displacement Illustration, Chapter 8.3.

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Note: If a new rotor or any replacement internals are being installed, it may be necessary to resize the thrust bearing shims. Refer to procedure "Internals or rotors Replacement - Axial shimming", in this chapter, before completing the bundle re-assembly.

4. The thrust disc 20 is heavy interference fit on the bushing and must be hydraulically installed and removed. In order to remove the thrust disc from the bushing proceed as follows:

   a. Use extreme care to keep dirt, etc..., away from the hydraulic oil and components used for thrust disc removal. Refer to the "Special Tools Section" and obtain the thrust disc removal components.

   b. Before installing the special tools, matchmark the thrust disc to the bushing so that they can be reassembled in the same circumferential position.

   c. Refer to User Manual, Part 7 Parts and Tools, Chapter 7.2 Special Tools and procure the thrust disc removal components.

   d. Carefully install the pusher nut and pusher ring on the shaft as illustrated. Confirm that the O-rings and back-up rings are in place. Insure the pusher nut is fully seated against the bushing shoulder.

   - PAGE 534-221-206 Hydraulic Fit Thrust Disc Tools - Tools List, Chapter 8.3.

   e. Install the adapter ring on the thrust disc outside diameter.

   f. Assemble the tubing/fittings between expansion and pusher pumps as illustrated, insure that the tubing/fittings are clean. The expansion pump is to be connected to external connector and pusher pump to the nut, as illustrated.

   - PAGE 000-028-952 Hydraulic Pump Assembly - Tools List, Chapter 8.3.

Warning: These tool assemblies incorporate high energy apparatus and extreme care must be taken in the assembly, operation and maintenance of the tools. Never substitute other tools for those provided. The hydraulic tubing and related parts are painted safety code color "yellow".

   g. Fill the pumps with hydraulic oil, and purge all the air from the lines.

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h. Pressurize the pusher ring with pump 2 so that the ring is just touching the thrust disc. Increase the pump pressure to approximately 500 psig (34.49 Bar) and hold. This is to allow the thrust disc to release slowly on the bushing taper and not over stress the pusher nut-shaft threads. Be certain that all air has been bled from the pumps and related piping.

Caution: Maintenance personnel should not stand in way of the pusher fixture while installing or removing the thrust disc as the high energy hydraulic pressure can generate dangerous forces.

i. Pressurize the thrust disc bore with the expander pump 1 to 10,000 psig (690 Bar) initially then in 1,000 psig (69 Bar) increments until the thrust disc is loose. As the bore pressure is increased, observe pusher pump gauge for an increase in pressure as the disc releases. Pusher pump pressure can now be bled off to move the thrust disc axially along the bushing taper. This reduction in pressure should be done "step-by-step", reducing pusher pressure and then reducing the bore pressure until the thrust disc is loose and pusher tools can be removed.

Warning: Do not exceed 35,000 psig (2413 bar) maximum bore pressure; it may be necessary to maintain the bore pressure for 15 to 20 minutes to allow for thrust disc expansion.

Note: Record the hydraulic bore pressure at which the thrust disc is "loose" on the shaft. For subsequent removal or assembly, the bore can be initially pressurized to this value to reduce maintenance time.

j. Release the bore pressure. Disconnect the tubing from the thrust disc external connector and pusher ring. Return the pump to storage. Remove the nut and pusher ring from the shaft and return to tools storage.

k. Pull the thrust disc from the shaft. The thrust disc is finish ground on both sides and warrants careful handling.

- 1000183031 Rotor Assembly, Serial No.: 10-01-050 - Drawing, Chapter 8.3.
- 1000183417 Rotor Assembly, Serial No.: 10-01-051 - Drawing, Chapter 8.3.

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6.1.6.4 JOURNAL BEARINGS DISASSEMBLY

It is recommended the bearing and seal be removed from one end and the rotor be supported before removing the opposite end journal bearing and seal.

6.1.6.4.1 1ST SECTION END

- 1000183032 Thrust & Journal Bearing Assembly, 1ST Section - Drawing, Chapter 8.3.
- PAGE 1000183032 Thrust & Journal Bearing Assembly, 1ST Section, Parts List, Chapter 8.3.

1. Refer to the above drawing. Loosen axial capscrews attaching the bearing housing to the head.
2. Install eyebolt in the lifter hole provided at the top of bearing housing and rig to hoist. Adjust hoist to just take the weight of bearing housing.
3. Position a jacking device to center the rotor, then remove two capscrews apart of the vertical center line, which attach the bearing housing to the head. Refer to special tools and install 2 bearing guide studs in capscrew holes.

   - PAGE 121-513-003 Bearing Guide Studs, Tool List, Chapter 8.3

4. Remove all capscrews which attach the bearing housing to the head. Slide the assembly axially until clear of the shaft end, taking care that sharp shaft ends do not damage the bearing pads.
5. At the workbench, the journal bearing can be removed from the housing as follows:

   a. Remove the housing splitline dowels and capscrews and lift the upper half off. The journal bearing cage can now be lifted out of the lower half.

   - 1000183032 Thrust & Journal Bearing Assembly, 1ST Section - Drawing, Chapter 8.3.

   b. Refer to DRESSER-RAND instructions for disassembly of the tilting pad journal bearings.

   - LHA-105170B-88 DRESSER-RAND Journal Bearings Instructions Manual, Chapter 8.1.
   - 1000183045 DRESSER-RAND Journal Bearing Assembly, 1ST Section, Drawing, Chapter 8.3.

   c. Coat all of the bearing parts with oil to protect against corrosion or other damage and store safely.

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d. Prior to place the bearing housing in storage, bolt the two halves together.

6. If seals are not going to be removed, install the support under the rotor to keep shaft centered while removing opposite end journal bearing assembly.

6.1.6.4.2 2ND SECTION END

- 1000183033 Journal Bearing Assembly, 2ND Section - Drawing, Chapter 8.3.
- PAGE 1000183033 Journal Bearing Assembly, 2ND Section - Parts List, Chapter 8.3.

1. At the coupling side, remove the baffle in two parts, taking care of the labyrinth teeth.

   Remove the cover in two parts, avoiding any damage to the labyrinth teeth

   At workbench, both half parts can be separated, for labyrinth inspection.

2. Install an eyebolt in the lifter hole provided in the bearing housing upper half and rig to hoisting equipment. Adjust to just take the weight of the housing.

3. Position a jacking device to center the rotor, then remove two capscrews apart of the vertical center line, attaching the bearing housing to the head. Refer to special tools and install 2 bearing guide studs in capscrew holes.

   - PAGE 121-513-003 Bearing Guide Studs, Tool List, Chapter 8.3

4. Remove all capscrews which attach the bearing housing to the head. The bearing can be removed as an assembly. With the slings just taking the weight, slide the assembly axially until clear of the shaft end**,** taking care that sharp shaft ends do not damage the bearing pads.

Note: To prevent damage to the labyrinth, avoid dragging it on the shaft.

5. At the workbench, the journal bearing can be removed from the housings as follows:

   a. Remove the housing splitline pull dowels and capscrews and lift the upper half off. The journal bearing can now be lifted out of the lower half.

   b. Prior to placing the bearing housing in storage, bolt the two halves together.

   c. Refer to DRESSER-RAND instructions for disassembly of the tilting pad journal bearings.

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- LHA-105170B-88 DRESSER-RAND Journal Bearings Instructions Manual, Chapter 8.1.

- 1000183051 DRESSER-RAND Journal Bearing Assembly, 2ND Section Drawing, Chapter 8.3.

d. Coat all of the bearing parts with oil to protect against corrosion or other damage and store safely.

6.1.6.5 BEARING INSPECTION

Journal Bearing Shoes: Inspect the babbitt lining for scratches, grooves and evidence of a good bond to steel backing. Radial scratches and grooves can be tolerated if the edges can be blended into the normal surface, and the load bearing area is not appreciably reduced. Axial grooves tend to disrupt formation of an oil film and they should not be reused. A poor bond can be identified by fine hairline cracks or portions of the babbitt lining broken away from the shoe. Reject such shoes.

Bearing Housings: Inspect rail fit to cage and 'marry' to mating piece. Entire seating surfaces must be free of nicks and burrs. Arc spring damper is not a maintenance item.Visual inspection and seating check could only be recommended.

Thrust bearings (outboard/inboard): The base ring assemblies are Kingsbury type. Refer to the Kingsbury instructions for recommended disassembly, cleaning, inspection and reassembly procedures.

- LHA-105170B-89 JC Sartorius Thrust Bearing Instructions Manual, Chapter 8.1.

- LHA-105170B-88 DRESSER-RAND Journal Bearings Instructions Manual, Chapter 8.1.

6.1.6.6 CHECKING JOURNAL BEARING CLEARANCES

Because of the type of construction of this bearing (tilting pad), a diametrical clearance check is difficult and not considered to be of any real value. The bearing shoes and housing ring are machined to fine tolerances and hand fitting to obtain a given clearance is not necessary. If required, the clearances can be determined at the workbench using a mandrel.

Careful visual inspection of each bearing shoe will indicate if replacement with a new shoe is required. Inspect each shoe for scoring, poor bonding or loss of babbitt, etc... If any doubt exists, it is recommended that replacement shoes be used.

Note: It is not always necessary to replace a complete set of bearing shoes. If for example, the babbitt bond is poor on one shoe, then replace that shoe and reuse the remaining shoes if visual inspection indicates their good condition.

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Do not intermix the shoes between sets at each end of the compressor.

1. Fabricate a mandrel and base. Diameter of mandrel must be the same as the rotor shaft bearing journal diameter.

2. Install the cage over cross-tightening mounting screws. Mark location of pins or retainer screws with a pencil to identify centerline of pivots.

3. Place the mandrel against each of the bearing shoes at the pivot point locations then tap the back of the cage/housing at each shoe to assure each one is properly seated.

4. Set up a dial indicator, to measure movement of the bearing cage/housing relative to the mandrel. The mandrel must remain stationary as the cage/housing is moved 180 degrees away from the pivot until contact is made between the two shoes across from the pivot.

5. Measure and record the clearance within 0.0001 inch from the pivot through the centerline of the bearing for each of the bearing shoes.

6. Average the clearance readings found and multiply result by .894 to obtain actual clearance of journal bearing.

6.1.6.7 JOURNAL BEARINGS REASSEMBLY

6.1.6.7.1 1ST SECTION

1. Insure that all components that comprise the journal bearing subassembly are available. Parts must cleaned, inspected and found serviceable.

2. Pre-assemble the journal bearing at a workbench. Refer to DRESSER-RAND instructions.

- LHA-105170B-88 DRESSER-RAND Journal Bearings Instructions Manual, Chapter 8.1.

- 1000183045 DRESSER-RAND Journal Bearing Assembly, 1ST Section, Drawing, Chapter 8.3.

3. Place the lower half journal bearing housing on the bench with the splitline surface up. Chock to prevent rolling.

- 1000183032 Thrust & Journal Bearing Assembly, 1ST Section - Drawing, Chapter 8.3.

4. Carefully place the journal bearing cage in the housing bore noting that the anti-rotation dowel on the cage is in the vertical center line.

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5. Apply a thin layer of gasket sealer to horizontal splitline surface of lower half journal bearing housing, spread sealer evenly with putty knife. Refer to the bearing assembly drawing for type of sealer required.

6. Carefully lower the journal bearing upper half housing onto the lower half, using the splitline dowel to guide the halves together.

7. Install horizontal splitline taper pins and capscrews. Tighten splitline capscrews to 17.4 ft.lbs (23.7 N m) torque. Remove safety hoist ring.

8. Thoroughly coat bearing shoes and shaft with oil.

9. The bearing assemblies can now be installed in the compressor.

Note: Insure that a rotor support is installed at the opposite end, when installing a journal bearing.

10. Install a new O-ring in the head face.

11. Install 2 bearing guide studs in the capscrew holes of the head. Temporarily install a jack under the shaft end to just take the weight of rotor.

- PAGE 121-513-003 Bearing Guide Studs, Tool List, Chapter 8.3

12. Apply a thin layer of gasket sealer on the inner face of bearing housing. Rig to hoisting equipment and install the bearing housing assembly over the shaft onto the guide studs. Seat the bearing assembly against the head. Insert several capscrews to attach the bearing assembly to the head.

13. Remove the installation tools. Install the remaining capscrews. To insure that the journal bearing is pulled up square on the head, alternately tighten all capscrews to 85 ft.lbs (115 N.m) torque.

14. After the journal bearing is completely installed, remove the rotor support at the opposite end of the compressor.

6.1.6.7.2 2ND SECTION

1. Repeat the same procedure that used for 1st Section End, and re-install the bearing assembly on coupling end.

2. Apply a thin layer of gasket sealer on the splitline surface. Install the journal bearing cage assembly in the bearing housing lower half . Using dowel, place the bearing housing upper half onto the lower half and secure with splitline capscrews.

3. Install 2 bearing guide studs in the capscrew holes of the head.

OFFICIAL COPY 00P39490100490.FB003--FB003 Reviewed but not (C)--05-Nov-2012 08:31 AM

- PAGE 121-513-003 Bearing Guide Studs, Tool List, Chapter 8.3

4. Apply a thin layer of gasket sealer on the inner face of bearing housing. Rig to hoisting equipment and install the bearing housing assembly over the guide studs and against the head, secure with axial capscrews.

5. Remove the guide studs and insert remaining capscrews. To insure that the journal bearing is pulled up square on the head, alternately tighten all capscrews to 85 ft.lbs (115 N.m) torque.

6. Install the labyrinth plate baffle, on the bearing housing outer face. Secure with capscrews.

6.1.6.8 HYDRAULIC FIT THRUST DISC REASSEMBLY

- 1000183031 Rotor Assembly, Serial No.: 10-01-050 - Drawing, Chapter 8.3.
- 1000183417 Rotor Assembly, Serial No.: 10-01-051 - Drawing, Chapter 8.3.
- LHC-000-124 Hydraulic Fit Thrust Disc Axial Displacement Illustration, Chapter 8.3.

1. Refer to Thrust Disc Axial Displacement Illustration, measure and record the thrust disc thickness dimension, 'T'.

2. Install the bushing on the shaft. Observe that the bushing is fully seated against the shaft shoulder and radially aligned to the matchmark on the shaft.

3. Very carefully measure the dimension between the outboard end of the bushing flange to the outboard end of the bushing, dimension 'B'.

4. Place the thrust disc onto the bushing without the O-rings (metal-to-metal). Thrust disc must be square with the shaft. Note the matchmarks, thrust disc to bushing and bushing to shaft, are in radial alignment.

5. Measure the dimension between the outboard face of thrust disc to the outboard end of the bushing, dimension **'B'**.

6. Remove the thrust disc; install the O-rings in the disc bore then reinstall the disc on the bushing. Use care not to damage the O-rings while installing the disc.

7. Temporarily install the split ring 23 halves, taping them in place if necessary, to secure the bushing axially.

   - 1000183031 Rotor Assembly, Serial No.: 10-01-050 - Drawing, Chapter 8.3.
   - 1000183417 Rotor Assembly, Serial No.: 10-01-051 - Drawing, Chapter 8.3.

8. Install the hydraulic pump components and thrust disc pusher components as outlined on the following illustrations:

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- PAGE 000-028-952 Hydraulic Pump Assembly - Tools List, Chapter 8.3.
- PAGE 534-221-206 Hydraulic Fit Thrust Disc Tools - Tools List, Chapter 8.3.

Note: If the hydraulic pump has not serviced for operation, fill the hydraulic pumps with oil, and then purge the hydraulic system of all air.

9. Use the pusher pump and move the thrust disc to the position measured in Step 5, (dimension 'B'). The o-ring seals in the disc bore will extrude if the bore is pressurized before the disc is metal-to-metal on the bushing.

- LHC-000-124 Hydraulic Fit Thrust Disc Axial Displacement Illustration, Chapter 8.3.

10. Install a dial indicator to observe the thrust disc axial travel, movement required, dimension 'D', equals 'A' minus ('B+T').

11. Set the pusher pump relief valve (part of pump) at 8000 psig (552 Bar).

12. Increase the pusher pump pressure to 500 to 1000 psig (35 to 69 Bar).

13. Pressurize the thrust disc bore to 10,000 psig (690 Bars) while maintaining the pusher pump pressure.

Caution: Do not allow the thrust disc to move off the shaft taper.

14. Gradually increase the bore pressure and pusher pressure to move the thrust disc to its assembled position 'E' (thrust disc is seated firmly against the bushing shoulder). The maximum allowable bore pressure is 35,000 psig (2413 Bar).

15. Measure and record dimension 'C' and axial travel dimension 'D'. Dimension 'C' should be finally, the same as recorded before thrust disc removal.

16. Release the expander pump bore pressure gradually at 5000 psig (345 Bar) intervals while maintaining the thrust disc position 'E' with the pusher pump pressure. When the bore pressure reads zero, maintain the pusher pressure for an additional five minutes, then release the pusher pressure.

17. Recheck the axial travel, dimension 'D'. If it is the same, remove the hydraulic tubing and pusher components.

18. Again, measure and record dimension 'C', it should be the same as previously recorded. As a final check, dimension 'C' plus 'T' will equal 'A'.

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- 1000183031 Rotor Assembly, Serial No.: 10-01-050 - Drawing, Chapter 8.3.
- 1000183417 Rotor Assembly, Serial No.: 10-01-051 - Drawing, Chapter 8.3.

19. Remove the tape, if used, from the split ring 23 (rotor assembly drawing). Clean and reinstall the split ring in the shaft groove outboard of the bushing. Ring can be made to adhere to the shaft with several dabs of grease until the retaining nut 22 is installed.

Note: If a replacement bushing has been installed, it will be necessary to supply or to grind the split ring thickness at assembly. Measure the dimension from the outboard end of bushing to back of split ring shaft groove. Grind the split ring to the dimension determined for a “metal to metal” fit . Finish grind must assure the sides are parallel and square with the bore within 0.0005 inches (0.013 mm) F.I.R (Fully Indicated Reading).

20. Install the retaining nut and lock in place with setscrews .

Caution: Retaining nut and shaft are RIGHT hand threads.

Note: If the thrust disc "run-out" exceeds 0.0005 inches (0.013 mm) F.I.R. (Full Indicator Reading), remove the thrust disc. Check each on surface plate and with micrometer for warpage or uneven wear. Replace with new parts if necessary. The thrust disc must be fully seated against shaft shoulder and the rotor must not drift axially while being rotated if this test is to be effective. Remove dial indicator from thrust bearing housing.

6.1.6.9 THRUST BEARING REASSEMBLY

- 1000183032 Thrust & Journal Bearing Assembly, 1ST Section - Drawing, Chapter 8.3.
- PAGE 1000183032 Thrust & Journal Bearing Assembly, 1ST Section, Parts List, Chapter 8.3.

1. Install the lower half thrust bearing housing. Apply gasket sealer to the inner face of the thrust housing. Using eyebolts in the tapped holes of the splitline surface, slowly install the thrust housing lower half under the rotor then against the journal bearing face. Secure with provided capscrews.

2. Install a dial indicator on the lower half thrust bearing housing with the stem contacting the thrust disc. Using the dial indicator set the rotor position at its mid point of travel.

3. Thoroughly coat the thrust bearing shoes and thrust disc end faces with oil.

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4. Roll lower half Kingsbury inboard base ring, into the lower half thrust bearing housing, use care not to damage instrumentation wiring. Set upper half inboard thrust bearing onto lower half and rotate into position. Place instrumentation wiring in groove in lower half and temporarily hold in place with putty.

Note: If a spare or replacement rotor and/or internals are being installed, it may be necessary to resize the inboard thrust bearing shim pack to correctly position the rotor assembly in relation to the stationary. For the procedure, refer to "Internals or Rotor replacement- Axial shimming " paragraph.

5. Now, push the rotor assembly against the Kingsbury inboard base ring. Carefully place the lower half KINGSBURY outboard base ring on the rotor shaft, correctly oriented, and rotate into the housing bore, between thrust disc and journal bearing housing. Use extreme care not to damage the temperature leads. Make the splitline surface horizontal.

6. Carefully place the mating upper half thrust bearing onto the lower half, then rotate the complete assembly around the housing bore to seat anti-rotation dowel in upper half in mating slot in housing splitline surface.

7. Check the thrust bearing running clearance. Resize KINGSBURY outboard base ring shim pack if necessary.

- LHA-105170B-89 JC Sartorius Thrust Bearing Instructions Manual, Chapter 8.1.

8. Apply gasket sealer to the lower half thrust bearing housing horizontal splitline surface. Spread sealer evenly.

9. Provide and install eyebolt at the top of thrust bearing housing. Using hoisting equipment, bring the upper half housing over the lower half. Insure the temperature detector wires do not make obstruction. Carefully and slowly bring the upper half down onto the lower half, keep the splitline aligned with the lower half splitline.

10. The splitline dowel pins can be used to help final installation. Insert the splitline capscrews and tighten to 17.4 ft.lbs (23.7 N.m). Insert capscrews through upper half and tighten capscrews to 85 ft lbs (115 N. m).

- 1000183032 Thrust & Journal Bearing Assembly, 1ST Section - Drawing, Chapter 8.3.

11. Re-install axial probes if they had been removed from thrust housing. Re-install the end cover and secure with provided capscrews.

12. If lube oil piping had been removed, re-install all pipes at proper location (excepted for drain and vent adapters).

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6.1.7 INNER GAS SEALS

It is necessary to remove the coupling hub and thrust & journal bearing assemblies to provide access to the inner gas seal assemblies.

- 1000183034 Gas Seal Assembly - 1st Section, Drawing, Chapter 8.3.
- PAGE 1000183034 Gas Seal Assembly - 1st Section, Parts List, Chapter 8.3.
- 1000183035 Gas Seal Assembly - 2nd Section, Drawing, Chapter 8.3.
- PAGE 1000183035 Gas Seal Assembly - 2nd Section, Parts List, Chapter 8.3.

This compressor is equipped with DRESSER-RAND gas seal cartridges.

The DRESSER-RAND gas seal cartridges are delivered as complete assemblies, “ready-to-install”. The cartridge is a tandem configuration of two seals, with an internal labyrinth. A barrier seal (FLOWSERVE) is also placed behind the seal. The barrier seal between the bearing and gas seal cartridge is placed to prevent any bearing oil vapors from passing along the shaft into the gas seal. Refer to User Manual, Part 3, Technical Description, Chapter 3.5 Gas Seal Cartridges, and Part 6 Maintenance, Chapter 6.5 Gas Seal Cartridges.

6.1.7.1 BARRIER SEAL DISASSEMBLY

It is necessary to remove thrust & journal bearings assemblies to provide access to the barrier seals. Seal assembly components rank with the journal and thrust bearings for optimum compressor operation. Handle and store all parts carefully, seal disassembly can start at either end. It is recommended that one assembly be removed before starting disassembly at the opposite end.

1. Remove retaining ring, and store safely.

- 1000183034 Gas Seal Assembly - 1st Section, Drawing, Chapter 8.3.
- 1000183035 Gas Seal Assembly - 2nd Section, Drawing, Chapter 8.3.
- 1000184143 Gas Seal Arrangement CCW - Intake End, Chapter 8.3.
- 1000184144 Gas Seal Arrangement CW - Discharge End, Chapter 8.3.
- 1000186418 Flowserve Barrier Seal Drawing, Chapter 8.3.

2. Refer to the User Manual, Part 7 Parts and Tools, Chapter 7.2 Special Tools, procure the gas seal special tools and mount on the case head as shown.

- 595-195-202 Tools Assembly for Mounting & Dismounting Gas Seal Assembly, Drawing, Chapter 8.3.
- PAGE 595-195-202 Tools Assembly for Mounting & Dismounting Gas Seal Assembly, Parts Lists, Chapter 8.3.

OFFICIAL COPY 00:39490100490.FB003--FB003 Reviewed with comments(C)--05-Nov-2012 08:31 AM

3. Installation plates are provided by FLOWSERVE and bolts to both the rotating sleeve and carbon ring stationary housing.

4. Bolt the settings plates with capscrews provided. Tapped holes are provided in the face of the installation plates for the adapters and puller rods and should be located at the horizontal centerline. Mark the setting plates with the gas seal and with the shaft in 3 angular locations to identify position before removal.

5. With studs 5 located at horizontal center, studbolts 2 would then be installed in the head using tapped holes for bearing housing. Lock studs in head and barrier seal using the nuts provided.

6. Carefully guide the puller plate over the shaft and onto the puller studs. Position the puller as shown and lock in position on studbolts with nut.

7. Now using hexagonal nut in the "puller" position, carefully pull the barrier seal from the head bore. Pull the barrier seal axially as far as possible, then remove the puller assembly.

8. Remove the barrier seal to the storage area. Protect the assembly against corrosion or other damage and store safely. Flowserve barriers seal should always be stored as complete units with the assembly plates (preventing movement or damage of the carbon segments and the thrust springs).

Caution: The barrier seal is not to be disassembled. The barrier seal must be returned to DRESSER-RAND for parts replacement and reassembly.

9. Take care of split ring located under the sleeve, which could drop when uncovered by the sleeve.

- 595-195-202 Tools Assembly for Mounting & Dismounting Gas Seal Assembly, Drawing, Chapter 8.3.

- PAGE 595-195-202 Tools Assembly for Mounting & Dismounting Gas Seal Assembly, Parts Lists, Chapter 8.3.

10. Repeat all this procedure and remove the opposite end barrier seal.

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6.1.7.2 GAS SEAL CARTRIDGE DISASSEMBLY

Seal assembly components rank with the journal and thrust bearings for optimum compressor operation. Handle and store all parts carefully. Seal disassembly can start at either end. It is recommended that one assembly be removed before starting disassembly at the opposite end.

The barrier seal assemblies have been removed. The gas seal cartridge is to be removed and installed as an assembly.

- 595-195-202 Tools Assembly for Mounting & Dismounting Gas Seal Assembly, Drawing, Chapter 8.3.
- PAGE 595-195-202 Tools Assembly for Mounting & Dismounting Gas Seal Assembly, Parts Lists, Chapter 8.3.
- 1000184143 DRESSER-RAND Gas Seal G.A - 1st Section, Drawing, Chapter 8.3.
- 1000184144 DRESSER-RAND Gas Seal G.A - 2nd Section, Drawing, Chapter 8.3.

1. Repeat the same procedure used for barrier seal extraction. Installation plates are provided by DRESSER-RAND and bolts to both the rotating shaft element and to the stationary housing.
2. Bolt the settings plates with capscrews provided. Tapped holes are provided in the face of installation plates for the adapters and puller rods and should be located at the horizontal centerline. Mark the setting plates with the gas seal and with the shaft in 3 angular locations to identify position before removal.
3. Refer to the User Manual, Part 7 Parts and Tools, Chapter 7.2 Special Tools, procure the gas seal special tools and mount on the case head as shown.

- 595-195-202 Tools Assembly for Mounting & Dismounting Gas Seal Assembly, Drawing, Chapter 8.3.
- PAGE 595-195-202 Tools Assembly for Mounting & Dismounting Gas Seal Assembly, Parts Lists, Chapter 8.3.

4. Repeat the same procedure, used for barrier seal extraction. With studs 5 located at horizontal center, studbolts 2 would then be installed in the head using tapped holes for bearing housing. Lock studs in head and setting plates.
5. Carefully guide the puller plate over the shaft and onto the puller studs. Position the puller as shown and lock in position on studbolts with nut.
6. Now using hexagonal nut in the "puller" position, carefully pull the gas seal cartridge from the head bore. Pull the cartridge axially as far as possible, then remove the puller assembly.

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7. Remove the gas seal cartridge (with installation plate) to storage area. Protect the assembly against corrosion or other damage and store safely.

Caution: The DRESSER-RAND gas seal cartridge is not to be disassembled. Only the O-rings are field replaceable. The gas seal cartridge must be returned to DRESSER-RAND for parts replacement and reassembly.

8. Now, the inner labyrinth is accessible. Use the same tools and the same procedure as previously done for gas seal cartridge (puller holes are provided in the face for the seal puller rods).

- 1000183034 Gas Seal Assembly - 1st Section, Drawing, Chapter 8.3.
- 1000183035 Gas Seal Assembly - 2nd Section, Drawing, Chapter 8.3.

9. When free, pull the labyrinth off the shaft. Use care not to damage the knife edges. Before pulling the labyrinth, provide a temporary support under the shaft to "just take the weight" of the rotor off the labyrinth.

10. Repeat the same procedure for the opposite end gas seal assembly.

6.1.7.3 INSPECTION

Labyrinths: Measure the shaft seal journal diameter and the ID (Inside Diameter) or each labyrinth to determine the clearance. If wear brings, the clearance above the maximum listed, the labyrinth should be replaced. Inspect the OD (Outside Diameter) and end faces for nicks and burrs. Dress off any found with a fine cut file.

Dowels: Be sure all dowels are tight.

O-Rings: It is recommended that all the o-rings used in the assemblies be replaced at each scheduled overhaul. Inspect and renew the o-rings if shaved, torn or material has deteriorated. A light coating of Dow # 4 lubricant (or equal) will facilitate assembly and reduce the possibility of damage to the o-rings.

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6.1.7.4 GAS SEAL CARTRIDGE REASSEMBLY

Caution: The DRESSER-RAND gas seal cartridge is to be axially adjusted after overlap adjustment. Refer to Chapter entitled ''Internals or rotor replacement-Axial shimming''. The position of the gas seal cartridge is determined by the''Fitting, or setting dimension that is the basic reference for the gas seal cartridge installation. This dimension is shown on the gas seal general arrangement drawing.

- 1000184143 DRESSER-RAND Gas Seal G.A - 1st Section, Drawing, Chapter 8.3.
- 1000184144 DRESSER-RAND Gas Seal G.A - 2nd Section, Drawing, Chapter 8.3.

Observe the head seal bore and shaft seal journal are wiped clean before starting reassembly.

The gas seal cartridges and inner labyrinths are not interchangeable between the First and Second Sections of the compressor. Refer to User Manual, Part 7 Parts and Tools, Chapter 7.1 Compressor Parts Lists, for the parts numbers, if any doubt exists.

- 1000183034 Gas Seal Assembly - 1st Section, Drawing, Chapter 8.3.
- 1000183035 Gas Seal Assembly - 2nd Section, Drawing, Chapter 8.3.

1. Refer to special tools and install the appropriate entering sleeve over the shaft end and against the shaft shoulder.

- PAGE 1000192229 Seal Entering Sleeve, Tool List, Chapter 8.3.

2. With O-ring and dowels in place, carefully guide the inner labyrinth over the shaft and into the head. The puller rods can be used to guide the labyrinth into the head (reverse the disassembly procedure). Push the labyrinth firmly against the head shoulder.

3. The gas seal cartridge is to be installed as an assembly. Refer to instructions to prepare the cartridge before installation in the seal cavity. Bolt the installation plates, if it had been removed or a replacement gas seal is being installed. All parts must be aligned as per match marks made at disassembly.

4. Install the gas seal entering sleeve over the shaft and against the shaft shoulder

- PAGE 1000192229 Seal Entering Sleeve, Tool List, Chapter 8.3.

5. Inspect the O-rings on the gas seal and replace them if shaved, torn or material has deteriorated. A light coating of Dow # 4 lubricant will facilitate assembly and reduce the possibility of damage to the O-rings.

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6. Carefully and slowly, push the cartridge “in” as far as possible by hand.

7. Procure the special tools supplied for gas seal cartridge and reverse the disassembly procedure. Use the rods to bottom the gas seal cartridge against the inner labyrinth , insuring the anti-rotation dowel enters its mating slot.

8. Remove the installation plates and store with the special tools for later use.

9. Temporarily, install shear ring segments.

10. Installation plates are provided by FLOWSERVE and bolts to both the rotating sleeve and carbon ring stationary housing.

11. Procure the barrier seal sleeve assembly from storage. Install the barrier seal over the shear ring sleeve and against the gas seal cartridge. Rotate slowly the barrier seal to have the anti-rotation dowel entering its mating hole. If necessary, install the special tools used for removal to push finally the barrier against the gas seal cartridge.

- 595-195-202 Tools Assembly for Mounting & Dismounting Gas Seal Assembly, Drawing, Chapter 8.3.

- PAGE 595-195-202 Tools Assembly for Mounting & Dismounting Gas Seal Assembly, Parts Lists, Chapter 8.3.

Repeat the same procedure to install the opposite end gas seal assembly. At this stage, the thrust and journal bearing assemblies can be installed.

6.1.8 BUNDLE DISASSEMBLY AND REASSEMBLY

The bundle disassembly and reassembly instructions are unique to the particular compressor furnished. The bundle assembly has been pulled out of the case onto the cradle or is resting on the bundle supports. The upper half of internal assembly has been removed, and clearances have been recorded at the splitline.

6.1.8.1 BUNDLE DISASSEMBLY

- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.

- PAGE 1000183025 Bundle Assembly, Parts List, s/n 1000181442, Chapter 8.3.

- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.

- PAGE 1000183414 Bundle Assembly, Parts List, s/n 1000181443, Chapter 8.3.

WARNING : The thrust and journal bearings and inner gas seals have been removed. The rotor assembly is being supported within the internal bundle by the inner labyrinths. Do not bump or rotate the shaft while so supported.

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1. **Head removal - first section end**

   a. Remove the O-ring and back-up rings from the outside diameter if not already done previously.

   b. Bundle clamping tools are installed on the head faces of the bundle, and tighten safely the whole assembly, to prevent any stress on the clamp segments attaching the head to return bend..

   - PAGE 121-593-202 Bundle Clamp, Tools List, Chapter 8.3.

   c. Install a safety hoist ring at the top of the head.. Attach overhead hoist to head lifter and take up slack.

   d. At the bottom of bundle assembly, remove the screws and clamp segments which attach the head to return bend assembly. Remove the clamp tooling, while tightening the head lifting slings.

   e. Keep the head centered, pull it straight back until clear of the rotor. Move the head to storage area and set down on wooden blocks in "as is" position, chock to prevent rolling or set down inboard face down .

2. **Head removal - Second section end**.

   a. Remove the O-rings and back-up rings. Install a safety hoist ring to second section head . Attach overhead hoist to head lifter and take up slack. At the bottom, remove the screws which attach the head to the return bend.

   b. Carefully slide the head carefully until clear of shaft end. Move to storage area and set down on wooden beams, chock to prevent rolling or set down inboard face down.

6.1.8.1.1 ROTOR REMOVAL:

- 1000183030 Internal Assembly, Drawing, s/n 1000181442, Chapter 8.3.
- 1000192540 Internal Assembly, Drawing, s/n 1000181443, Chapter 8.3.
- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.

   a. Division wall labyrinth carrier ring should be removed before rotor removal. Matchmark the ring with the rotor and proceed as follows: Refer to internals assembly drawing, rotate the carrier ring (02) around the rotor by 90° until its splitline is horizontal and the splitline capscrews facing up. Remove the splitline capscrews. Lift the upper half straight up and move to storage. Rotate lower half around the rotor and lift up. Attach the half parts together and store safely.

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b. Rig sound slings fabricated from laminated nylon belting around the rotor seal journals and into the hoist. Center the hook carefully and provide a spreader bar between sling parts to prevent axial loading on adjacent impellers.

- LHC-105170B-04 Weights Record - D06RXB Compressor, Chapter 8.3.

c. Hoist the rotor slowly and carefully. Set it down in 'V' block supports prepared beforehand. Line the cradle notches with well-greased leather or sheet lead and set the rotor down on the bearing journals .

NOTE : DO NOT SET ROTOR DOWN ON IMPELLERS. USE 'V' BLOCKS DESIGNED TO SUPPORT ROTOR AT BEARING JOURNALS.

- LHC-105170B-02 Rotor Dimensions, D06R8B Compressors, Chapter 8.3.

This is the normal extent of the bundle disassembly. Refer to following step if complete disassembly is required.

6.1.8.1.2 INTERNAL PARTS DISASSEMBLY:

Unless it becomes necessary for cleaning, there is no particular reason for disassembling the bundle halves. If necessary, the following is recommended.

a. The interstage labyrinth seals can be rotated out when the bundle halves are separated. The upper half labyrinths are secured with two anti-rotation screws at the splitline.

- 1000183030 Internal Assembly, Drawing, s/n 1000181442, Chapter 8.3.

- 1000192540 Internal Assembly, Drawing, s/n 1000181443, Chapter 8.3.

b. The 1st section inlet guide and diaphragms can be lifted directly out of the return bend by installing 2 safety hoist rings in the splitline surface. The upper halves are secured with anti-rotation screws at the splitline. Second section diaphragms and inlet ring can be removed in same way. The upper halves are also secured with anti-rotation screws at the splitline.

c. If complete bundle removal is required, proceed as follows : Turn the bundle half over on wedges or wooden beams so that the splitline is down. Matchmark the division wall at each side with the return bends.

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d. Start removing the axial nuts located in the small openings at the outer periphery of the return bend. Rig hoisting equipment to the second section return bend and carefully slide the return bend (1st Section) on the studs until the nuts can be moved of the stud ends. The return bend (1st Section) can be removed axially on the studs. Avoid to damage the threading of the studs.

e. The division wall half can be separated from the return bend, taking care of the sealing devices. Matchmark the parts before removal. USE ONLY SAFETY HOIST RING.

6.1.8.2 BUNDLE REASSEMBLY INSTRUCTIONS

Determine wear in the interstage labyrinths and clearances to the rotating component. Record the clearance. The rotor assembly must be carefully cleaned and inspected. The shaft bearing and seal journal surfaces must be clean and smooth. The shaft tapers must be clean, smooth and protected with taped-on sleeves. Determine that all bundle components are clean and near at hand in order of assembly. Clean the case bore and determine it is free of nicks or burrs.

Have all new O-rings on hand. Use extreme care when cleaning the casing that tools, wiping cloths, etc, are not dropped into the nozzle openings. Shear ring annulus of the casing bore should be clean and free of roughened areas which could tear or abrade the bundle O-rings.

The bundle reassembly instructions are unique to the particular compressor being furnished. It is necessary to measure the axial dimension of the bundle components and overall assembled bundle length. This is required to assure correct bundle components have been assembled if any replacement parts are being installed. This is also a check to determine any loss of bundle length due to corrosion and/or cleaning procedures.

- 1000183030 Internal Assembly, Drawing, s/n 1000181442, Chapter 8.3.
- 1000192540 Internal Assembly, Drawing, s/n 1000181443, Chapter 8.3.
- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.

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Bundle assembly - Axial dimensions

Parts	Inches	Millimeters
Head - First section End	7.659 +0/-0.002	194.54 +0/-0.05
First section Return Bend	16.544 +0/-0.004	420.23 +0/-0.10
Division Wall	7.123 +0/-0.002	180.95 +0/-0.05
Second section Return Bend	14.451 +0/-0.004	367.06 +0/-0.10
Head - Second section End	1.860 +0/-0.002	47.25 +0/-0.05
Total bundle length	47.639 **+0/-0.014**	1210.03 **+0/-0.35**

6.1.8.2.1 BUNDLE PARTS REASSEMBLY

All components of the bundle assembly have been cleaned, inspected and found serviceable.

NOTE: Due to high discharge temperature around the division wall, an important thermal expansion is applied to the bundle and division wall area. In order to insure that the parts are correctly secured in position, but also to allow the bundle for a normal thermal expansion, the following sequence must be used for bundle reassembly .

a. Obtain the 2nd section end return bend lower half. Install on solid level planks with splitline facing up. Using bundle lifter and hoisting equipment, install the upper half onto the lower half, insert splitline dowels and capscrews. Tighten the splitline capscrews firmly.

b. Rig hoisting equipment to the second section return bend assembly, and install on a flat and level surface with axis in vertical position, seating on the suction end face. Thread and tighten the studs in their holes facing the division wall.

- 1000183030 Internal Assembly, Drawing, s/n 1000181442, Chapter 8.3.
- 1000192540 Internal Assembly, Drawing, s/n 1000181443, Chapter 8.3.

c. Obtain the division wall halves. Preassemble both division wall halves together with splitline dowels and capscrews. Using hoisting equipment lift and bring the division wall over the studs and onto the second section return bend, in correct position, with the shunt hole facing up. Install the division wall half against the return bend, sliding the studs through the division wall holes.

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d. Obtain the 1st section end return bend lower half. Install on solid level planks with splitline facing up. Using bundle lifter and hoisting equipment, install the upper half onto the lower half, insert splitline dowels and capscrews. Tighten the splitline capscrews firmly.

e. Rig hoisting equipment to the 1st section return bend assembly, lift in vertical position, and install over the studs and onto the division wall first section face.

f. Keep the hoisting equipment in place. Slightly lift the return bend to insert the nuts on the stud ends, then tighten the nuts as follows :

g. First, tighten the nuts in cross pattern, 1 'O' clock, 7, 11, 5, 10, 4, 8, 2. at a torque value of 85 ft.lbs.(115 N.m).

h. Then, one by one, in the same cross order as above, untighten each nut completely, and retighten using stretching dimension only, for final tightening. Stretching will be checked with a dial indicator installed on the division wall and the tip on the stud end. Stretching value will be : 0.00068 x 361,67 mm = 0.25 mm (with tol.= 0.049 mm).

NOTE: The stretching value will be determined by using the following formula : stretching value = 0.00068 x L, with L : dim between the threading face receiving the stud, and nut) with stretching tolerance of +/- stretch.value x 0.2.

i. Now, Install the bundle assembly on 'V' blocks, and separate the two half parts, in order to fill each half with internal components.

j. Lift and place the lower half return bend assembly with splitline up. Turn the upper half return bend assembly over on its wooden planking so that the splitline is up. Chock to prevent rolling.

- 1000183030 Internal Assembly, Drawing, s/n 1000181442, Chapter 8.3.
- 1000192540 Internal Assembly, Drawing, s/n 1000181443, Chapter 8.3.

6.1.8.2.2 INTERNALS REASSEMBLY

Refer to Internal assembly drawing

- 1000183030 Internal Assembly, Drawing, s/n 1000181442, Chapter 8.3.
- 1000192540 Internal Assembly, Drawing, s/n 1000181443, Chapter 8.3.

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a. Obtain the first section inlet guide halves (04). Attach to overhead hoist and carefully install inlet guide halves into upper and lower halves of the internal assembly. The upper halves of the inlet guides are held in upper half internal assembly with two anti-rotation screws.

b. Obtain the 1st and 2nd stage diaphragm halves. Using safety eyebolts in threaded holes at split, attach overhead hoist to diaphragm halves and carefully place diaphragm halves into upper and lower internal assembly halves. The upper half diaphragm is held in the upper half internal assembly with two anti-rotation screws.

c. Repeat step b and install the remaining second section diaphragms and inlet support ring.

d. The interstage labyrinths may now be rolled into the inlet guides, and diaphragms. The labyrinths are held in the upper half internal assembly by anti-rotation screws.

e. Division wall labyrinth and carrier will be fitted after the rotor.

f. Place O-ring cordage in the lower half internal assembly splitline surface.

6.1.8.2.3 ROTOR INSTALLATION

- 1000183031 Rotor Assembly, Serial No.: 10-01-050 - Drawing, Chapter 8.3.
- 1000183417 Rotor Assembly, Serial No.: 10-01-051 - Drawing, Chapter 8.3.

a. Lower half internal assembly is completely assembled and resting on the bundle stands (or bundle cradle).

b. Rig to the rotor as described for removal. Rotor assembly is clean and correctly oriented to the bundle. Lower carefully and set down on the internal labyrinth halves. Keep the slings tight to support the rotor. **Do not turn rotor while it is supported by the labyrinths.**

c. The division wall labyrinth and carrier ring 2 can be placed onto the rotor. Carefully install the labyrinth carrier ring lower half, and centering rings 21 & 22 on the rotor, and roll around the rotor in the division wall lower half. Install the labyrinth carrier ring upper half and centering rings onto the lower half, aligning the upper half to bottom half carrier ring. Insert and firmly tighten the capscrews to attach the halves. Insure the anti-rotation dowel is fitted.

NOTE : Be sure the labyrinths carrier ring is correctly oriented. Shunt holes should be imperatively on second section side.

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d. Rotate the carrier ring to have its splitline 90° from the division wall splitline and to place its dowel in the division wall groove.

e. Be sure the teflon insert is placed in bottom half division wall. Do not reinstall the upper half of bundle at this time.

Note: If a new rotor or any replacement internals are being installed, it may be necessary to check the thrust disc axial position. Refer to procedure "Internals or rotor Replacement - Axial shimming " before completing the bundle re-assembly.

Caution: To prevent damage, do not turn the rotor while it is supported by the labyrinths.

6.1.8.2.4 BUNDLE COMPLETION

1. **Head installation - first section end**

a. Install a safety hoist ring at the top of the First Section head. The head must be equipped with new O rings and back up rings on the outside diameter. Place the O-ring in the inner groove of the head.

b. Attach overhead hoist equipment to the head and move the head close to the internal assembly first section side, over the shaft end. Taking care to keep the head correctly oriented, push it slowly against the internal assembly, observing the shrouded end of return bend and inlet guide are coming against their mating face of the head.

c. Keep hoisting equipment in place, install the clamping tools, to tighten the head against the internal assembly

- PAGE 121-593-202 Bundle Clamp, Tools List, Chapter 8.3.

d. Install the lower clamp segments 9 straddling the splitline between head and return bend. Secure the clamp with provided capscrews.

- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.

2. **Head installation - Second section :**

a. Install lifter at the top of head and rig to hoisting equipment. The head must be equipped with new O rings and back up rings on the outside diameter. Carefully lift and drive the head onto and over the shaft end, correctly oriented, and push it slowly against the Second Section face of return bend. Drive the capscrews through the lower holes only of the head and tighten firmly to insure the head is pulled up square on the return bend.

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b. With the bundle assembled, carefully measure the overall length. Refer to table "Bundle axial dimensions" for theoretical dimensions of bundle drawing. The dimensional check is to assure that the correct components have been assembled and is also to check for loss of axial length due to corrosion and clean procedure.

- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.

c. Overlap checking. With the rotor installed in the bundle and temporarily supported, refer to overlap records made at disassembly and theoretical dimensions. Adjust the rotor at its final position.

3. Refer to "Inner Gas Seals" chapter and install the inner gas seals, while keeping the overlaps as required.

4. Refer to "Thrust & Journal Bearing Assembly" chapter and install the bearing assemblies. Thrust shimming can be finallized, without changing the rotor position .

6.1.8.3 BUNDLE ASSEMBLY - UPPER HALF INSTALLATION

- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.

The upper half internal assembly is now preassembled with division wall, inlet guides and diaphragms. Lift and install upper half on solid and level surface then turn it over to have its splitine facing down. Check axial dimension.

1. Install the bundle guide studs in the threaded holes of the lower half splitline surface. Install safety hoisting rings on the return bend upper half assembly in the same way as done for removal. Rig to hoisting equipment. Lift the return bend upper half off and move over the bundle bottom half.

- PAGE 121-513-002 Bundle Guide Stud, Tool List, Chapter 8.3.

2. With the splitline surface horizontal, carefully lower the upper half over the rotor and onto the bottom half, between the heads using the guide studs and dowels to help final assembly. Observe that division wall upper half match correctly the carrier ring already installed after the rotor.

3. Install the bundle clamp tightening tools. At first section end install the two remaining upper clamp segments. Secure with capscrews. At the second section head, insert remaining capscrews attaching the head to the internal assembly. With the upper half return bend installed, insert the splitline capscrews and tighten to torque value 42.6 ft.lbs (58 N.m).

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4. Secure the capscrews with locking wire as shown on bundle drawing.

- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.

5. Install new O-rings and back-up rings on all the grooves of bundle outside diameter.

6. The bundle can now be inserted into the compressor casing.

6.1.9 INTERNALS OR ROTORS REPLACEMENT- AXIAL SHIMMING

When a spare or replacement rotor and/or internals are being installed it may be necessary to make adjustment to the thrust bearing to properly position the rotor within the internal module. The design of bearings does not require to resize the thrust disc bushing flange on the rotor. If using a new bushing, its flange needs only to have the same thickness as the bushing replaced.

Caution: The thrust disc bushing flange faces after finish grind must be parallel and square with the bore within 0.0005 in. (0.01 mm) F.I.R. (Fully Indicated Runout). Grind all metal from the surface of the bushing seating against the shaft shoulder. Use extreme care not to deform the bushing when anchoring it for grinding.

The shim packs affixed to the thrust bearing backing rings are used to position the rotor. The outboard thrust bearing shim pack positions the rotor assembly in relation to the stationary internal components. The inboard shim pack is sized for the thrust bearing running clearance.

SIZING THRUST BEARING SHIM PACKS

1. During internal bundle reassembly complete the following checks.

2. Assemble the lower half internal assembly components.

3. Carefully lower the rotor into the lower half internals until it is resting on the internal assembly halves. Leave the slings rigged so that they support most of the rotor weight.

- LHC-000-008 Impeller/Diffuser Overlap Illustration, Chapter 8.3.
- 1000183025 Bundle Assembly Drawing, s/n 1000181442, Chapter 8.3.
- 1000183414 Bundle Assembly Drawing, s/n 1000181443, Chapter 8.3.

4. Refer to illustration and position the rotor axially so that the overlaps are as near optimum as possible. Refer to bundle assembly drawing for theoretical overlap limits. Install a dial indicator on static parts and with the stem on the rotor, adjust to zero.

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5. Now, with the optimum overlap determined, carefully push the rotor toward the inlet end its limit of travel. Using the dial indicator, measure this distance and record the dimension found.

6. Now, push the rotor its limit of travel toward the Second Section, and using a dial indicator, measure and record the dimension found. The total axial movement (end float) of the rotor should be 0.130" (3.302 mm).

Caution: Do not apply excessive load to the end of rotor when determining the axial movement.

7. Complete the compressor reassembly (without bundle upper half) through installing the thrust disc and the lower half thrust bearing housing.

8. Remove the filler plates and shim pack from the outboard thrust bearing.

9. With the bundle upper half removed, position the rotor axially as recorded in Step 4, to obtain optimum overlap.

10. Install the outboard thrust bearing in the thrust housing, insuring it is fully seated against the thrust bearing housing face.

11. Carefully measure the distance, from the face of the outboard thrust bearing shoes to the outboard face of the thrust disc. This is the required thickness of the outboard thrust bearing total shimming.

12. Install (in halves) the correct thickness of shims.

Note: Shim thickness in the upper and lower halves of the outboard thrust bearing must be exactly equal.

13. Reassemble the inboard thrust bearing.

Caution: Be sure rotor assembly is positioned axially with outboard face of disc against outboard thrust shoes.

14. Install the lower half of the inboard thrust bearing omitting the shim pack. Position the bearing axially so that the thrust shoes are against the thrust disc.

15. Now measure the gap between the inboard thrust bearing and the seating face of the housing. Record the dimension found.

16. The inboard thrust bearing shim pack thickness will then be the dimension found in Step 15. minus the design thrust bearing running clearance.

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17. Remove the lower half inboard thrust bearing and install the correctly sized shim pack (Step 16) in both the upper and lower halves.

18. Refer and complete the thrust bearing reassembly.

6.1.10 RADIAL & AXIAL PROXIMITY PROBES

Refer to User Manual, Part 7, Parts and Tools, Chapter 7.1 Compressor Parts Lists, for the probe assemblies being furnished for this compressor.

System used : BENTLY NEVADA- 3300 SERIES.

- LHA-00-3483 Bently Nevada Proximity Transducer System Manual, Chapter 8.1, Manufacturer Documentation.

6.1.10.1 DISASSEMBLY

Removal of the probe and its mounting hardware from the compressor is readily discernible from the drawing, observing the following precautions:

1. Identify each probe and its mounting hardware to its location in the compressor before removal by tagging.

2. Exercise care in handling as the probe tip is easily damaged-do not allow tip the contact any rough or sharp surface.

3. It is recommended the probe tip be taped over for protection when in storage.

4. Remove axial and radial proximity probes from the bearing housings. Radial probes can be threaded directly out of the thrust bearing housing. The axial probes are contained within the thrust bearing housing and “read” directly on the shaft end. Refer to Bundle Assembly drawing and loosen capscrews which attach the thrust end cover to the thrust bearing housing. Carefully try to remove the axial probes out of the housing, using extreme care not to damage the probe tips against the housing. Tape the probe tip over for storage and store safely.

6.1.10.2 REASSEMBLY

For recommended alarm and/or shutdown settings for monitoring equipment, refer to:

- LHC-000-005 “Rainbow”, Operational Guide Lines for Vibration on Centrifugal Compressor, Chapter 8.3.

Refer to Proximity Probe Setup and Calibration Procedure.

Reinstall each probe in the compressor from which it was removed, observing the following:

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1. Do not measure actual clearance between shaft and end of probe with metal feeler gauge to do so may result in destruction of the probe. Probes should be positioned electrically.

2. Once probe is set, loosen connector and remove any twist put in leads when adjusting probe. It may be necessary to do this once or twice white adjusting probe be certain connector is tight when actually setting the probe.

3. When adjusting axial probes, the rotor assembly must be in its "thrusted" position. Determine rotor total axial travel within the thrust bearings then use a dial indicator to center the rotor. Position jam nut on probe so that tip cannot touch if location is such that probe is threaded into a "blind" hole.

6.1.10.3 PROXIMITY PROBE SETUP AND CALIBRATION PROCEDURE

This procedure is designed to enable those responsible for setting up a Compressor Vibration Detection System to do so without the use of the contract vibration monitor. This procedure will also allow for early detection of mechanical interferences that cause inaccurate proximity measurements and minimize improper probe calibrations. This procedure is intended for factory and field use.

RADIAL AND AXIAL PROBE CALIBRATION

1. Equipment required :

- Contract skid electrical components.
- Battery - powered digital voltmeter, 3 1/2 digits.
- DC, power supply source; 20 V DC (Volts Direct Current) at 1 ampere.
- Dial indicator; range 0.030 -0- 0.030 in. (30 -0- 30 mils)
- Lever.

2. Radial probe calibration :

a. Connect the power supply to the proximiter COM (Common) and minus 20 V DC terminals. Connect the voltmeter to the proximiter COM and signal terminals.

b. Using the contract extension cable and probe, screw the probe in toward the shaft until it touches. Do not overtighten the probe to avoid any damage. When touching, the voltmeter should read between zero and minus one volt.

c. Back the probe out until a reading of minus 7.5 +/-0.5 volts is obtained. Lock the probe in place.

d. In certain circumstances, as outlined below, a decreased (voltage) gap is acceptable. Voltage gap may be as low as minus 4.0 V DC as long as the shut-down setting can be obtained within the calibrated linear operating range.

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- Where distance between probes or clearance around a probe affects the signal (crosstalk).

- Where adjusting probes externally is difficult (i.e., overhung rotors on P-line compressors), the gap voltage may be between 5.5 volts and 9.5 volts as long as the shutdown setting can be obtained within the calibrated linear operating range.

3. Axial probe calibration:

a. With the compressor uncoupled, thrust the rotor inboard and outboard several times. Establish the total rotor travel with the dial indicator.

b. Locate the rotor as closely as possible to its mid-travel or center position.

c. Connect the power supply to the proximiter COM and minus 20 V DC terminals. Connect the voltmeter to proximiter COM and signal terminals.

d. Using the contract extension cable and probe, screw the probe in toward, the thrust disc until it touches. Do not overtighten the probe to avoid any damage. When touching, the voltmeter should read between zero and minus one volt.

e. Back the probe out until a reading of minus 10.0 +/- 1.0 V DC is obtained. Lock the probe in place.

f. Thrust the rotor (as in Step a.) and ensure that the travel measured by the dial indicator agrees with the travel measured with the voltmeter.

- The proximiter output is 200 mv/mil (millivolts/0.001 in.). As an example, if a rotor has a total travel of 0.020 in., it should cause the voltmeter to read between minus 8.0 and minus 12.0 volts. The further the thrust disc is from the probe, the greater the voltage reading magnitude will be.

- Deviation from: Minus 10 V $= \dfrac{(\text{Total Travel in mils})\,(0.200)}{2}$

- Discrepancies can generally be attributed to side-wall influences on the probes magnetic field. These side-wall influences should be eliminated. Consult Dresser-Rand Service Engineering for assistance.

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CONTRACT VIBRATION MONITOR CALIBRATION

1. **Equipment required:**

   - Contract monitor and associated support equipment.
   - Bently Nevada TK3-2 or equal.
   - Proximiter, extension cable and probe (typically Bently Nevada 7200 or 3300 series).
   - Digital Voltmeter, 3 1/2 digits.
   - Single channel oscilloscope.

2. **Radial probe calibration:**

   a. Power up the contract monitor.

   b. Connect the proximiter, extension cable and probe to the TK3 "wobulator" disc. Connect power to the proximiter and make monitor connections.

   c. Adjust the probe distance from the center of the wobulator disc to develop a sine wave of 0.2 volts peak-to-peak, 0.4 volts peak-to-peak and 0.6 volts peak-to-peak. Adjust monitor span and zero for readings of 0.001, 0.002 and 0.003 in.

3. **Axial probe setup procedure:**

   a. Power up the contract monitor.

   b. Connect the proximiter, extension cable and probe to the TK3 micrometer/disc assembly. Connect power to the proximitor and make monitor connections.

   c. Adjust the micrometer to allow for a voltage reading of minus 10 volts. Note the reading. Zero the monitor. Rotate the micrometer +/- 0.020 in and ensure that the voltmeter indicates minus to 10 +/- 4.0 volts. Adjust monitor span to indicate 0 +/- 20 mils travel.

6.1.11 EMBEDDED TEMPERATURE DETECTORS

Refer to User Manual, Part 7 parts and tools, Chapter 7.1 spare parts, for the T.I.D. assemblies being furnished for this compressor.

The compressor is equipped with thermocouples or resistance temperature detectors embedded in the thrust bearing and/or journal bearing shoes, special precautions must be observed when removing or installing the bearings to avoid damaging the delicate lead wire.

OFFICIAL COPY 00P39490100490.FB003 Reviewed with comments (C) 05-Nov-2012 08:31 AM

6.1.11.1 DISASSEMBLY

- LHC-000-003 Embedded Temperature Detectors Illustrations, FIG. A & B, Chapter 8.3.

1. On installations that employ connection heads attached to the casing or cover, remove cap from head and disconnect lead wire at that point. Loosen nut on bushing inside connector then carefully pull leads to the thermocouple or detector into the bearing chamber. Remove clamp which secures leads to housing.

2. If it is necessary to remove the outboard base ring assembly from the housing, first disassemble packing gland then carefully pull assembly out of cover while gently guiding lead wire through packing gland.

3. The thrust bearing assembly is provided with embedded temperature detectors, disassemble packing gland at outside of bearing housing. Carefully pull thrust bearing from housing bore while gently guiding the leads through the housing. Lead wires are retained in housing split grooves.

4. Embedded temperature detectors are provided in the journal bearing shoes, disassemble the bearing, except when rotating the shoe containing the temperature detector. Refer to Thrust & Journal Bearings maintenance.

   a. Disassemble the packing gland and remove clamp holding leads to housing or case.

   b. Carefully push leads through bearing housing while removing housing.

5. If lead wire is secured by a holding screw and washer, they must be removed to allow replacement of a thrust or journal bearing shoe. When reassembling the holding screw, apply Loctite 242 or similar material and tighten screw and washer to secure, but not crush, lead wire.

6.1.11.2 REASSEMBLY

When reassembling units, reverse the disassembly procedure. Be sure to tighten nuts on bushing and packing glands and to install clamps where provided. When installing the journal bearing shoe assembly, it will be necessary to start the leads through the hole in the housing and then to pull the leads through while the shoe is being rotated into the housing.

6.1.11.3 TEMPERATURE DETECTOR REPLACEMENT

- LHC-000-003 Embedded Temperature Detectors Illustrations, FIG. C, Chapter 8.3.

Normally there is no need to remove the temperature detector from the thrust and/or journal bearing shoes. If replacement is necessary, complete the following at the workbench with the bearing assembly completely dismantled.

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1. Remove the original temperature detector, compression spring and retaining ring from the bearing shoe. Use extreme care not to damage the babbitt face on the shoe when handling.

2. Procure the replacement temperature detector and slip the compression spring over the leads and locate on the detector case.

3. Slip the retaining ring over the leads and locate against the compression spring. The angle of the self-locking prongs in the retaining ring must be facing away from the spring and detector case.

4. Slip a short length of metal tubing over the leads and place against the retaining ring. The metal tube will be used to seat the temperature detector, spring and retaining ring in the bearing shoe. It is suggested the pusher tube be fabricated from 3/16" O.D. thin wall stainless steel tubing.

5. Insert the detector, spring and retaining ring in the appropriate hole in the bearing shoe. Use the metal tube to bottom the detector in the shoe then remove the tube. Note the retaining ring prongs securely fasten the detector in the shoe.

6. Complete Journal Bearings Reassembly.

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6.2 DRIVER

6.2.1 GENERAL INFORMATION

Prior to performing any maintenance review the User Manual, Part 2 main data, Chapter 2.2 safety warnings and precautions. Also pay attention to warnings listed within WEG documentation. Preventive maintenance through inspection is the best way to detect problems before damage occurs.

- LHA-105170B-01 WEG Instruction Manual, Motor MGW900A, Chapter 8.1.

6.2.2 MOTOR MAINTENANCE

Under normal operating conditions, only preventive maintenance should be necessary for the WEG motor, as requested in the documentation, typically during operating routine inspection periods, or when time permits. For additional information see the motor drawings and maintenance instructions.

Note: Prior to start maintenance activities, insure that the power supply is disconnected, and locked “off” on the panel, with lube oil system stopped, and casing at cold temperature. Refer to WEG instructions for additional requirements.

Note: After the complete stop of the motor, it is recommended to perform a “hot” alignment check. Record the results for reference during re-installation, especially if some mechanical and/or vibration problems have been met, previously.

It is recommended to remove, first, the coupling guard between the motor and the gearbox, then to remove the coupling. Disconnect cooling piping and lubricating piping to free the access to the maintenance parts. Remove carefully the vibration and temperature detector leads, if any risk of damage occurs during maintenance activities.

Prepare a space around the motor, large enough for easy access to the motor parts and maintenance activities. Proceed with motor inspection, following instructions described in WEG documentation, User Manual, Part 8 documents and drawings, Chapter 8.1, supplier’s manufacturer documents.

Even so a common lube oil system is feeding lubricating oil to motor, gear and compressor, check with regular intervals that the lubricating oil pressure and temperature are in accordance with the WEG requirements, as shown in their manual. It is recommended to take a sample of oil and make an analysis.

OFFICIAL COPY 00P39490100490.FB003 Reviewed but not (C) 05-Nov-2012 08:11 AM

6.3 GEARBOX

6.3.1 GENERAL INFORMATION

Prior to performing any maintenance review the User Manual, Part 2 main data, Chapter 2.2 safety warnings and precautions. Also pay attention to warnings listed within FLENDER documentation. Preventive maintenance through inspection is the best way to detect problems before damage occurs.

- LHA-105170B-56 FLENDER Operating & Maintenance Instructions, Chapter 8.1.

6.3.2 GEARBOX MAINTENANCE

Under normal operating conditions, only preventive maintenance should be necessary for the gearbox, as requested in the FLENDER documentation, typically during operating routine inspection periods, or when time permits. For additional information see gearbox drawing and maintenance instructions.

- LHA-105170B-56 FLENDER Operating & Maintenance Instructions, Chapter 8.1.

Warning : Prior to start maintenance activities, insure that gearbox is in safety conditions. Driving power and lube oil pumps must be locked in “off” position. All electrical instruments must be disconnected from power supply. Gearbox casing should be at cold temperature.

Note: Once the gearbox is shut down, it is recommended to perform a “hot” alignment check. Record the results for reference during re-installation, especially if some mechanical and /or vibration problems have been met, previously.

It is recommended to start removal with the coupling guards at both sides of the gearbox. Refer to chapter “Couplings” for coupling guards removal procedure. Then remove the couplings (See next paragraph, if coupling hub must be removed, and is hydraulic fit type). Remove the instrumentation and associated piping around. Then proceed with gearbox removal and inspection, refer to procedure described in FLENDER documentation, User Manual, Part 8 documents and drawings, Chapter 8.1 supplier’s manufacturer documents.

- LHA-105170B-56 FLENDER Operating & Maintenance Instructions, Chapter 8.1.

Even if a common lube oil system is feeding lubricating oil to motor, gearbox, and compressor, check with regular intervals that the lubricating oil pressure and temperature are in accordance with the FLENDER requirements, as shown in their manual. It is recommended to take a sample of oil (from gearbox casing) and make an analysis as required.

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6.4 COUPLINGS

6.4.1 GENERAL INFORMATION

Prior to performing any maintenance review the User Manual, Part 2 main data, Chapter 2.2 safety warnings and precautions. Also pay attention to warnings listed within J.CRANE's documentation. Preventive maintenance through inspection is the best way to detect problems before damage occurs.

- LHA-105170B-02 J.CRANE Couplings Fitting and Maintenance Instructions, Between Driver & Gearbox, Chapter 8.1.
- LHA-105170B-03 J.CRANE Couplings Fitting and Maintenance Instructions, Between Gearbox & Compressor, Chapter 8.1.

6.4.2 COUPLING GUARDS MAINTENANCE

6.4.2.1 DRIVER / GEARBOX COUPLING GUARD

- LHA-105170B-91 Coupling Guard Drawing & Parts List, between Driver & Gearbox, Chapter 8.3.

6.4.2.1.1 DISASSEMBLY

Disconnect all piping attached to the coupling guard. There is normally no need to remove adapter, except in case of coupling hub installation or removal.

1. Remove screws 18 and washers 20 at split.
2. Remove screws 18 and washers 20 on top halves of coupling guard 2 / outer clamping ring 4 assembly.
3. Remove screws 13 and washers 14 on top halves of coupling guard 2 / adapter-motor side 1 assembly. Lift and remove top half of coupling guard 2, using hoisting bolts 12. Temporarily, wire together gasket 8 and adapter 1, driver side.
4. Repeat Steps 2. and 3., for remove bottom halves coupling guard 2. Detach and slide gasket 8 on J.CRANE coupling spacer (model: HLFE-0320-0533-05080).
5. Remove screws 19 and washers 20 on top halves of inner clamping ring 5 / adapter 6, gearbox side assembly. Temporarily, wire together expansion joint 9 and adapter 6.
6. Repeat Step 5., for remove bottom half. Then, detach and slide expansion joint 9 and inner clamping ring 5, on J.CRANE coupling spacer.
7. If necessary, remove adapter 6, gearbox side.

OFFICIAL COPY 00P39490100490.FB003--FB003 Reviewed but not (C)--05-Nov-2012 08:31 AM

8. At this stage the J.CRANE's coupling spacer can be removed. Refer to J.CRANE's drawing and documentation.

- LHA-105170B-02 J.CRANE Couplings Fitting and Maintenance Instructions, Between Driver & Gearbox, Chapter 8.1.
- HI-39490-AD-006 J.CRANE Low Speed Coupling Drawing (Driver/Gearbox), Chapter 8.2.

Note: Expansion joint 9 and gasket 8 are in one piece and can only be moved out with coupling spacer.

6.4.2.1.2 REASSEMBLY

For reassembly, reverse disassembly procedure, observing following : the gasket 8 and thermal expansion gasket 9 must be passed over the coupling spacer. Thermal expansion gasket and sealing gaskets should be replaced, if found not serviceable.

6.4.2.2 GEARBOX / D06R8B COMPRESSOR COUPLING GUARD

- LHA-105170B-92 Coupling Guard Drawing & Parts List, between Gearbox & D06R8B Compressor, Chapter 8.3.

6.4.2.2.1 DISASSEMBLY

Disconnect all piping attached to the coupling guard. There is normally no need to remove adapter, except in case of coupling hub installation or removal.

1. Remove screws 22, washers 24 and dowels 25 at split.
2. Remove screws 22 and washers 24 on top halves of coupling guard 3 / outer clamping ring 5 assembly.
3. Remove screws 22 and washers 24 on top halves of coupling guard 3 / adapter-compressor side 1 assembly. Lift and remove top half of coupling guard 3, using hoisting rings 16. Temporarily, wire together gasket 9 and adapter 1, compressor side.
4. Repeat Steps 2. and 3., for remove bottom halves. Detach and slide gasket 9 on J.CRANE coupling spacer (model: HLCE-0159-H533-06100).
5. Remove screws 23 and washers 24 on top halves of inner clamping ring 6 / adapter 7, gearbox side assembly. Temporarily, wire together expansion joint 11 and adapter 7.
6. Repeat Step 5., for remove bottom half. Then, detach and slide expansion joint 11, on J.CRANE coupling spacer.

OFFICIAL COPY 00P39490100490.FB003--FB003 Reviewed but not (C)--05-Nov-2012 08:31 AM

7. If necessary, remove adapter 1 & 2, compressor side.

8. At this stage the J.CRANE's coupling spacer can be removed. Refer to J.CRANE's drawing and documentation.

- LHA-105170B-03 J.CRANE Couplings Fitting and Maintenance Instructions, Between Gearbox & Compressor, Chapter 8.1.
- HI-39490-AD-010 J.CRANE High Speed Coupling Drawing (Gearbox/Compressor), Chapter 8.2.

Note: Expansion joint 11 and gasket 10 are in one piece and can only be moved out with coupling spacer.

6.4.2.2.2 REASSEMBLY

For reassembly, reverse disassembly procedure, observing following : the gasket 11 and thermal expansion gasket 10 must be passed over the coupling spacer. Thermal expansion gasket and sealing gaskets should be replaced, if found not serviceable.

6.4.3 COUPLINGS MAINTENANCE

6.4.3.1 DRIVER / GEARBOX COUPLING

A J.CRANE high speed coupling, model HLFE-0320-0533-05080, is provided for mounting between the driver and the gearbox. Although typically under normal operating conditions no servicing or maintenance should be necessary for the coupling, the bolts should be checked for tightness and the membranes inspected periodically, typically during compressor inspection periods, or when time permits. For additional information see the coupling drawing and maintenance instructions.

- LHA-105170B-02 J.CRANE Couplings Fitting and Maintenance Instructions, Between Driver & Gearbox, Chapter 8.1.
- HI-39490-AD-006 J.CRANE Low Speed Coupling Drawing (Driver/Gearbox), Chapter 8.2.

Note: Dismantling the coupling and membrane units will invalidate the dynamic balance and thus is left at the discretion of the user.

Note: It is recommended, that on all bolted assemblies using self locking nuts, the nuts be replaced after dismantlings.

Torque values for the coupling bolts are provided in J.CRANE's documentation, User Manual, Part 8 documents and drawings, Chapter 8.1 Suppliers Manufacturer Documents.

OFFICIAL COPY 00P39490100490.FB003 Reviewed with comments (C) 05-Nov-2012 08:31 AM

6.4.3.2 GEARBOX / D06R8B COMPRESSOR COUPLING

A J.CRANE high speed coupling, model HLCE-0159-H533-06100, is provided for mounting between the gearbox and the D06R8B compressor. Although typically under normal operating conditions no servicing or maintenance should be necessary for the coupling, the bolts should be checked for tightness and the membranes inspected periodically, typically during compressor inspection periods, or when time permits. For additional information see the coupling drawing and maintenance instructions.

- LHA-105170B-03 J.CRANE Couplings Fitting and Maintenance Instructions, Between Gearbox & Compressor, Chapter 8.1.
- HI-39490-AD-010 J.CRANE High Speed Coupling Drawing (Gearbox/Compressor), Chapter 8.2.

Note: Dismantling the coupling and membrane units will invalidate the dynamic balance and thus is left at the discretion of the user.

Note: It is recommended, that on all bolted assemblies using self locking nuts, the nuts be replaced after dismantlings.

Torque values for the coupling bolts are provided in J.CRANE's documentation, User Manual, Part 8 documents and drawings, Chapter 8.1 Suppliers Manufacturer Documents.

OFFICIAL COPY 00P39490100490.FB003 Reviewed with comments (C) - 05-Nov-2012 08:31 AM

6.5 GAS SEAL CARTRIDGE

6.5.1 GENERAL INFORMATION

Prior to performing any maintenance review the User Manual, Part 2 main data, Chapter 2.2 safety warnings and precautions. Preventive maintenance through inspection is the best way to detect problems before damage occurs.

This chapter describes the preparation, installation and removal of the gas seal from the compressor. Seal cartridge disassembly is not covered, as the complete cartridge should be returned to Dresser-Rand S.A. Le Havre when refurbishment is required.

- 1000184143 DRESSER-RAND Gas Seal G.A - 1st Section, Drawing, Chapter 8.3.
- 1000184144 DRESSER-RAND Gas Seal G.A - 2nd Section, Drawing, Chapter 8.3.

6.5.2 GAS SEAL CARTRIDGE MAINTENANCE

6.5.2.1 INTRODUCTION

The seals are delivered packed separately. Prior to unpacking the seals, the conditions of storage must be checked to ensure the bag used to protect each seal has not been opened during the storage period. If the bag is not in a good shape or if the seal cartridge installation takes place later than 24 months after shipment, it is recommended that Dresser-Rand inspect the seals before installation.

Prior to starting the gas seal installation, confirm that:

1. All necessary tools are available.
2. The thrust adjustment has been completed and that the compressor shaft is secured in its running position.
3. The area, into which the gas seal cartridge is to fit, is clean and free from dirt, grease, oil, etc...
4. If the gas seals have not been previously fit into the case, (i.e. new gas seal, new shaft, etc...), check that the cavity dimensions are in accordance with installation drawings.
5. Prior to the seal installation, exercise the seal cartridge to check for free axial and rotational movements.

- INST-GS-199 REV1 Specification - Checking for the Free Axial Movement of the Gas Seal, Chapter 8.3.
- INST-GS-200 Specification - Checking for the Free Rotational Movement of the Gas Seal, Chapter 8.3.

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6.5.2.2 GAS SEAL CARTRIDGE INSTALLATION

Warning: Absence of liquids and cleanliness are mandatory to ensure reliable operation of the gas seals. Attention is hereby drawn on dryness and cleanliness of all lines connected to the gas seals immediately prior to all start-ups. Refer to inner gas seal function, Chapter 5.1 operating procedures in the manual and associated P&IDs for details and sequence for draining the gas seal supply lines, gas seal vent lines (primary and secondary) and seal gas supply filter bodies. Any intervention on the seal system/piping or inside the compressor seal cavities must be carried out in such a way as to prevent the introduction of dirt or foreign particles.

Note: Due to the configuration of the compressor, it is not necessary to adjust the axial position of the gas seal. No shims are provided with the gas seal.

Note: Dresser-Rand gas seals are designed for one direction rotation, therefore reverse rotation should be avoided. However, for alignment and maintenance procedures, hand-rotation in either direction is possible.

During operation, the gas seal stator is pushed against the head shear ring, in the outboard direction, by the gas pressure. The gas pressure also pushes the gas seal rotor in the outboard direction, against the shaft shear ring. It is held in place by 2 anti-rotation keys.

Note: All gas seals are packaged (and to be stored) with its installation plates. The installation plates secure the rotating and stationary components of the gas seal in the correct position relative to each other.

For a new installation of a gas seal cartridge, for either a new rotor or new gas seal, proceed as follows to determine the axial position of the seal cartridge:

Caution: Ensure that the direction of the rotation shown on the seal matches with actual shaft rotation for the end being fitted.

1. Prior to installing the gas seal, secure the compressor rotor axially, in its running position.

2. Apply a light coating of high temperature silicon lubricant to all interface O-rings. Apply a light coating of copper anti-seize compound (Recommended: Rocol 166) inside the bore of the gas seal main sleeve in the region of the tolerance rings.

OFFICIAL COPY 00P39490100490.FB003--FB003 Reviewed with comments(C)--05-Nov-2012 08:31 AM

3. Turn the rotor such that the anti-rotation keyways are in the horizontal position. Engage the seal as far as possible by hand. Adjust the angular position of the seal stator into the head cavity. To rotate the seal stator versus the seal rotor, it is necessary to slightly loosen the setting plates.

4. Refer to User Manual, Part 6 maintenance, Chapter 6.1 compressor. Install gas seal special tools, supplied by Dresser-Rand, and jack the seal into its position on the rotor.

- 595-195-202 Tools Assembly for Mounting & Dismounting Gas Seal Assembly Drawing, Chapter 8.3.

Note: Dowels in the gas seal cartridge must be kept aligned with the dowel holes in the counter part.

Note: NEVER ATTEMPT TO ROTATE A MISALIGNED SEAL AS IT WILL DAMAGE THE INTERNAL COMPONENTS.

5. Remove the gas seals tools and the installation plates. Store the setting plates and screws with the gas seal tools.

6. Check the offset dimension prior to install the shear ring.

Note: The actual offset dimension should not deviate from the theoretical value by more than +/- 0.04 in.

- 1000184143 DRESSER-RAND Gas Seal G.A - 1st Section, Drawing, Chapter 8.3.
- 1000184144 DRESSER-RAND Gas Seal G.A - 2nd Section, Drawing, Chapter 8.3.

7. With the gas seal installed, place the anti rotation keys, and the shear ring in the shaft groove.

8. Install the sleeve cover shear ring. Secure the sleeve cover shear ring with the provided screws. In order to finalise the proper axial positioning, in some cases, it may be necessary to push/pull slightly the seal rotor on the compressor shaft in order to finalise the proper axial positioning.

9. Repeat this installation procedure for the opposite gas seal cartridge.

OFFICIAL COPY 00P394901000490.FB003 -- FB003 Reviewed but not commented (C) -- 05-Nov-2012 08:31 AM

6.5.2.3 BARRIER SEALS MAINTENANCE

The barrier seals (supplied Flowserve, model : L200W 160.2) are designed to cover the widest range of operating parameters and require virtually no maintenance.

In the compressor, the barriers seals are to be inserted and extracted as an assembly. The barrier seals are not dismantled. Only the O-rings between the barrier seals and compressors heads are field replaceable. Refer to the User Manual, Part 6 maintenance, Chapter 6.1 compressor.

- 1000183034 Gas Seal Assembly, 1st Section Drawing, Chapter 8.3.
- 1000183035 Gas Seal Assembly, 2nd Section Drawing, Chapter 8.3.

To protect the carbon segments during storage and transportation, the barrier seals must be shipped with the transit sleeves supplied by Flowserve.

- 1000186418 Flowserve Barrier Seal CW and CCW Drawings, Chapter 8.3.

Although typically under normal operating conditions, no servicing or maintenance should be necessary for the barrier seals, the external o-rings should be inspected and changed periodically, typically during compressor inspection periods, or when time permits. For additional informations see Flowserve documentation, User Manual, Part 8 documents and drawings, Chapter 8.1 supplier's manufacturer documents.

- LHA-105170B-04 Flowserve Barrier Seal Instructions, Chapter 8.1.

6.5.2.4 GAS SEAL CARTRIDGE REMOVAL

Warning: The compressor must be out of service, fully depressurized and the rotor axially secured before the dry gas seals can be removed.

- 1000184143 DRESSER-RAND Gas Seal G.A - 1st Section, Drawing, Chapter 8.3.
- 1000184144 DRESSER-RAND Gas Seal G.A - 2nd Section, Drawing, Chapter 8.3.

1. Remove the retaining ring and store it in its gas seal box.
2. Secure the installation plates using the appropriate screws.
3. Refer to User Manual, Part 6 maintenance, Chapter 6.1 compressor. Install gas seal special tools, supplied by Dresser-Rand, and remove the seal cartridge carefully from the compressor head.
   - 595-195-202 Tools Assembly for Mounting & Dismounting Gas Seal Assembly Drawing, Chapter 8.3.
4. Record status and any abnormality.

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5. Protect the seal assembly (including the installation plate) against corrosion, other damage or dust and store it in its shipping box.

Note: Light axial marks on the rotor are normal and acceptable.

6.5.2.5 COMMINSSIONING PROCEDURE

6.5.2.5.1 STATIC TEST

If a barrier seal is installed, commission the buffer gas system to ensure that it is functioning correctly. Pressurize the gas seal incrementally up to maximum working pressure. Record seal leakage against each pressure increment. If excessive leakage is observed, the compressor must be depressurized and the cause of excessive seal leakage rectified. If the static results are acceptable, depressurize the compressor and proceed to the dynamic test.

- 1000186418 Flowserve Barrier Seal CW and CCW Drawings, Chapter 8.3.
- LHA-105170B-04 Flowserve Barrier Seal Instructions, Chapter 8.1.

6.5.2.5.2 DYNAMIC TEST

Ensure that separation gas is fed to the barrier seals prior to admit oil to the bearings. For each start-up, record inboard seal leakage rates up to the maximum continuous speed is attained. Compare these values with those guaranteed. Continue to record the inboard seal leakage rates every 30 minutes for a period of 4 hours continuous operation.

Table 1

Seal pressure	Speed	Guaranteed leakage L/min	
Bar g	RPM	Inboard	Outboard
109	12126	204	N/A
109	0	59	N/A
0.2	12126	N/A	Negligible
0.2	0	N/A	Negligible

6.5.2.6 STORAGE

Gas seals should always be stored as complete cartridge units with the installation plates properly fitted, and wrapped in a clean sealed plastic bag. Prior to dispatch, every seal is protected by a bag and packaged in a wooden box. This box is suitable for shipment and storage in an enclosed, dry and tempered area over the maintenance period. Seals should be stored indoors and away from direct source of heat.

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6.6 LUBE OIL SYSTEM

6.6.1 GENERAL INFORMATION

Prior to performing any maintenance, review the User Manual, Part 2 main data, Chapter 2.2 safety warnings and precautions. Also pay attention to warnings listed within OELTECHNIK lube oil console documentation. Preventive maintenance through inspection is the best way to detect problems before damage occurs.

- LHA-00-3677 Oeltechnik Lube Oil Console Instruction Manual, Chapter 8.1.
- LHA-105170B-41 Oeltechnik Lube Oil Console Instruction Manual, Chapter 8.1.

6.6.2 LUBE OIL SYSTEM PREVENTIVE INSPECTION

6.6.2.1 COMPRESSOR LUBE OIL SYSTEM PREVENTIVE MAINTENANCE

Check lube oil supply piping connections and compressor return drain oil piping to the reservoir

Check lube oil instrumentation fitted on compressor and piping around

Record all deviations from normal operation and corrective actions to be carried out after compressor shut down.

Lube oil rundown tank : Record the oil level, and temperatures. Check for leaks. Lube oil rundown tank check valve integrity will have to be checked once the unit is in stop position.

6.6.2.2 LUBE OIL CONSOLE PREVENTIVE MAINTENANCE

Following recommendations must be read as general guidelines and should be used with OELTECHNIK instructions.

While the unit is running, it is recommended to record, on a periodic basis, all parameters which could help the operator to detect any progressive loss of equipment performance and conduct to maintenance removals and inspections. As a minimum, the following preventive maintenance inspections are recommended for the main components of OELTECHNIK lube oil system:

Lube oil reservoir: record the oil level and temperatures. Check for leaks and electric heater connections. Check temperature regulation of heater. Check presence of water that could be accumulated at the bottom of the tank. The bottom of the reservoir is designed such that the water can be eliminated by opening slightely the drain valve.

- LHA-105170B-20 Elmess Heater Specification & Drawing, Chapter 8.1.

Lube oil pumps: record the oil discharge pressure. Check for leaks. Observe that there is no abnormal vibration or abnormal temperature on pump bearings. Check electric motors.

- LHA-105170B-39 Leistritz Oil Pump Instructions, Chapter 8.1.

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- LHA-105170B-08 ABB Motors Instruction Manual, Chapter 8.1.

Lube oil coolers : Check for oil leaks. Record the upstream and downstream temperatures on the operating oil cooler. Check water cooling circuit, for clogging. To avoid contamination of oil system by water, the water pressure should ALWAYS be lower than the oil pressure. If oil cooler switch over is required, do not forget to open the vent valve to purge the air.

- LHA-105170B-45 Thermowave Oil Coolers Instruction Manual, Chapter 8.1.

Lube oil filters : Record the differential pressure. If too high, switch over to the other filter, and remove the cartridge for cleaning, then re-install the cartridge or new one, to have it available for operation if necessary. When placing the filters in operation, vent the filter at the top, to avoid damage to the cartridge, while changing filter selection during compressor operation.

- LHA-105170B-34 Indufil Duplex Oil Filters Instructions, Chapter 8.1.

Instrumentation: Check for leaks. Compare indicators readings from one to each others to confirm accuracy. Vent and drain all instruments and manifolds which have drain / vent valves.

When the oil unit is out of service (or in stand-by) position: if the unit needs to be kept in stand-by position for a long period, the oil must be circulated for 8 hours, once every 2 weeks.

6.6.3 PREPARATION OF MAINTENANCE ACTIVITIES

With the unit secured in shutdown position, prior to starting any maintenance, develop and implement a safety action plan, for all activities including compressor lube oil system and lube oil console equipment; based on observations made during the operating routine inspections.

6.6.3.1 COMPRESSOR LUBE OIL SYSTEM MAINTENANCE

All surfaces around the equipments requiring maintenance should be clean and free of grease.

Inspect the lube oil supply lines to the compressor and return draining lines from compressor to the lube oil reservoir for leaks .

Inspect instrumentation fitted on, such that flow glass, temperature and pressure indicators, etc...Check for leaks, and connections.

It is recommended to check the lube oil rundown tank integrity, in hot oil condition, to insure a correct operation during the shut down sequence. Refer to the operating routine instructions, in user’s manual, Part 5, chapter 5.1, for the procedure to be applied.

6.6.3.2 LUBE OIL CONSOLE MAINTENANCE

Prior to major equipment removal, power must be shut-off on main oil pump driver. All electrical power must be **locked in the “off” position** for stand-by and emergency pump motors. Operators should be equipped with protection clothes, gloves, safety shoes and hard hats.

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The oil console equipment and lube oil tank will be easily drained with oil at a hot temperature. When draining oil, open the vent valves at top points for air entry, and open instrument manifold drain valves as well.

Warning: The lubricating oils, particularly those of synthetic origin, may be toxic in nature and could be resulting in skin injury. Therefore, protective gloves and clothing should be worn when dealing with leakage or spillage of lubricant.

Prepare all necessary tools and space around the oil console.

Inspect all piping connections, for oil leaks, and bolting. Check the hand valves for proper operation. Check the air and oil connections of control valves and check actuation as well .

Inspect electrical connections, cable tray, and lead wires connections to junction boxes.

Check the oil pump to driver alignments in ‘hot’ condition.

In case of any doubt about a pressure or temperature transmitter, take the opportunity to test the instrument against a temperature / pressure reference instrument of known accuracy. Test the instrument by increasing and decreasing signal in the both ways, 2 to 3 times, for consistency

Using preventive observations noted during operation of the oil console, prepare maintenance works to be carried out on equipment . Refer to lube oil console OELTECHNIK and other major component’s instructions for maintenance procedures of specific equipment. See User Manual, Part 8 documents and drawings, Chapter 8.1 Suppliers Manufacturers Documents .

- LHA-105170B-21 Elmess Heater Instructions, Chapter 8.1.
- LHA-105170B-39 Leistritz Oil Pump Instructions, Chapter 8.1.
- LHA-105170B-08 ABB Motors Instruction Manual, Chapter 8.1.
- LHA-105170B-45 Thermowave Oil Coolers Instruction Manual, Chapter 8.1.
- LHA-105170B-34 Indufil Duplex Oil Filters Instructions, Chapter 8.1.

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PART 7 - PARTS AND TOOLS

7.1 COMPRESSORS PARTS LISTS 1-1

7.1.1 GENERAL INFORMATION 1-1

7.1.2 RECOMMENDED ORDERING PROCEDURE 1-1

7.1.3 PARTS ORDERS 1-2

7.1.4 COMPRESSORS TABLE OF CONTENTS 1-3

7.1.4.1 D06R8B COMPRESSOR - TAG NO. K-T7131/32 1-3

7.1.4.2 D06R8B COMPRESSOR - TAG NO. K-T7141/42 1-4

7.2 SPECIAL TOOLS 2-1

7.2.1 GENERAL INFORMATION 2-1

7.2.2 TOOLS FOR D06R8B COMPRESSOR 2-1

OFFICIAL COPY 00P39490100490.FB003 Reviewed but not submitted (C) 05-Nov-2012 08:15 AM

7.1 COMPRESSORS PARTS LISTS

7.1.1 GENERAL INFORMATION

The compressors parts lists are provided in the following Tables of Contents. Parts lists for auxiliary equipment are provided in the supplier's manufacturers documents, Chapter 8.1.

7.1.2 RECOMMENDED ORDERING PROCEDURE

The parts lists contained in this manual have been prepared to assist you in ordering spare replacement parts. Your order will be filled promptly and accurately when the following information is included:

1. The model and serial number of the unit for which the parts are intended. This specifically identifies the equipment you have and permits Dresser-Rand to verify the part numbers in your order.

2. The parts list page number and revision number. These numbers appear at the bottom of the page.

3. The item number for the part. Item numbers on the parts list page correspond to the number shown on illustration to picture the part and its location in the assembly.

4. Use the description for the part as it appears on the parts list page.

5. Use the part number shown in the parts list, column " Part No.".

6. When a complete assembly is required, use the assembly number given in the parts list. If no assembly number is given, order by main assembly title and show only the item numbers you want.

Note: Dresser-Rand reserves the right to substitute existing spare parts either of later design or alternative procurement, if this in Dresser-Rand's opinion will result in improved performance or expedited deliver.

Warning: The use of repair parts other than those included within the Dresser-Rand approved parts list may create hazardous condition over which the company has no control. Such hazardous condition can lead to accidents that my be life-threatening, can cause substantial bodily injury or result in damage to the equipment.

If the part ordered has a new part number assigned to it, or if the part has been replaced by a newer design with a different part number, the purchase order acknowledgment, and shipping papers as well as invoices will be written with the current part number first, and the old part number be included in the part description.

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For example: A new Inlet Guide is ordered from a parts list as part 84-7640. The new part delivered with the shipping papers is marked as “ 421-386-201 Inlet Guide (84-7640) ”.

The reference to the old part number is to confirm to you that the parts are fully interchangeable. This will avoid any unnecessary correspondence or delay in processing your order. Dresser-Rand suggests that you post this new number to your parts lists for future use.

- LHC-000-024 Example of Parts List Usage, Chapter 8.3.

7.1.3 PARTS ORDERS

Parts orders may be placed through the Le Havre Spare Parts Department.

Le Havre Parts Department
31 blvd, Winston Churchill
76600 Le Havre
FRANCE

Tel: 33-(0)-2-35-25-5446
Fax: 33-(0)-2-35-25-5364

Note: When ordering parts, it is essential that a purchase order be issued.

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7.1.4 COMPRESSORS TABLE OF CONTENTS

7.1.4.1 D06R8B COMPRESSOR - TAG NO. K-T7131/32

D06R8B Compressor
Tag No. K-T7131/32
Serial Number 1000181442

1000181442 Rev.OR.1

Document Title	Qty. per Unit	Dwg. No.	Part List No.
Unit Assembly	-	1000183022	PAGE 1000183022
Case Assembly	1	1000183026	PAGE 1000183026
Bundle Assembly	1	1000183025	PAGE 1000183025
Internal Assembly	1	1000183030	PAGE 1000183030
Rotor Assembly **, Serial No.: 10-01-050	1	1000183031	PAGE 1000183031
Thrust & Journal Bearing Assembly, 1ST Section End	1	1000183032	PAGE 1000183032
Journal Bearing Assembly, 2ND Section End	1	1000183033	PAGE 1000183033
Gas Seal Assembly - 1ST Section End	1	1000183034	PAGE 1000183034
Gas Seal Assembly - 2ND Section End	1	1000183035	PAGE 1000183035
Radial Vibration Probe Assembly,	4	1000183596	PAGE 1000183596
T.I.D. Installation Assembly	1	1000183039	PAGE 1000183039

Warning: For final “Painted Case” use Part No. 1000183024.

Note : ** This rotor, identified by its Serial No, has been originally allocated by engineering to this compressor. However, any of the rotors supplied for this type of compressor, within this project, can be installed instead .

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7.1.4.2 D06R8B COMPRESSOR - TAG NO. K-T7141/42

D06R8B Compressor
Tag No. K-T7141/42
Serial Number 1000181443

1000181443 Rev.OR.1

Document Title	Qty. per Unit	Dwg. No.	Part List No.
Unit Assembly	-	1000183412	PAGE 1000183412
Case Assembly	1	1000183415	PAGE 1000183415
Bundle Assembly	1	1000183414	PAGE 1000183414
Internal Assembly	1	1000192540	PAGE 1000192540
Rotor Assembly **, Serial No.: 10-01-051	1	1000183417	PAGE 1000183417
Thrust & Journal Bearing Assembly, 1ST Section End	1	1000183032	PAGE 1000183032
Journal Bearing Assembly, 2ND Section End	1	1000183033	PAGE 1000183033
Gas Seal Assembly - 1ST Section End	1	1000183034	PAGE 1000183034
Gas Seal Assembly - 2ND Section End	1	1000183035	PAGE 1000183035
Radial Vibration Probe Assembly,	4	1000183596	PAGE 1000183596
T.I.D. Installation Assembly	1	1000183039	PAGE 1000183039

Warning: For final "Painted Case" use Part No. 1000183413.

Note : ** This rotor, identified by its Serial No, has been originally allocated by engineering to this compressor. However, any of the rotors supplied for this type of compressor, within this project, can be installed instead .

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7.2 SPECIAL TOOLS

7.2.1 GENERAL INFORMATION

Most of the compressor and plant system maintenance can be performed by means of standard tools.

Prior to performing any maintenance, plant personnel should read all applicable user manual sections, particularly Chapter 2.2 Safety, Warnings and Precautions, Chapter 6.1 D06R8B Compressors Maintenance, and Chapter 8.1 Supplier's Manufacturer Documents. Attention should also be noted to the warnings listed under each manufacturer's documentation.

Note: Failure to use the recommended special tools may result in damage and/or unsatisfactory operation of components. Any warranty currently in force may be invalidated if correct maintenance procedures are not followed.

7.2.2 TOOLS FOR D06R8B COMPRESSOR

Index 1000192550 Rev. OR.1

Tool Description	* Qty.	Part No.	List No.	Drawing No.
Hydraulic Pump Assembly Kit	1	000-028-952	PAGE 000-028-952	-
Bundle Guide Stud	2	121-513-002	PAGE 121-513-002	-
Hydraulic Fit Thrust Disc Tool	1	534-221-206	PAGE 534-221-206	-
Pusher Ring Holder	1	446-613-207	PAGE 446-613-207	-
Hydraulic Fit Coupling	1	595-230-216	PAGE 595-230-216	-
Inlet Guide Support Sleeve	1	1000192228	PAGE 1000192228	-
Bearing Entering Sleeve *(2ND Section End)*	1	1000192226	PAGE 1000192226	-
Bearing Entering Sleeve *(1ST Section End)*	1	1000192227	PAGE 1000192227	-
Seal Entering Sleeve	1	1000192229	PAGE 1000192229	-
Spanner Wrench	1	491-574-204	PAGE 491-574-204	-
Bearing Guide Stud	2	121-513-003	PAGE 121-513-003	-
Grease[ab]	2	091-015-005	-	-

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Index 1000192550 Rev. OR.1

Tool Description	* Qty.	Part No.	List No.	Drawing No.
Tools Assembly For Mounting & Dismounting Gas Seal Assembly	1	595-195-202	PAGE 595-195-202	595-195-202
Gas Seal Tool Plate	1	1000190568	PAGE 1000190568	-
Screw[a]	1	000-005-318	-	-
Threaded Rod Assembly *(for Heads)*	2	524-560-208	PAGE 524-560-208	-
Threaded Rod Assembly *(for Gas Seal)*	2	595-205-203	PAGE 595-205-203	-
Bundle Cradle Assembly	1	685-325-203	PAGE 685-325-203	685-325-203
Roller Assembly	1	732-187-203	PAGE 732-187-203	-
Bundle Pusher Block	1	139-670-007	PAGE 139-670-007	-
Bundle Pusher Assembly	1	1000176623	PAGE 1000176623	-
Extension Case	1	1000192219	PAGE 1000192219	-
Bundle Clamp	2	121-593-202	PAGE 121-593-202	-
Cradle Bench Top Bundle Support	1	123-985-201	PAGE 123-985-201	-
Bundle Lifter	1	1000190567	PAGE 1000190567	-
Screw[a]	4	000-005-144	-	-
Bundle Extension Assembly	1	1000192221	PAGE 1000192221	-

a. Not illustrated.
b. Quantity = box.

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PART 8 - DOCUMENTS AND DRAWINGS

8.1 MANUFACTURER DOCUMENTATION 1-1

8.1.1 GENERAL INFORMATION 1-1

8.1.2 TABLE OF CONTENTS 1-1

8.2 PROJECT DOCUMENTS 2-1

8.2.1 GENERAL INFORMATION 2-1

8.2.1.1 PAPER FORMAT 2-1

8.2.1.2 ELECTRONIC FORMAT 2-1

8.2.2 TABLE OF CONTENTS 2-1

8.3 TECHNICAL SUPPORT DOCUMENTS 3-1

8.3.1 GENERAL INFORMATION 3-1

8.3.1.1 PAPER FORMAT 3-1

8.3.1.2 ELECTRONIC FORMAT 3-1

8.3.2 TABLE OF CONTENTS 3-1

OFFICIAL COPY 00P39490100490.FB003--FB003 Reviewed but not (C)--05-Nov-2012 08:15 AM

8.1 MANUFACTURER DOCUMENTATION

8.1.1 GENERAL INFORMATION

This chapter includes all documentation considered of value to maintenance personnel for the correct maintenance of that specific product.

The documentation inserted in this section will be supplied, as presented by the original supplier, retaining their original document structure. Should official project drawings be part of that supplied documentation they will be removed and a page inserted, annotated with the information as to the reference and location of that drawing(s). Similarly should component documentation be duplicated, this will be removed and suitable annotation inserted.

Note: Project P. & I.D. component lists (or Bill of Materials) include all necessary information relating to vendor make, model, size, description and/or tag numbers. These documents and the corresponding P. & I.Ds are the primary source for this information.

The following table of contents lists manufacturers and products with reference to :

1. Alphabetical tab,
2. Manufacturer's name listed alphabetically,
3. Dresser-Rand Le Havre Doc No.,
4. Item description,
5. Project tag or item number.

8.1.2 TABLE OF CONTENTS

Manufacturer's documentation

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 2				
A	**ABB**	LHA-105170B-07	**MOTOR** **(Lube Oil Console)** Model : M3KP 160 Drawing : 3GZF500116-28A	PM7131A PM7131B PM7141A PM7141B
	ABB	LHA-105170B-08	**MOTOR** **(Lube Oil Console)** Model : M3KP 160 Manual : Ex-motors 01-2004	PM7131A PM7131B PM7141A PM7141B

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Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 2				
	ACHECH	LHA-105170B-09	**VALVES** **(Lube Oil Console)** Model : Globe / Gate Drawing : 1GLF-12	Item 146 Item 153 Item 140 Item 141 Item 148 Item 144 Item 152 Item 150 Item 151 Item 170 Item 143
	ACHECH	LHA-105170B-10	**VALVES** **(Lube Oil Console)** Model : Globe / Gate Drawing : 1GAF-12	Item 146 Item 153 Item 140 Item 141 Item 148 Item 144 Item 152 Item 150 Item 151 Item 170 Item 143
	ACHECH	LHA-105170B-11	**VALVES** **(Lube Oil Console)** Model : Globe / Gate Manual : Gate, Globe & Check valves	Item 146 Item 153 Item 140 Item 141 Item 148 Item 144 Item 152 Item 150 Item 151 Item 170 Item 143
	AMAL	LHA-105170B-12	**FLAME ARRESTOR** **(Lube Oil Console)** Model : LEFC Drawing : 373227/2	N/A
	AMAL	LHA-105170B-13	**FLAME ARRESTOR** **(Lube Oil Console)** Model : LEFC Manual : ES/0/0/129	N/A

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Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 2				
	ASHCROFT	LHA-00-3531	**PRESSURE GAUGE** **(Lube Oil Console)** Model : T6500 Technical Datas : G1.T5500/E	T71-PI-3856 T71-PI-4856
	ASHCROFT	LHA-00-3657	**THERMOMETER** **(Lube Oil Console)** Model : E Technical Datas : G2.BIM/E	T71-TI-3847 T71-TI-4847
	ASHCROFT	LHA-00-3398	**THERMOWELL** **(Lube Oil Console)** Model :HF Technical Datas : G2.TWFLG/E	N/A
B	**BECEP**	LHA-105170B-48	**RUPTUR DISK** **(Gas Seal Panel)** Model : HPX Drawing : 91011063301	T71-PSE-3775 T71-PSS-3775 T71-PSE-3774 T71-PSS-3774 T71-PSE-4775 T71-PSS-4775 T71-PSE-4774 T71-PSS-4774
	BECEP	LHA-105170B-49	**RUPTUR DISK** **(Gas Seal Panel)** Model : HPX Drawing : 91011063302	T71-PSE-3775 T71-PSS-3775 T71-PSE-3774 T71-PSS-3774 T71-PSE-4775 T71-PSS-4775 T71-PSE-4774 T71-PSS-4774
	BECEP	LHA-105170B-50	**RUPTUR DISK** **(Gas Seal Panel)** Model : HPX Drawing : CD14087	T71-PSE-3775 T71-PSS-3775 T71-PSE-3774 T71-PSS-3774 T71-PSE-4775 T71-PSS-4775 T71-PSE-4774 T71-PSS-4774

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Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 2				
	BECEP	LHA-105170B-51	**RUPTUR DISK** **(Gas Seal Panel)** Model : HPX Drawing : SC14537	T71-PSE-3775 T71-PSS-3775 T71-PSE-3774 T71-PSS-3774 T71-PSE-4775 T71-PSS-4775 T71-PSE-4774 T71-PSS-4774
	BECEP	LHA-105170B-52	**RUPTUR DISK** **(Gas Seal Panel)** Model : HPX Manual : GEP-6069	T71-PSE-3775 T71-PSS-3775 T71-PSE-3774 T71-PSS-3774 T71-PSE-4775 T71-PSS-4775 T71-PSE-4774 T71-PSS-4774
	BECEP	LHA-105170B-53	**RUPTUR DISK** **(Gas Seal Panel)** Model : HPX Manual : GEP-6400	T71-PSE-3775 T71-PSS-3775 T71-PSE-3774 T71-PSS-3774 T71-PSE-4775 T71-PSS-4775 T71-PSE-4774 T71-PSS-4774
	BECEP	LHA-105170B-54	**RUPTUR DISK** **(Gas Seal Panel)** Model : HPX Manual : GEP-6100	T71-PSE-3775 T71-PSS-3775 T71-PSE-3774 T71-PSS-3774 T71-PSE-4775 T71-PSS-4775 T71-PSE-4774 T71-PSS-4774
	BECEP	LHA-105170B-55	**RUPTUR DISK** **(Gas Seal Panel)** Model : HPX Manual : SC14537-II	T71-PSE-3775 T71-PSS-3775 T71-PSE-3774 T71-PSS-3774 T71-PSE-4775 T71-PSS-4775 T71-PSE-4774 T71-PSS-4774

OFFICIAL COPY 00P39490100490.FB003 Reviewed with comments (C) 05-Nov-2012 08:15 AM

Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 2				
	BENTLY NEVADA	LHA-00-3483	**PROXIMITY TRANSDUCER** **(Vibration Probe)** Model 3300 & 3300 XL Manual :141078-01 Rev. K (11/07)	N/A
	BONNEY FORGE	LHA-105170B-14	**VALVES** **(Gas Seal Panel)** Model : Gate & Globe Drawing : HL100	Item 146 Item 153 Item 140 Item 141 Item 148 Item 144 Item 152 Item 150 Item 151 Item 170 Item 143 Item 227 Item 220
	BONNEY FORGE	LHA-105170B-15	**VALVES** **(Lube Oil Console)** Model : Gate / globe Drawing : L1 100	Item 146 Item 153 Item 140 Item 141 Item 148 Item 144 Item 152 Item 150 Item 151 Item 170 Item 143
	BONNEY FORGE	LHA-105170B-16	**VALVES** **(Lube Oil Console)** Model : Gate / globe Drawing : HL 300	Item 146 Item 153 Item 140 Item 141 Item 148 Item 144 Item 152 Item 150 Item 151 Item 170 Item 143 Item 227 Item 220

OFFICIAL COPY 00.39490100490.00003--FB003 Reviewed but not (C)--05-Nov-2012 08:15 AM

Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 2				
	BONNEY FORGE	LHA-105170B-17	**GATE VALVES** **(Lube Oil Console)** **(Package)** Model : 3/4" RTJ 2500 Manual : STD-UM-ENG Ed. 2007 rev.3	Item 146 Item 153 Item 140 Item 141 Item 148 Item 144 Item 152 Item 150 Item 151 Item 170 Item 143 Item 502
D	**DEDA TECNICA**	LHA-105170B-91	**COUPLING GUARD** **(Motor / Gear)** Drawing : 10.116.12 rev.A	
	DEDA TECNICA	LHA-105170B-92	**COUPLING GUARD** **(Gear / Compressor)** Drawing : 10.116.14	
	DERVAL	LHA-105170B-93	**Gate Valve** **(Package)** Model : 1" #150 Model : 3/4" #150 Model : 1/2" #150 Data Sheet : RD/02/IF01	Item 503 Item 504 Item 505 Item 520
	DERVAL	LHA-105170B-94	**Gate Valve** **(Package)** Model : 1" #150 Model : 3/4" #150 Model : 1/2" #150 Manual : RD0282en/2006	Item 503 Item 504 Item 505 Item 520
	DRESSER-RAND	LHA-105170B-88	**JOURNAL BEARING** Model : P400NX00 CW & CCW Instructions for Installing & Operating	N/A
SEE SEPARATE CD				
	DRESSER-RAND HOUSTON	LHA-105170B-05	**UNIT CONTROL PANEL (UCP)**	N/A

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Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 2				
E	**ELMESS**	LHA-105170B-20	**HEATER** **(Lube Oil Console)** Model : DHFB Drawing : 50020564	H-T7131 H-T7141
	ELMESS	LHA-105170B-21	**HEATER** **(Lube Oil Console)** Model : DHFB Manual : DH_B_Q_T_DE_EN	H-T7131 H-T7141
	EMERSON	LHA-105170B-22	**PRESSURE TRANSMITTER** **(Lube Oil Console)** Model : 3051 Drawing : 3051-001	T71-PDIT-3853 T71-PDIT-4853 T71-PIT-3851 T71-PIT-4851 T71-PDIT-3859 T71-PDIT-4859 T71-PIT-3860 T71-PIT-4860 T71-PIT-3843 T71-PIT-4843 T71-PIT-3844 T71-PIT-4844
	EMERSON	LHA-105170B-23	**DIFFERENTIAL PRESSURE TRANSMITTER** **(Lube Oil Console)** Model : 3051 Drawing : 3051-002	T71-PDIT-3853 T71-PDIT-4853 T71-PIT-3851 T71-PIT-4851 T71-PDIT-3859 T71-PDIT-4859 T71-PIT-3860 T71-PIT-4860 T71-PIT-3843 T71-PIT-4843 T71-PIT-3844 T71-PIT-4844

OFFICIAL COPY 00P39490100490.FB003 Reviewed with (C) 05-Nov-2012 08:15 AM

Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 2				
	EMERSON	LHA-105170B-24	**DIFFERENTIAL PRESSURE TRANSMITTER (Lube Oil Console)** Model : 3051 Drawing : 3051-004	T71-PDIT-3853 T71-PDIT-4853 T71-PIT-3851 T71-PIT-4851 T71-PDIT-3859 T71-PDIT-4859 T71-PIT-3860 T71-PIT-4860 T71-PIT-3843 T71-PIT-4843 T71-PIT-3844 T71-PIT-4844
	EMERSON	LHA-105170B-25	**DIFFERENTIAL PRESSURE TRANSMITTER (Lube Oil Console)** Model : 3051 Drawing : 3051-005	T71-PDIT-3853 T71-PDIT-4853 T71-PIT-3851 T71-PIT-4851 T71-PDIT-3859 T71-PDIT-4859 T71-PIT-3860 T71-PIT-4860 T71-PIT-3843 T71-PIT-4843 T71-PIT-3844 T71-PIT-4844
	EMERSON	LHA-105170B-26	**PRESSURE TRANSMITTER (Lube Oil Console)** Model : 3051 Drawing : 3051-003	T71-PDIT-3853 T71-PDIT-4853 T71-PIT-3851 T71-PIT-4851 T71-PDIT-3859 T71-PDIT-4859 T71-PIT-3860 T71-PIT-4860 T71-PIT-3843 T71-PIT-4843 T71-PIT-3844 T71-PIT-4844

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Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 2				
	EMERSON	LHA-105170B-27	**VALVE MANIFOLD** **(Lube Oil Console)** Model : 3051 Drawing : 0306RT-001	T71-PDIT-3853 T71-PDIT-4853 T71-PIT-3851 T71-PIT-4851 T71-PDIT-3859 T71-PDIT-4859 T71-PIT-3860 T71-PIT-4860 T71-PIT-3843 T71-PIT-4843 T71-PIT-3844 T71-PIT-4844
	EMERSON	LHA-105170B-28	**TEMPERATURE TRANSMITTER** **(Lube Oil Console)** Model : 3144 Drawing : 3144-001	T71-TIT-3852 T71-TIT-4852 T71-TIT-3842 T71-TIT-4842
	EMERSON	LHA-105170B-29	**TEMPERATURE TRANSMITTER** **(Lube Oil Console)** Model : 3144 Drawing : 0065-001	T71-TIT-3852 T71-TIT-4852 T71-TIT-3842 T71-TIT-4842
	EMERSON	LHA-105170B-30	**TEMPERATURE TRANSMITTER** **(Lube Oil Console)** Model : 3144 Drawing : 0065-002	T71-TIT-3852 T71-TIT-4852 T71-TIT-3842 T71-TIT-4842
	EMERSON	LHA-105170B-31	**LEVEL TRANSMITTER** **(Lube Oil Console)** Model : 3301 Drawing : 3301-001	T71-LIT-3840 T71-LIT-4840

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Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 3				
	EMERSON	LHA-00-1031	**PRESSURE TRANSMITTER (Lube Oil Console)** Model : 3051 Manual : 00809-0100-4001 rev EA december 2002	T71-PDIT-3853 T71-PDIT-4853 T71-PIT-3851 T71-PIT-4851 T71-PDIT-3859 T71-PDIT-4859 T71-PIT-3860 T71-PIT-4860 T71-PIT-3843 T71-PIT-4843 T71-PIT-3844 T71-PIT-4844
	EMERSON	LHA-00-3517	**TEMPERATURE TRANSMITTER (Lube Oil Console)** Model : 3144 Manual : 00809-0100-4021 rev EB May 2007	T71-TIT-3852 T71-TIT-4852 T71-TIT-3842 T71-TIT-4842
VOLUME 4				
	EMERSON	LHA-00-3340	**LEVEL TRANSMITTER (Lube Oil Console)** Model : 3301 Manual : 00809-0100-4811 rev CA february 2006	T71-LIT-3840 T71-LIT-4840
	EMERSON	LHA-105170B-60	**DIFFERENTIAL PRESSURE TRANSMITTER (Gas Seal Panel)** Model : 3051 Drawing : 3051CD-002	T71-PDIT-3765 T71-PDIT-4765 T71-PDIT-3768 T71-PDIT-4768 T71-PDIT-3776 T71-PDIT-4776 T71-PDIT-3777 T71-PDIT-4777
	EMERSON	LHA-105170B-61	**DIFFERENTIAL PRESSURE TRANSMITTER (Gas Seal Panel)** Model : 3051 Drawing : 3051CD-001	T71-PDIT-3765 T71-PDIT-4765 T71-PDIT-3768 T71-PDIT-4768 T71-PDIT-3776 T71-PDIT-4776 T71-PDIT-3777 T71-PDIT-4777

OFFICIAL COPY 00P39490100490.FB003 Reviewed with (C) 05-Nov-2012 08:15 AM

Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 4				
	EMERSON	LHA-105170B-62	**PRESSURE TRANSMITTER (Gas Seal Panel)** Model : 3051 Drawing : 1195-001	T71-PIT-3767 T71-PIT-4767 T71-FIT-3770 T71-FIT-4770 T71-FIT-3771 T71-FIT-4771 T71-PIT-3778 T71-PIT-4778 T71-PIT-3779 T71-PIT-4779 T71-PIT-3750 T71-PIT-4750 T71-PIT-3751 T71-PIT-4751
	EMERSON	LHA-105170B-63	**PRESSURE TRANSMITTER (Gas Seal Panel)** Model : 3051 Drawing : 3051CG-001	T71-PIT-3767 T71-PIT-4767 T71-FIT-3770 T71-FIT-4770 T71-FIT-3771 T71-FIT-4771 T71-PIT-3778 T71-PIT-4778 T71-PIT-3779 T71-PIT-4779 T71-PIT-3750 T71-PIT-4750 T71-PIT-3751 T71-PIT-4751

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Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 4				
	EMERSON	LHA-105170B-64	**DIFFENTIAL PRESSURE TRANSMITTER** **(Gas Seal Panel)** Model : 3051 Manual : 00825-0100-4001 rev HA	T71-PIT-3767 T71-PIT-4767 T71-FIT-3770 T71-FIT-4770 T71-FIT-3771 T71-FIT-4771 T71-PIT-3778 T71-PIT-4778 T71-PIT-3779 T71-PIT-4779 T71-PIT-3750 T71-PIT-4750 T71-PIT-3751 T71-PIT-4751 T71-PDIT-3765 T71-PDIT-4765 T71-PDIT-3768 T71-PDIT-4768 T71-PDIT-3776 T71-PDIT-4776 T71-PDIT-3777 T71-PDIT-4777
	EMERSON	LHA-105170B-65	**INTEGRAL ORIFICE PRIMARY ELEMENT** **(Gas Seal Panel)** Model : 3051 Manual : 00825-0100-4686 rev BA	T71-PIT-3767 T71-PIT-4767 T71-FIT-3770 T71-FIT-4770 T71-FIT-3771 T71-FIT-4771 T71-PIT-3778 T71-PIT-4778 T71-PIT-3779 T71-PIT-4779 T71-PIT-3750 T71-PIT-4750 T71-PIT-3751 T71-PIT-4751
F	**FLENDER GRAFFENSTADEN**	LHA-105170B-56	**GEARBOX** Model : TX71/6C Installation, operating and maintenance instructions.	N/A

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Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 4				
	FLOWSERVE	LHA-105170B-04	**BARRIER SEAL** Gaspac Model L Installation, Operating & Maintenance Instructions	N/A
	FLOWSERVE	LHA-105170B-95	**Check Valve** **(Package)** Model : RK86A Data Sheet : 810587-02/707cm	Item 501 Item 502
	FLOWSERVE	LHA-105170B-96	**Check Valve** **(Package)** Model : RK86A Manual : 808519-07/1106cs	Item 501 Item 502
VOLUME 5				
G	**GEORGIN**	LHA-105170B-66	**VALVES** **(Gas Seal Panel)** Model : Block and bleed Drawing : A3300-DBB-009	Item 228 Item 218 Item 217 Item 226
	GEORGIN	LHA-105170B-67	**VALVES** **(Gas Seal Panel)** Model : Block and bleed Drawing : A3300-DBB-010	Item 228 Item 218 Item 217 Item 226
	GEORGIN	LHA-105170B-68	**VALVES** **(Gas Seal Panel)** Model : Block and bleed Drawing : A3300-DBB-008	Item 228 Item 218 Item 217 Item 226
H	**HYDAC**	LHA-105170B-32	**FLOW LIMITER** **(Lube Oil Console)** Model : DRV 12 Drawing: DRV	N/A
	HYDAC	LHA-00-3142	**FLOW LIMITER** **(Lube Oil Console)** Model : DRV 12 Manual : DV/DRV	N/A
I	**INDUFIL**	LHA-105170B-33	**FILTERS** **(Lube Oil Console)** Model : IDGL Drawing : 045.21.350.99	ST-7133A ST-7133B ST-7143A ST-7143B

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Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 5				
	INDUFIL	LHA-105170B-34	**FILTERS** **(Lube Oil Console)** Model : IDGL Manual :4000256122.1/2	ST-7133A ST-7133B ST-7143A ST-7143B
	INDUFIL	LHA-105170B-69	**FILTER** **(Gas Seal Panel)** Model : IDGS / IDSH Drawing : FFA1-085-0426 rev 1	S-T7131A S-T7141A S-T7131B S-T7141B S-T7132A S-T7142B S-T7132A S-T7142B
	INDUFIL	LHA-105170B-70	**FILTER** **(Gas Seal Panel)** Model : IDGS / IDSH Drawing : FFA1-085-0423 Rev 2	S-T7131A S-T7141A S-T7131B S-T7141B S-T7132A S-T7142B S-T7132A S-T7142B
	INDUFIL	LHA-105170B-71	**FILTER** **(Gas Seal Panel)** Model : IDGS / IDSH Manual : SF-10.74E Rev.D	S-T7131A S-T7141A S-T7131B S-T7141B S-T7132A S-T7142B S-T7132A S-T7142B
	INDUFIL	LHA-105170B-72	**FILTER** **(Gas Seal Panel)** Model : IDGS / IDSH Manual : SF-10.84E Rev.G	S-T7131A S-T7141A S-T7131B S-T7141B S-T7132A S-T7142B S-T7132A S-T7142B
J	**JC SARTORIUS**	LHA-105170B-89	**TRHUST BEARING**	N/A

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Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 5				
	JOHN CRANE	LHA-105170B-02	**COUPLING** **(Package)** Model : H_FE Fitting, Operation, and Maintenance Instructions : 87962/07 - Issue b	N/A
	JOHN CRANE	LHA-105170B-03	**COUPLING** **(Package)** Model : T-Series User Manual : BE040-195	N/A
K	**KLUBER**	LHA-105170B-47	**LEVEL GAUGE** **(Lube Oil Console)** Model : MRA L = 890 Drawing : 10 PL 715	T71-LG-3841 T71-LG-4841
L	**LEISTRITZ**	LHA-105170B-35	**PUMP** **(Lube Oil Console)** Model : L3MF Drawing : E1624949	P-T7131A P-T7131B P-T7141A P-T7141B
	LEISTRITZ	LHA-105170B-36	**PUMP** **(Lube Oil Console)** Model : L3MF Drawing : E1606257	P-T7131A P-T7131B P-T7141A P-T7141B
	LEISTRITZ	LHA-105170B-37	**PUMP** **(Lube Oil Console)** Model : L3MF Drawing : E1606724	P-T7131A P-T7131B P-T7141A P-T7141B
	LEISTRITZ	LHA-105170B-38	**PUMP** **(Lube Oil Console)** Model : L3MF Drawing : 35800/1	P-T7131A P-T7131B P-T7141A P-T7141B
	LEISTRITZ	LHA-105170B-39	**PUMP** **(Lube Oil Console)** Model : L3MF Manual :E185 5690	P-T7131A P-T7131B P-T7141A P-T7141B
	LESER	LHA-105170B-40	**PRESSURE SAFETY VALVES** **(Lube Oil Console)** Model : 5264 Drawing :5262.1102	T71-PSV-3845 T71-PSV-4845 T71-PSV-3846 T71-PSV-4846

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Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 5				
	LESER	LHA-00-3525	**PRESSURE SAFETY VALVES** **(Lube Oil Console)** Model : 5264 Manual : OPERATING october 2005	T71-PSV-3845 T71-PSV-4845 T71-PSV-3846 T71-PSV-4846
M	**MASONEILAN**	LHA-000-377	**CONTROL VALVES** **(Lube Oil Console)** Model : 535H / 526 Manual : EY 52004E	T71-PCV-3848 T71-PCV-4848 T71-PCV-3855 T71-PCV-4855
	MASONEILAN	LHA-000-133	**CONTROL VALVES** **(Lube Oil Console)** Model : 535H / 526 Manual : 176418E	T71-PCV-3848 T71-PCV-4848 T71-PCV-3855 T71-PCV-4855
	MASONEILAN	LHA-105170B-18	**CONTROL VALVES** **(Lube Oil Console)** Model : 535H / 526 Drawing : 21-02470-01	T71-PCV-3848 T71-PCV-4848 T71-PCV-3855 T71-PCV-4855
	MASONEILAN	LHA-105170B-19	**CONTROL VALVES** **(Lube Oil Console)** Model : 535H / 526 Drawing : 21-02470-02	T71-PCV-3848 T71-PCV-4848 T71-PCV-3855 T71-PCV-4855
	MIOFILTRE	LHA-105170B-97	**BREATHER** **(Package)** Model : FC175EC097S601 Manual	Item 531
O	**OELTECHNIK**	LHA-00-3677	**LUBE OIL CONSOLE** Model API 614 Manual : LOC.START.01.E.rev0	A-T7131 A-T7141
	OELTECHNIK	LHA-105170B-41	**LUBE OIL CONSOLE** Model API 614 Manual : LOD.START.01.E.rev0	A-T7131 A-T7141
P	**PARKER**	LHA-105170B-86	**NEEDLE VALVE** **(Gas Seal Panel)** Model : HNV*8FF	Item 218
R	**RIVERHAWK**	LHA-00-3463	**PUMP ASSEMBLY KIT** Instruction Manual IM122 for Hydraulic Hub Installation and Removal	N/A

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Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 5				
S	**SAMSON**	LHA-105170B-73	**CONTROL VALVES** **(Gas Seal Panel)** Model : 3252 Drawing : A044-0427 rev 0	T71-PCV-3766 T71-FCV-4769 T71-PCV-3766 T71-FCV-4769
	SAMSON	LHA-105170B-74	**CONTROL VALVES** **(Gas Seal Panel)** Model : 3252 Drawing : A252-0428 rev 2	T71-PCV-3766 T71-FCV-4769 T71-PCV-3766 T71-FCV-4769
	SAMSON	LHA-105170B-75	**CONTROL VALVES** **(Gas Seal Panel)** Model : 3252 Technical Datas : T2626EN	T71-PCV-3766 T71-FCV-4769 T71-PCV-3766 T71-FCV-4769
	SAMSON	LHA-105170B-76	**CONTROL VALVES** **(Gas Seal Panel)** Model : 3252 Technical Datas : T8353EN	T71-PCV-3766 T71-FCV-4769 T71-PCV-3766 T71-FCV-4769
	SAMSON	LHA-105170B-77	**CONTROL VALVES** **(Gas Seal Panel)** Model : 3252 Technical Datas : T8310-1EN	T71-PCV-3766 T71-FCV-4769 T71-PCV-3766 T71-FCV-4769
	SAMSON	LHA-105170B-78	**CONTROL VALVES** **(Gas Seal Panel)** Model : 3252 Technical Datas : T8384-2/3EN	T71-PCV-3766 T71-FCV-4769 T71-PCV-3766 T71-FCV-4769
	SAMSON	LHA-105170B-79	**CONTROL VALVES** **(Gas Seal Panel)** Model : 3252 Technical Datas : T8546EN	T71-PCV-3766 T71-FCV-4769 T71-PCV-3766 T71-FCV-4769
	SAMSON	LHA-105170B-80	**CONTROL VALVES** **(Gas Seal Panel)** Model : 3252 Mounting and Operating Instructions : EB2626-1 EN	T71-PCV-3766 T71-FCV-4769 T71-PCV-3766 T71-FCV-4769
	SAMSON	LHA-105170B-81	**CONTROL VALVES** **(Gas Seal Panel)** Model : 3252 Mounting and Operating Instructions : EB8311EN	T71-PCV-3766 T71-FCV-4769 T71-PCV-3766 T71-FCV-4769

OFFICIAL COPY 00P39490100490.FB003 Reviewed but not commented 05-Nov-2012 08:15 AM

Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 6				
	SAMSON	LHA-105170B-82	**CONTROL VALVES** **(Gas Seal Panel)** Model : 3252 Mounting and Operating Instructions : EB8384-3EN	T71-PCV-3766 T71-FCV-4769 T71-PCV-3766 T71-FCV-4769
	SAMSON	LHA-105170B-83	**CONTROL VALVES** **(Gas Seal Panel)** Model : 3252 Technical Datas : EB8546EN	T71-PCV-3766 T71-FCV-4769 T71-PCV-3766 T71-FCV-4769
	SAMSON	LHA-105170B-84	**CONTROL VALVES** **(Gas Seal Panel)** Model : 3252 Technical Datas : EB8053N	T71-PCV-3766 T71-FCV-4769 T71-PCV-3766 T71-FCV-4769
	SCA	LHA-105170B-06	**ANTI-VIBRATION MOUNTS** Doc no. : SCA21594-6000 Unpacking, Installation & Maintenance Procedures	N/A
	SWAGELOK	LHA-105170B-87	**CHECK VALVE** **(Gas Seal Panel)** Drawing : MS-01-176	Item 203 Item 221 Item 232
T	**TECSAS**	LHA-105170B-85	**ORIFICE METER** **(Gas Seal Panel)** Model : Orifice meter Drawing : 10-404-2C	Item 223 Iem 224
	THERMOREG	LHA-105170B-42	**THERMOSTATIC VALVES** **(Lube Oil Console)** Model : M80T Drawing : 3141	T71-TCV-3858 T71-TCV-4858
	THERMOREG	LHA-105170B-43	**THERMOSTATIC VALVES** **(Lube Oil Console)** Model : M80T Drawing : 3143	T71-TCV-3858 T71-TCV-4858
	THERMOREG	LHA-105170B-44	**THERMOSTATIC VALVES** **(Lube Oil Console)** Model : M80T Manual :TH10/90.1	T71-TCV-3858 T71-TCV-4858

OFFICIAL COPY 00P39490100490.FB003 Reviewed with comments (C) 05-Nov-2012 08:15 AM

Manufacturer's documentation (Continued)

Tab	Manufacturer	Doc. N°	Item description	Tag N°
VOLUME 6				
	THERMOWAVE	LHA-105170B-45	**COOLER** **(Lube Oil Console)** Model : TL200 Manual : BA AA 000 EN 00 04	E-T7133 E-T7143
VOLUME 7				
V	**VALVAR**	LHA-105170B-46	**VALVES** **(Lube Oil Console)** Model : Butterfly valves Manual : VF-73	N/A
	VIMEC	LHA-105170B-98	**DOUBLE BLOCK & BLEED VALVE** **(Package)** Model : 3/4"#1500 RTJ Drawing : VMC-655	Item 501
	VIMEC	LHA-105170B-99	**DOUBLE BLOCK & BLEED VALVE** **(Package)** Model : 3/4"#1500 RTJ Manual : OM-09-2009	Item 501
W	**WEG**	LHA-105170B-01	**MOTOR** Type : MGW900 Installation, Operation and Maintenance Manual, Rev.0	A-T7130 A-T7140
	WIKA	LHA-105170B-100	**THERMOMETER** **(Package)** Model : S5550 1/2"NPT Drawing & Data Sheet TM 55.01	TI1755 TI1754 TI1756 TI1757 TI1758
	WIKA	LHA-105170B-101	**THERMOMETER** **(Package)** Model : S5550 1/2"NPT Manual : 12968065 02/2009	TI1755 TI1754 TI1756 TI1757 TI1758

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8.2 PROJECT DOCUMENTS

8.2.1 GENERAL INFORMATION

This chapter contains the engineering drawings and documents supplied for this contract.

The following table of contents lists all project documents organized after Project Document number.

The table of contents provides the following information :

1. Project Dresser-Rand Document number.
2. Document title.

8.2.1.1 PAPER FORMAT

The paper copy drawing file consists of several covers. The drawing file table of contents lists all official Project drawings.

8.2.1.2 ELECTRONIC FORMAT

From the drawing file table of contents links are established to each individual document.

8.2.2 TABLE OF CONTENTS

Project Documents

Document N°	Title
Volume 8	
HI.39490-AB-002 *105170-CPC-B*	Performance Curves - main B
HI.39490-AC-002 *1000182929*	Compressor Train General Arrangement
HI.39490-AC-011 *1000182933*	Compressor Train Customer Connections
HI.39490-AC-015 *1000182936*	Lube Oil Rundown Tank General Arrangement
HI.39490-AC-017 *1000181666*	Lube Oil Console General Arrangement
HI.39490-AD-002 *1000190602*	Gas Seal General Arrangement

OFFICIAL COPY 00P39490100490.FB003--FB003 Reviewed but not (C)--05-Nov-2012 08:31 AM

Project Documents (Continued)

Document N°	Title
Volume 8	
HI.39490-AD-006 *F-N0259-00*	General Arrangement Drawing Low Speed Coupling
HI.39490-AD-010 *F-N0151-00*	General Arrangement Drawing High Speed Coupling
HI.39490-AD-015 *02-N-621*	Outline Drawing
HI.39490-AD-017 *10000882797*	Motor Outline Drawing
HI.39490-AD-025 *1000190689*	Demister General Arrangement
HI.39490-AD-027 *FA DS&GA1*	Flame Arrestor Data sheet and General Arrangement - Main A & B
HI.39490-AF-005 *1000182555*	Temperature & Vibration P&ID
HI.39490-AF-006 *1000181639*	Lube Oil Console P&ID
HI.39490-AF-007 *1000181647*	Lube Oil System P&ID
HI.39490-AF-008 *1000181655*	Gas Seal P&ID
HI.39490-AG-002 *660-445*	Control Narrative
HI.39490-AJ-002 *1000186867*	Anti-Surge And Suction Throttle Valves Data Sheets
HI.39490-AK-002 *105170-CDS-B*	Compressor Data Sheet - main B
HI.39490-AK-007 *105170-CPL-DS2*	Data Sheet for Low speed coupling for Motor-Gear
HI.39490-AK-011 *105170-CPL-DS6*	Data Sheet for High speed coupling for Gear-Compressor
HI.39490-AK-015 *36330-B*	Gearbox - Data Sheet - main A Future & B (12126 rpm)

OFFICIAL COPY 00P39490100490.FB003--FB003 Reviewed but not (C)--05-Nov-2012 08:31 AM

Project Documents (Continued)

Document N°	Title
Volume 8	
HI.39490-AL-002 *660-444*	UCP - Cause and Effect Chart
HI.39490-AS-003 *1000182559*	Compressor Train Temperature & Vibration Bill Of Material
HI.39490-AS-007 *1000181640*	Lube Oil Console List Of Materials
HI.39490-AS-010 *1000181648*	Lube Oil System List Of Materials
HI.39490-AS-014 *1000181656*	Gas Seal System List Of Materials
HI.39490-AT-001 *1000183682*	Piping Specification
HI.39490-AT-003 *105170-MWS*	Compressor E&I Installation Procedure
HI.39490-AT-004 *MPS758-771 2010-07*	Motor Expo Purge System Spec
Volume 9	
HI.39490-BA-002 *660-448*	UCP - Logic Diagrams
HI.39490-BB-002 *660-443*	UCP - I/O Listing
HI.39490-BB-006 *660-446*	Unit Control Panel OPC Data List
HI.39490-BF-002 *660-440*	Face Layout Unit Control Panel
HI.39490-BG-002 *1000184239*	On Skid Wiring Diagram
HI.39490-BG-006 *660-441*	Schematic Diagrams Unit Control Panel
HI.39490-BG-010 *10000895006*	Motor Wiring Diagram/Connection Diagram
HI.39490-BI-002 *660-442*	Terminal Block Layout Unit Control Panel

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Project Documents (Continued)

Document N°	Title
Volume 9	
HI.39490-BK-002 *1000181662*	Utilities List Main B
HI.39490-BL-001 *105170-DSM-01*	Motor Data Sheet
HI.39490-BL-002 *CM-13696-0-2010*	Torque & Power vs Speed, Stall Protection Curves
HI.39490-BL-003 *CM-14082-0-2010*	Thermal Limit Curves
HI.39490-BL-010 *CM-14081-0-2010*	Performance vs Output Curve
HI.39490-CS-002 *TA-105170-MainB*	Compressor Mechanical & Performance Test Procedure - Type 2 - Main B
HI.39490-EE-001 *105170-PS01-1*	Packing and Preservation Procedure - Compressor Skid - Main A & B, Injection
HI.39490-HB *RT00437-10*	Motor, Predicted Noise Levels
HI.39490-HB-002 *1000189157*	Noise Data Sheet

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8.3 TECHNICAL SUPPORT DOCUMENTS

8.3.1 GENERAL INFORMATION

This chapter contains Technical Support documents not included in the official Project Drawings table of contents.

The table of contents gives the following information :

1. Dresser-Rand document number,
2. Technical support document title.

8.3.1.1 PAPER FORMAT

The technical support documents table of contents lists all technical support documents.

8.3.1.2 ELECTRONIC FORMAT

From the technical support documents table of contents links are established to each individual document.

8.3.2 TABLE OF CONTENTS

Technical support documents

Document N°	Title
Volume 10	
003-021-001R13	Export Packaging Specification
003-067-001R39	Installation & Commissioning Requirements for Dresser-Rand Turbomachinery
003-074-001R6	Protection of non Running Compressors and Systems
003-171-001R22	Preparation for Shipment and Short-term Storage
523-264-344	Specification Plate Stamping - D06R8B Compressor Tag No K-T7141/42 - Serial No 1000181443
523-264-346	Specification Plate Stamping - D06R8B Compressor Tag No K-T7131/32 - Serial No 1000181442
595-195-202	Tools Assembly for Mounting & Dismounting Gas Seal Assembly, Drawing
685-325-203	Bundle Cradle Assembly, Drawing

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Technical support documents (Continued)

Document N°	Title
Volume 10	
105170B198	Revision Record Sheet
105170B199	User Manuel Structure
1000183022	Unit Assembly, Drawing, s/n 1000181442
1000183025	Bundle Assembly Drawing, s/n 1000181442
1000183026	Case Assembly, s/n 100181442, Drawing
1000183030	Internal Assembly, Drawing, s/n 1000181442
1000183031	Rotor Assembly, Serial No.: 10-01-050, Drawing
1000183032	Thrust & Journal Bearing Assembly, 1ST Section End, Drawing
1000183033	Journal Bearing Assembly, 2ND Section End, Drawing
1000183034	Gas Seal Assembly - 1ST Section End, Drawing
1000183035	Gas Seal Assembly - 2ND Section End, Drawing
1000183039	T.I.D. Installation Assembly, Drawing
1000183045	DRESSER-RAND Journal Bearing Assembly, 1ST Section, Drawing
1000183046	JC SARTORIUS Thrust Bearing General Arrangement
1000183051	DRESSER-RAND Journal Bearing Assembly, 2ND Section, Drawing
1000183412	Unit Assembly, Drawing, s/n 1000181443
1000183414	Bundle Assembly Drawing, s/n 1000181443
1000183415	Case Assembly, s/n 100181443, Drawing
1000183417	Rotor Assembly, Serial No.: 10-01-051 - Drawing
1000183596	Radial Vibration Probe Assembly, Drawing
1000184143	DRESSER-RAND Gas Seal General Arrangement - 1ST Section, Drawing

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Technical support documents (Continued)

Document N°	Title
Volume 10	
1000184144	DRESSER-RAND Gas Seal General Arrangement - 2ND Section, Drawing
1000186418	Flowserve CW & CWW Barrier Seal.
1000192540	Internal Assembly, Drawing, s/n 1000181443
ALN-1000194939	Shaft Alignment Drawing - Main Gas Compressor Package A
INST-GS-199 REV1	Specification - Checking for the Free Axial Movement of the Gas Seal.
INST-GS-200	Specification - Checking for the Free Rotational Movement of the Gas Seal.
LHC-000-001	Hydraulic Fit Type Coupling Tools Illustration
LHC-000-003	Embedded Temperature Detectors Illustrations, FIG. A & B
LHC-000-005	“Rainbow”, Operational Guide Lines for Vibration on Centrifugal Compressor
LHC-000-006	Centrifugal Compressor Surge Illustrations
LHC-000-007	Puller blocks, retaining and shear rings illustrations
LHC-000-008	Impeller/Diffuser Overlap Illustration
LHC-000-009	Safety Labels For Operators
LHC-000-010	Safety Labels For Container
LHC-000-024	Example Of Parts List Usage
LHC-000-029	Safety Labels For Special Gas
LHC-000-037	Lubricating Oil Recommendations
LHC-000-069	Components of a Gas Seal
LHC-000-070	Description Of Gas And Groove Operation
LHC-000-072	Gas Seal Arrangements
LHC-000-073	Pressure Profile In The Groove Area
LHC-000-075	O-Ring Location

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Technical support documents (Continued)

Document N°	Title
Volume 10	
LHC-000-076	Shrouded Seat Design
LHC-000-124	Hydraulic Fit Thrust Disc Axial Displacement Illustration
LHC-000-292	Rotor Containers (N_2 Storage), Instructions Manual
LHC-000-473	Maximum Allowable Misalignment Chart
LHC-105170B-01	Hydraulic Fit Type Coupling Dimensions
LHC-105170B-02	Rotor Dimensions, D06R8B Compressors
LHC-105170B-03	Head connections, D06R8B Compressors
LHC-105170B-04	Weight Record, D06R8B Compressors
PAGE 000-028-952	Hydraulic Pump Assembly, Tools List
PAGE 121-513-002	Bundle Guide Stud, Tool Lis
PAGE 121-513-003	Bearing Guide Studs, Tool List
PAGE 121-593-202	Bundle Clamp, Tools List
PAGE 123-985-201	Craddle Bench Top Bundle Support, Tool list
PAGE 139-670-007	Bundle Pusher Block, Tool List
PAGE 446-613-207	Pusher Ring Holder, Tool List
PAGE 491-574-204	Spanner Wrench, Tool List
PAGE 524-560-208	Threaded Rod Assembly *(for Heads)*, Tool List
PAGE 534-221-206	Hydraulic Fit Thrust Disc Tools, Tools List
PAGE 595-195-202	Tools Assembly for Mounting & Dismounting Gas Seal Assembly, Tool Lists
PAGE 595-205-203	Threaded Rod Assembly *(for Gas Seal)*, Tool List
PAGE 595-230-216	Hydraulic Fit Coupling, Tool List
PAGE 685-325-203	Bundle Cradle Assembly, Tool List
PAGE 732-187-203	Roller Assembly, Tool List

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Technical support documents (Continued)

Document N°	Title
Volume 10	
PAGE 1000176623	Bundle Pusher Assembly, Tool List
PAGE 1000183022	Unit Assembly, Parts List, s/n 1000181442
PAGE 1000183025	Bundle Assembly, Parts List, s/n 1000181442
PAGE 1000183026	Case Assembly, s/n 100181442, Parts List
PAGE 1000183030	Internal Assembly, Part List, s/n 1000181442
PAGE 1000183031	Rotor Assembly, Serial No.: 10-01-050 - Parts List
PAGE 1000183032	Thrust & Journal Bearing Assembly, 1ST Section, Parts List
PAGE 1000183033	Journal Bearing Assembly, 2ND Section - Parts List
PAGE 1000183034	Gas Seal Assembly - 1st Section, Parts List
PAGE 1000183035	Gas Seal Assembly - 2nd Section, Parts List
PAGE 1000183039	T.I.D. Installation Assembly, Parts List
PAGE 1000183412	Unit Assembly, Parts List, s/n 1000181443
PAGE 1000183414	Bundle Assembly, Parts List, s/n 1000181443
PAGE 1000183415	Case Assembly, s/n 100181443, Parts List
PAGE 1000183417	Rotor Assembly, Serial No.: 10-01-051, Parts List
PAGE 1000183596	Radial Vibration Probe Assembly, Parts List
PAGE 1000190567	Bundle Lifter, Tool List
PAGE 1000190568	Gas Seal Tool Plate, Tool List
PAGE 1000192219	Extension Case, Tool List
PAGE 1000192221	Bundle Extension Assembly, Tool List
PAGE 1000192226	Bearing Entering Sleeve *(2ND Section End)*, Tool List
PAGE 1000192227	Bearing Entering Sleeve *(1ST Section End)*, Tool List
PAGE 1000192228	Inlet Guide Support Sleeve, Tool List
PAGE 1000192229	Seal Entering Sleeve, Tool List

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Technical support documents (Continued)

Document N°	Title
Volume 10	
PAGE 1000192540	Internal Assembly, Part List, s/n 1000181443
SD105170B01-D6R8B	Contract Seal Diagram - D06R8B Compressors